

pybind11——C++11 Python

pybind11 是一个 C++ 库，它允许 C++ 代码与 Python 代码无缝集成。它是由 David Abrahams 开发的，是 Boost.Python 的一个分支。它提供了与 Boost.Python 类似的功能，但更易于使用，并且支持 C++11 的许多新特性。

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1.1 安装

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- 支持 C++11 的许多新特性
- 易于使用
- 支持移动语义
- 支持智能指针
- 支持 lambda 表达式
- 支持移动语义
- 支持智能指针
- 支持 ranges
- 支持移动语义
- 支持智能指针
- STL 容器
- 支持移动语义
- Internal references with correct reference counting

- Python C++

1.2

pybind11

- Python 2.7, 3.5+, PyPy/PyPy3 7.3
- lambda
- pybind11 C++11 (pybind11 uses C++11 move constructors and move assignment operators whenever possible to efficiently transfer custom data types.)
- Python buffer C++ Eigen NumPy
- pybind11 NumPy
- Python
- Boost.Python
- `constexpr`
- C++ Python pickle/unpickle

1.3

1. Clang/LLVM 3.3 (Apple Xcode's clang 5.0.0)
2. GCC 4.8
3. Microsoft Visual Studio 2015 Update 3
4. Intel classic C++ compiler 18 or newer (ICC 20.2 tested in CI)
5. Cygwin/GCC (previously tested on 2.5.1)
6. NVCC (CUDA 11.0 tested in CI)
7. NVIDIA PGI (20.9 tested in CI)

1.4 ☐☐

This project was created by Wenzel Jakob. Significant features and/or improvements to the code were contributed by Jonas Adler, Lori A. Burns, Sylvain Corlay, Eric Cousineau, Aaron Gokaslan, Ralf Grosse-Kunstleve, Trent Houlston, Axel Huebl, @hulucc, Yannick Jadoul, Sergey Lyskov Johan Mabilie, Tomasz Miąsko, Dean Moldovan, Ben Pritchard, Jason Rhineland, Boris Schäling, Pim Schellart, Henry Schreiner, Ivan Smirnov, Boris Staletic, and Patrick Stewart.

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1.5 ☐☐

See the [contributing guide](#) for information on building and contributing to pybind11.

1.6 License

pybind11 is provided under a BSD-style license that can be found in the [LICENSE](#) file. By using, distributing, or contributing to this project, you agree to the terms and conditions of this license.

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3. 安装

在[pybind/pybind11 on GitHub](https://github.com/pybind/pybind11)上找到pybind11的仓库，并克隆到本地。

3.1 使用Git子模块

在根目录下使用Git子模块添加pybind11仓库。



```
git submodule add -b stable ../../pybind/pybind11 extern/pybind11
git submodule update --init
```

在根目录下使用`extern`子模块添加GitHub仓库。使用`https`或`ssh` URL。
在根目录下使用`git`命令添加子模块。`.git`子模块。

在根目录下使用`include`子模块添加`extern/pybind11/include`子模块。使用`Build System`子模块添加`pybind11`子模块。

3.2 使用PyPI

使用`pip`安装PyPI上的Pybind11 Python包。



```
pip install pybind11
```

在根目录下使用`Python`子模块添加`root`子模块。使用`pybind11`子模块。



```
pip install "pybind11[global]"
```

```
Python root /usr/local/include/pybind11
/usr/local/share/cmake/pybind11
pyproject.toml
```

3.3 使用conda-forge

You can use pybind11 with conda packaging via [conda-forge](#):



```
conda install -c conda-forge pybind11
```

3.4 使用vcpkg

Microsoft [vcpkg](#) pybind11

```
git clone https://github.com/Microsoft/vcpkg.git
cd vcpkg
./bootstrap-vcpkg.sh
./vcpkg integrate install
vcpkg install pybind11
```

3.5 使用brew

brew Homebrew on macOS, or Linuxbrew on Linux pybind11



```
brew install pybind11
```

3.6 其他位置

Other locations you can find pybind11 are [listed here](#); these are maintained by various packagers and the community.

pybind11

1. 安装

1.1 安装依赖

在python3-dev安装pybind11头文件include pybind11python3头文件
cmake



```
set(PYTHON_TARGET_VER 3.6)
find_package(PythonInterp ${PYTHON_TARGET_VER} EXACT)
find_package(PythonLibs ${PYTHON_TARGET_VER} EXACT REQUIRED)

include_directories(pybind11_include_path)
include_directories(${PYTHON_INCLUDE_DIRS})
```

1.2 编写代码

```
#include <pybind11/pybind11.h>

int add(int i, int j) {
    return i + j;
}

PYBIND11_MODULE(example, m) {
    m.doc() = "pybind11 example plugin"; // optional module docstring
    m.def("add", &add, "A function which adds two numbers");
}
```

```

PYBIND11_MODULE example Python import
py::module_ m(module_::def())

```

1.2.1

`py::arg` Python

```

m.def("add", &add, "A function which adds two numbers",
      py::arg("i"), py::arg("j"));

```

```

// regular notation
m.def("add1", &add, py::arg("i"), py::arg("j"));
// shorthand
using namespace pybind11::literals;
m.def("add2", &add, "i"_a, "j"_a);

```

Python



```

import example
example.add(i=1, j=2) #3L

```

1.2.2

pybind11 `arg`

```

m.def("add", &add, "A function which adds two numbers",
      py::arg("i") = 1, py::arg("j") = 2);

```

```
// regular notation
m.def("add1", &add, py::arg("i") = 1, py::arg("j") = 2);
// shorthand
m.def("add2", &add, "i"_a=1, "j"_a=2);
```

1.2.3 静态绑定

静态绑定（Static Binding）是指编译时就能确定调用的函数。

在 C++ 中，静态绑定通常通过函数重载（Function Overloading）实现。Python 解释器通过静态绑定来调用 C++ 函数。

```
m.def("add", static_cast<int(*)>(int, int>(&add), "A function which adds
two int numbers");
m.def("add", static_cast<double(*)>(double, double>(&add), "A function
which adds two double numbers");
```

在 C++14 中，静态绑定可以通过 `py::overload_cast` 实现。

```
m.def("add", py::overload_cast<int, int>(&add), "A function which adds two
int numbers");
m.def("add", py::overload_cast<double, double>(&add), "A function which
adds two double numbers");
```

在 Python 中，静态绑定可以通过 `py::overload_cast` 实现。在 C++ 中，静态绑定可以通过 `py::const` 实现。

1.3 属性

在 Python 中，属性（Attribute）是指对象的属性。在 C++ 中，属性可以通过 `py::attr` 实现。在 Python 中，属性可以通过 `py::cast` 实现。

```
PYBIND11_MODULE(example, m) {
    m.attr("the_answer") = 42;
    py::object world = py::cast("World");
    m.attr("what") = world;
}
```

Python

```
```python
>>> import example
>>> example.the_answer
42
>>> example.what
'World'
```

## 1.4

C++ Pet

```
struct Pet {
 Pet(const std::string &name) : name(name) { }
 void setName(const std::string &name_) { name = name_; }
 const std::string &getName() const { return name; }

 std::string name;
};
```



```
#include <pybind11/pybind11.h>
namespace py = pybind11;

PYBIND11_MODULE(example, m) {
 py::class_<Pet>(m, "Pet")
 .def(py::init<const std::string &>())
 .def("setName", &Pet::setName)
 .def("getName", &Pet::getName)
 .def("__repr__",
 [](const Pet &a) {
 return "<example.Pet named '" + a.name + "'>";
 });
}
```

`class_` 是 C++ `class` 或 `struct` 的初始化函数，用于初始化类。

`print(p)` 会调用 `__repr__` 方法，返回类的字符串表示。

Python 测试：



```
>>> import example
>>> p = example.Pet("Molly")
>>> print(p)
<example.Pet named 'Molly'>
>>> p.getName()
u'Molly'
>>> p.setName("Charly")
>>> p.getName()
u'Charly'
```

还可以使用 `class_::def_static` 来定义静态方法。

### 1.4.1 属性

使用 `class_::def_readwrite` 或 `class_::def_readonly` 来定义属性。

```
py::class_<Pet>(m, "Pet")
 .def(py::init<const std::string &>())
 .def_readwrite("name", &Pet::name)
 // ... remainder ...
```

Python



```
>>> p = example.Pet("Molly")
>>> p.name
u'Molly'
>>> p.name = "Charly"
>>> p.name
u'Charly'
```

`Pet::name` setter/getters

```
class Pet {
public:
 Pet(const std::string &name) : name(name) { }
 void setName(const std::string &name_) { name = name_; }
 const std::string &getName() const { return name; }
private:
 std::string name;
};
```

`class_::def_property()` (`class_::def_property_readonly()`)  
`setter`/`getter`

```
py::class_<Pet>(m, "Pet")
 .def(py::init<const std::string &>())
 .def_property("name", &Pet::getName, &Pet::setName)
 // ... remainder ...
```

`readonly`/`nullptr`

`class_::def_readwrite_static()`, `class_::def_readonly_static()`  
`class_::def_property_static()`, `class_::def_property_readonly_static()`



```
>>> p = example.Pet()
>>> p.name = "Charly" # OK, overwrite value in C++
>>> p.age = 2 # OK, dynamically add a new attribute
>>> p.__dict__ # just like a native Python class
{'age': 2}
```

\_\_\_\_\_ \_\_dict\_\_ \_\_\_\_\_  
 \_\_\_\_\_Python\_\_\_\_\_pybind11\_\_\_\_\_Python\_\_\_\_\_  
 \_\_\_\_\_

### 1.4.3

\_\_\_\_\_

```
struct Pet {
 Pet(const std::string &name, int age) : name(name), age(age) { }

 void set(int age_) { age = age_; }
 void set(const std::string &name_) { name = name_; }

 std::string name;
 int age;
};

// method 1
py::class_<Pet>(m, "Pet")
 .def(py::init<const std::string &, int>())
 .def("set", static_cast<void (Pet::*)(int)>(&Pet::set), "Set the pet's age")
 .def("set", static_cast<void (Pet::*)(const std::string &)>(&Pet::set), "Set the pet's name");

// method 2
py::class_<Pet>(m, "Pet")
 .def("set", py::overload_cast<int>(&Pet::set), "Set the pet's age")
 .def("set", py::overload_cast<const std::string &(&Pet::set), "Set the pet's name");
```

## 1.5 枚举

在C++中枚举类型

```
enum Flags {
 Read = 4,
 Write = 2,
 Execute = 1
};

py::enum_<Flags>(m, "Flags", py::arithmetic())
 .value("Read", Flags::Read)
 .value("Write", Flags::Write)
 .value("Execute", Flags::Execute)
 .export_values();
```

`enum_::export_values()` 在C++11中引入

在Python中，枚举类型 `__members__` 属性返回一个字典，其中键是枚举成员的名称，值是枚举成员的值。使用 `str(enum)` 可以获取枚举类型的字符串表示。

## 1.6 \*args 和 \*\*kwargs

Python中函数可以接受任意数量的位置参数和关键字参数。



```
def generic(*args, **kwargs):
 ... # do something with args and kwargs
```

在Python中，`pybind11`库支持使用 `*args` 和 `**kwargs` 来定义函数。

```
void generic(py::args args, const py::kwargs& kwargs) {
 /// .. do something with args
 if (kwargs)
 /// .. do something with kwargs
}

/// Binding code
m.def("generic", &generic);
```

py::args 是 py::tuple 和 py::kwargs 的 py::dict

## 2. 非平凡返回

### 2.1 非平凡返回

Python C++ 非平凡返回类型 (non-trivial return type) 的函数，在 Python 中调用时，需要返回一个 Python 对象。在 C++ 中，可以使用 `pybind11::return_value_policy::automatic` 来指定返回策略。

在 C++ 中，可以使用 `pybind11::return_value_policy::automatic` 来指定返回策略。

```
/* Function declaration */
Data *get_data() { return _data; /* (pointer to a static data structure)
*/ }
...

/* Binding code */
m.def("get_data", &get_data); // <-- KABOOM, will cause crash when called
from Python
```

在 Python 中，调用 `get_data()` 时，返回的是一个 C++ 指针。在 Python 中，需要返回一个 Python 对象。在 C++ 中，可以使用 `pybind11::return_value_policy::automatic` 来指定返回策略。

Python 的 `_data` Python 的 `pybind11` C++ 的 `operator delete()` 的  
的的的的的的的的的的的的的的的的的的的的的的的的的的的

的的的的的的的的的的的的 `return_value_policy::reference` 的的的的的的的的的的的的  
的的的的的的

```
m.def("get_data", &get_data, py::return_value_policy::reference);
```

的的的的的的的的的的的的的的的的的的的的的的的的的的的 `pybind11` 的的的的的的的的  
的的的的的的的的的的的的的的的的的的的的的的的的的的的

的的的的	的
<code>return_value_policy::take_ownership</code>	的的的的的的的的的的的的的的的的的的的的的的的的的的的 Python 的的的的的的 <code>delete</code> 的的的
<code>return_value_policy::copy</code>	的的的的的 Python 的的的的的的的的的的 的的的的的的的的的的的的的的
<code>return_value_policy::move</code>	的 <code>std::move</code> 的的的的的的的的的的的 Python 的的的的的的的的的的的的的的 的的
<code>return_value_policy::reference</code>	的的的的的的的的的的的 C++ 的的的的的的 的的的的的的的的的的的的的的的 Python 的的的 C++ 的的的的的的的的的的的
<code>return_value_policy::reference_internal</code>	的的的的的的的的的的的的的的的的的的的的的的的的的的的 <code>this</code> 的 <code>self</code> 的的的的的的 <code>reference</code> 的 <code>keep_alive&lt;0, 1&gt;</code> 的的的的的的 Python 的的的的的的的的的的的的的的的的 <code>def_property</code> 的 <code>def_readwrite</code> <code>getter</code> 的的的的的的的的的的
<code>return_value_policy::automatic</code>	的的的的的的的的的的的 <code>return_value_policy::take_ov</code> 的的的的的的的的的的 <code>return_value_policy::copy</code> 的 的的的的的的的的的的的的的的的 <code>py::cl</code> 的的的的的

返回值	策略
<code>return_value_policy::automatic_reference</code>	pybind11 默认策略，返回的 C++ 对象在 Python 中拥有自己的副本，生命周期独立于 C++ 对象。   <code>pybind11/stl.h</code> 中 <code>casters</code> 部分有相关实现。

使用 `return_value_policy::copy`

```
class_<MyClass>(m, "MyClass")
 .def_property("data", &MyClass::getData, &MyClass::setData,
 py::return_value_policy::copy);
```

使用 `return_value_policy::reference` 和 `return_value_policy::reference_internal` 时，需要指定 `cpp_function` 属性。

```
class_<MyClass>(m, "MyClass")
 .def_property("data"
 py::cpp_function(&MyClass::getData,
 py::return_value_policy::copy),
 py::cpp_function(&MyClass::setData)
);
```

使用 `return_value_policy::move` 时，需要指定 `move` 属性。

总结

- pybind11 默认策略是 `return_value_policy::automatic_reference`，返回的 C++ 对象在 Python 中拥有自己的副本，生命周期独立于 C++ 对象。
- 使用 `return_value_policy::copy` 时，需要指定 `cpp_function` 属性。
- 使用 `return_value_policy::reference` 和 `return_value_policy::reference_internal` 时，需要指定 `cpp_function` 属性。



## 2.2 保持存活

保持存活是指一个对象在垃圾回收之前，先被通知并有机会做一些清理工作。

### keep alive

C++ 保持存活 C++ 保持存活 `keep_alive<Nurse, Patient>` 保持存活  
Nurse 保持存活 Patient 保持存活 0 保持存活 1 保持存活 1 保持存活 this 保持存活 2 保持  
Nurse 保持存活 None 保持存活

nurse 保持存活 pybind11 保持存活 nurse 保持存活 nurse 保持存活 pybind11 保持  
保持存活

保持存活 "Could not cativate keep\_alive!" 保持存活 review  
保持存活

保持存活 list append 保持存活

```
py::class_<List>(m, "List").def("append", &List::append, py::keep_alive<1, 2>());
```

保持存活 1 this 保持存活 0 保持存活 void 保持存活  
保持存活

```
py::class_<Nurse>(m, "Nurse").def(py::init<Patient &>(), py::keep_alive<1, 2>());
```

Note: keep\_alive 在 Boost.Python 中 with\_custodian\_and\_ward 和  
with\_custodian\_and\_ward\_postcall 保持存活

### Call guard

call\_guard<T> 保持存活 T 保持存活 scope guard 保持存活

```
m.def("foo", foo, py::call_guard<T>());
```

□□□□□□□□

```
m.def("foo", [](args...) {
 T scope_guard;
 return foo(args...); // forwarded arguments
});
```

□□□□□□□□T□□□□□□□□ gil\_scoped\_release □□□□□□□□□□□□

call\_guard □□□□□□□□□□□□ call\_guard<T1, T2, T3 ...> □□□□□□□□□□□□□□□□□□□□

See also: test/test\_call\_policies.cpp □□□□□□□□□□□□ keep\_alive □  
call\_guard □□□□

## 2.3 Keyword-only□□

Python3□□□keyword-only□□□□□□□□□□□□ \* □□□□□□□□



```
def f(a, *, b): # a can be positional or via keyword; b must be via
 keyword
 pass

f(a=1, b=2) # good
f(b=2, a=1) # good
f(1, b=2) # good
f(1, 2) # TypeError: f() takes 1 positional argument but 2 were given
```

pybind11□□□ py::kw\_only □□□□□□□□□□□□

```
m.def("f", [](int a, int b) { /* ... */ },
 py::arg("a"), py::kw_only(), py::arg("b"));
```

py::args

## 2.4 Positional-only

python3.8 Positional-only pybind11 `py::pos_only()`

```
m.def("f", [](int a, int b) { /* ... */ },
 py::arg("a"), py::pos_only(), py::arg("b"));
```

`a` keyword-only

## 2.5 Non-converting

- `py::implicitly_convertible<A,B>()`
- 
- `std::complex<float>`
- Calling a function taking an Eigen matrix reference with a numpy array of the wrong type or of an incompatible data layout.

`py::arg` `.noconvert()`

```
m.def("floats_only", [](double f) { return 0.5 * f; },
 py::arg("f").noconvert());
m.def("floats_preferred", [](double f) { return 0.5 * f; }, py::arg("f"));
```

`TypeError`



```
>>> floats_preferred(4)
2.0
>>> floats_only(4)
Traceback (most recent call last):
 File "<stdin>", line 1, in <module>
TypeError: floats_only(): incompatible function arguments. The following
argument types are supported:
 1. (f: float) -> float

Invoked with: 4
```

```
PyObject* _a py::arg().noconvert()
```

### 3. 类型转换

#### 3.1 类型转换

```
PyObject*
```

```
struct Pet {
 Pet(const std::string &name) : name(name) { }
 std::string name;
};

struct Dog : Pet {
 Dog(const std::string &name) : Pet(name) { }
 std::string bark() const { return "woof!"; }
};
```

```
pybind11::class_<Pet, std::string>("Pet", py::module::import("pet").attr("Pet"))
 .def(py::init<const std::string>())
 .def("bark", &Pet::bark)
```

```
py::class_<Pet>(m, "Pet")
 .def(py::init<const std::string &>())
 .def_readwrite("name", &Pet::name);

// Method 1: template parameter:
py::class_<Dog, Pet /* ← specify C++ parent type */>(m, "Dog")
 .def(py::init<const std::string &>())
 .def("bark", &Dog::bark);

// Method 2: pass parent class_ object:
py::class_<Dog>(m, "Dog", pet /* ← specify Python parent type */)
 .def(py::init<const std::string &>())
 .def("bark", &Dog::bark);
```

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```
>>> p = example.Dog("Molly")
>>> p.name
u'Molly'
>>> p.bark()
u'woof!'
```

# Python

```
// 编译选项
m.def("pet_store", []() { return std::unique_ptr<Pet>(new Dog("Molly"));
});
```



```
>>> p = example.pet_store()
>>> type(p) # `Dog` instance behind `Pet` pointer
Pet # no pointer downcasting for regular non-polymorphic types
>>> p.bark()
AttributeError: 'Pet' object has no attribute 'bark'
```

```
pet_store Dog Python Pet C++
pybind11
```

```

struct PolymorphicPet {
 virtual ~PolymorphicPet() = default;
};

struct PolymorphicDog : PolymorphicPet {
 std::string bark() const { return "woof!"; }
};

// Same binding code
py::class_<PolymorphicPet>(m, "PolymorphicPet");
py::class_<PolymorphicDog, PolymorphicPet>(m, "PolymorphicDog")
 .def(py::init<>())
 .def("bark", &PolymorphicDog::bark);

// Again, return a base pointer to a derived instance
m.def("pet_store2", []() { return std::unique_ptr<PolymorphicPet>(new
PolymorphicDog); });

```



```

>>> p = example.pet_store2()
>>> type(p)
PolymorphicDog # automatically downcast
>>> p.bark()
u'woof!'

```

pybind11 C++ Python

## 3.2 Python C++

Python C++

```

class Animal {
public:
 virtual ~Animal() { }
 virtual std::string go(int n_times) = 0;
};

class Dog : public Animal {
public:
 std::string go(int n_times) override {
 std::string result;
 for (int i=0; i<n_times; ++i)
 result += "woof! ";
 return result;
 }
};

std::string call_go(Animal *animal) {
 return animal->go(3);
}

PYBIND11_MODULE(example, m) {
 py::class_<Animal>(m, "Animal")
 .def("go", &Animal::go);

 py::class_<Dog, Animal>(m, "Dog")
 .def(py::init<>());

 m.def("call_go", &call_go);
}

```

PythonAnimal“No constructor defined!”

Animal

```

class PyAnimal : public Animal {
public:
 /* Inherit the constructors */
 using Animal::Animal;

 /* Trampoline (need one for each virtual function) */
 std::string go(int n_times) override {
 PYBIND11_OVERRIDE_PURE(
 std::string, /* Return type */
 Animal, /* Parent class */
 go, /* Name of function in C++ (must match Python
name) */
 n_times /* Argument(s) */
);
 }
};

```

PYBIND11\_OVERRIDE\_PURE PYBIND11\_OVERRIDE  
 PYBIND11\_OVERRIDE\_PURE\_NAME PYBIND11\_OVERRIDE\_NAME  
 Python \_\_str\_\_

```

std::string toString() override {
 PYBIND11_OVERRIDE_NAME(
 std::string, // Return type (ret_type)
 Animal, // Parent class (cname)
 "__str__", // Name of method in Python (name)
 toString, // Name of function in C++ (fn)
);
}

```

Animal



```
PYBIND11_MODULE(example, m) {
 py::class_<Animal, PyAnimal /* <--- trampoline*/>(m, "Animal")
 .def(py::init<>())
 .def("go", &Animal::go);

 py::class_<Dog, Animal>(m, "Dog")
 .def(py::init<>());

 m.def("call_go", &call_go);
}
```

pybind11 的 `class_` 函数需要 `PyAnimal` 和 `Python` 的 `Animal` 类

但是 `Animal` 类在 `Python` 中并没有定义

```
py::class_<Animal, PyAnimal /* <--- trampoline*/>(m, "Animal");
 .def(py::init<>())
 .def("go", &PyAnimal::go); /* <--- THIS IS WRONG, use &Animal::go */
```

在 `Python` 中定义 `Animal::go` 函数



```
from example import *
d = Dog()
call_go(d) # u'woof! woof! woof! '
class Cat(Animal):
 def go(self, n_times):
 return "meow! " * n_times

c = Cat()
call_go(c) # u'meow! meow! meow! '
```

在 `Python` 中定义 `Animal` 类时，需要调用 `C++` 的 `__init__` 函数，这会导致 `pybind11 2.6` 版本出现 `TypeError` 错误



```
class Dachshund(Dog):
 def __init__(self, name):
 Dog.__init__(self) # Without this, a TypeError is raised.
 self.name = name

 def bark(self):
 return "yap!"
```

Python 的 `__init__` 方法必须调用 `super().__init__()` 方法，否则在 Python 3 中会引发 `TypeError`。这是因为 Python 3 的类继承机制要求子类在初始化时必须调用父类的 `__init__` 方法。

### 3.3 类继承

在 Python 中，类继承是通过 `class` 语句中的类名来实现的。例如，我们可以定义一个 `Animal` 类，然后定义一个 `Dog` 类，使其继承 `Animal` 类。

```
class Animal {
public:
 virtual std::string go(int n_times) = 0;
 virtual std::string name() { return "unknown"; }
};

class Dog : public Animal {
public:
 std::string go(int n_times) override {
 std::string result;
 for (int i=0; i<n_times; ++i)
 result += bark() + " ";
 return result;
 }
 virtual std::string bark() { return "woof!"; }
};
```

在 Python 中，`Dog` 类继承 `Animal` 类，因此 `Dog` 类中的 `bark()` 方法会覆盖 `Animal` 类中的 `go()` 方法。而 `Dog` 类中的 `name()` 方法则会调用 `Dog` 类自己的 `name()` 方法。

```

class PyAnimal : public Animal {
public:
 using Animal::Animal; // Inherit constructors
 std::string go(int n_times) override {
PYBIND11_OVERRIDE_PURE(std::string, Animal, go, n_times); }
 std::string name() override { PYBIND11_OVERRIDE(std::string, Animal,
name,); }
};
class PyDog : public Dog {
public:
 using Dog::Dog; // Inherit constructors
 std::string go(int n_times) override { PYBIND11_OVERRIDE(std::string,
Dog, go, n_times); }
 std::string name() override { PYBIND11_OVERRIDE(std::string, Dog,
name,); }
 std::string bark() override { PYBIND11_OVERRIDE(std::string, Dog,
bark,); }
};

```

name() bark()

pybind11

```

class Husky : public Dog {};
class PyHusky : public Husky {
public:
 using Husky::Husky; // Inherit constructors
 std::string go(int n_times) override {
PYBIND11_OVERRIDE_PURE(std::string, Husky, go, n_times); }
 std::string name() override { PYBIND11_OVERRIDE(std::string, Husky,
name,); }
 std::string bark() override { PYBIND11_OVERRIDE(std::string, Husky,
bark,); }
};

```



### 3.4 pybind11

pybind11 是一个 C++ 绑定库，它允许 C++ 类库被 Python 调用。它使用 `std::unique_ptr` 和 `py::nodelete` 来管理内存。C++ 代码示例如下：

```
/* ... definition ... */

class MyClass {
private:
 ~MyClass() { }
};

/* ... binding code ... */

py::class_<MyClass, std::unique_ptr<MyClass, py::nodelete>>(m, "MyClass")
 .def(py::init<>())
```

### 3.5 函数

定义一个函数 `A` 和 `B`，其中 `A` 是 `B` 的基类。

```
py::class_<A>(m, "A")
 /// ... members ...

py::class_(m, "B")
 .def(py::init<A>())
 /// ... members ...

m.def("func",
 [](const B &) { /* */ }
);
```

在 Python 中调用 `func(B(a))`，在 C++ 中调用 `func(a)`，其中 `A` 是 `B` 的基类。

在 Python 中调用 `B` 的构造函数，在 C++ 中调用 `B` 的构造函数。

```
py::implicitly_convertible<A, B>();
```

### 3.6 自定义

自定义 `Vector2` 数据类型

```
class Vector2 {
public:
 Vector2(float x, float y) : x(x), y(y) { }

 Vector2 operator+(const Vector2 &v) const { return Vector2(x + v.x, y
+ v.y); }
 Vector2 operator*(float value) const { return Vector2(x * value, y *
value); }
 Vector2& operator+=(const Vector2 &v) { x += v.x; y += v.y; return
*this; }
 Vector2& operator*=(float v) { x *= v; y *= v; return *this; }

 friend Vector2 operator*(float f, const Vector2 &v) {
 return Vector2(f * v.x, f * v.y);
 }

 std::string toString() const {
 return "[" + std::to_string(x) + ", " + std::to_string(y) + "]";
 }
private:
 float x, y;
};
```

自定义



```
#include <pybind11/operators.h>

PYBIND11_MODULE(example, m) {
 py::class_<Vector2>(m, "Vector2")
 .def(py::init<float, float>())
 .def(py::self + py::self)
 .def(py::self += py::self)
 .def(py::self *= float())
 .def(float() * py::self)
 .def(py::self * float())
 .def(-py::self)
 .def("__repr__", &Vector2::toString);
}
```

```
.def(py::self * float())
```

```
.def("__mul__", [](<const Vector2 &a, float b) {
 return a * b;
}, py::is_operator())
```

### 3.7

Python `copy`

Python3 `pickle` `__copy__` `__deepcopy__`  
 Python2.7 `pybind11` `cPickle`

```
py::class_<Copyable>(m, "Copyable")
 .def("__copy__", [](<const Copyable &self) {
 return Copyable(self);
 })
 .def("__deepcopy__", [](<const Copyable &self, py::dict) {
 return Copyable(self);
 }, "memo"_a);
```

Note:

### 3.8

pybind11 holder pybind11 class\_

```
py::class_<MyType, BaseType1, BaseType2, BaseType3>(m, "MyType")
...
```

holder pybind11

Python C++ Python

pybind11 multiple\_inheritance

```
py::class_<MyType, BaseType2>(m, "MyType", py::multiple_inheritance());
```

### 3.9 protected

Python protected

```
class A {
protected:
 int foo() const { return 42; }
};
```

```
py::class_<A>(m, "A")
 .def("foo", &A::foo); // error: 'foo' is a protected member of 'A'
```



Python protected publicist pattern trampoline

```
class A {
protected:
 int foo() const { return 42; }
};

class Publicist : public A { // helper type for exposing protected
functions
public:
 using A::foo; // inherited with different access modifier
};

py::class_<A>(m, "A") // bind the primary class
 .def("foo", &Publicist::foo); // expose protected methods via the
publicist
```

&Publicist::foo &A::foo Publicist  
public

Python protected publicist pattern trampoline

```

class A {
public:
 virtual ~A() = default;

protected:
 virtual int foo() const { return 42; }
};

class Trampoline : public A {
public:
 int foo() const override { PYBIND11_OVERRIDE(int, A, foo,); }
};

class Publicist : public A {
public:
 using A::foo;
};

py::class_<A, Trampoline>(m, "A") // <-- `Trampoline` here
 .def("foo", &Publicist::foo); // <-- `Publicist` here, not
 `Trampoline`!

```

### 3.10 `final`

C++11 提供了 `final` 关键字，Python 提供了 `py::is_final` 宏。Python 的 `final` 关键字在 C++ 中是无效的。

```

class IsFinal final {};

py::class_<IsFinal>(m, "IsFinal", py::is_final());

```

Python 的 `final` 关键字在 C++ 中是无效的。



```

class PyFinalChild(IsFinal):
 pass

```

```

TypeError: type 'IsFinal' is not an acceptable base type

```



Exception thrown by C++	Translated to Python exception type
<code>pybind11::buffer_error</code>	<code>BufferError</code>
<code>pybind11::import_error</code>	<code>ImportError</code>
<code>pybind11::attribute_error</code>	<code>AttributeError</code>
Any other exception	<code>RuntimeError</code>

```

PythonPython
pybind11::error_already_set

```

```

Python handle::call() cast_error

```

## 4.2

```

pybind11pybind11 class
globalC++what()C++Python

```

```

py::register_exception<CppExp>(module, "PyExp");

```

```

PyExpPythonCppExpPyExp

```

```


```

```

py::register_local_exception<CppExp>(module, "PyExp");

```

```

handle

```

```

py::register_exception<CppExp>(module, "PyExp", PyExc_RuntimeError);
py::register_local_exception<CppExp>(module, "PyExp", PyExc_RuntimeError);

```

```

PyExpPyExpRuntimeError

```

```

PythonPythonStandard ExceptionsPyExc_Exception

```

```

py::register_exception_translator(translator) {}
py::register_local_exception_translator(translator) {}
void(std::exception_ptr) {}

```

### 9.3 C++ Python

C++ Python Python Python pybind11 Python  
 pybind11::error\_already\_set C++ Python  
 error\_already\_set Python Python C++

Exception raised in Python	Thrown as C++ exception type
Any Python Exception	pybind11::error_already_set

Python

```

try {
 // open("missing.txt", "r")
 auto file = py::module_::import("io").attr("open")("missing.txt",
 "r");
 auto text = file.attr("read")();
 file.attr("close")();
} catch (py::error_already_set &e) {
 if (e.matches(PyExc_FileNotFoundError)) {
 py::print("missing.txt not found");
 } else if (e.matches(PyExc_PermissionError)) {
 py::print("missing.txt found but not accessible");
 } else {
 throw;
 }
}

```

C++ Python Python error\_already\_set.

```

try {
 py::eval("raise ValueError('The Ring')");
} catch (py::value_error &boromir) {
 // Boromir never gets the ring
 assert(false);
} catch (py::error_already_set &frodo) {
 // Frodo gets the ring
 py::print("I will take the ring");
}

try {
 // py::value_error is a request for pybind11 to raise a Python
 exception
 throw py::value_error("The ball");
} catch (py::error_already_set &cat) {
 // cat won't catch the ball since
 // py::value_error is not a Python exception
 assert(false);
} catch (py::value_error &dog) {
 // dog will catch the ball
 py::print("Run Spot run");
 throw; // Throw it again (pybind11 will raise ValueError)
}

```

## 9.4 Python C API

pybind11 wrappers Python C API Python C API  
 pybind11

Python C API Python `throw py::error_already_set();`  
 pybind11 Python `PyErr_SetString`

```

PyErr_SetString(PyExc_TypeError, "C API type error demo");
throw py::error_already_set();

// But it would be easier to simply...
throw py::type_error("pybind11 wrapper type error");

```

`PyErr_Clear`

PythonPython/pybind11

## 9.5 `unraisable`

PythonC++ `noexcept(true)` `std::terminate()` C++ Python `error_already_set` `error_already_set::discard_as_unraisable()` Python

`__del__` Python Python `unraisable` Python 3.8+ `system hook` `auditing event`

`noexcept` `try-catch` `error_already_set` `pybind11` Python Python `pybind11` C++ `noexcept` Python `discard_as_unraisable`

```
void nonthrowing_func() noexcept(true) {
 try {
 // ...
 } catch (py::error_already_set &eas) {
 // Discard the Python error using Python APIs, using the C++ magic
 // variable __func__. Python already knows the type and value and
of the
 // exception object.
 eas.discard_as_unraisable(__func__);
 } catch (const std::exception &e) {
 // Log and discard C++ exceptions.
 third_party::log(e);
 }
}
```

**5. □□□□**

## 6. python C++

7.  

## 7.1 □□□□□□□□

```
pybind11::PYBIND11_DECLARE_HOLDER_TYPE(, PYBIND11_OVERRIDE_*
 (PyObject*, PyObject*))
```

PYBIND11\_OVERRIDE(MyReturnType<T1, T2>, Class<T3, T4>, func)

```
000000000050000000000000003000000000000000000000000 PYBIND11_TYPE 00
000
```

```
// Version 1: using a type alias
using ReturnType = MyReturnType<T1, T2>;
using ClassType = Class<T3, T4>;
PYBIND11_OVERRIDE(ReturnType, ClassType, func);

// Version 2: using the PYBIND11_TYPE macro:
PYBIND11_OVERRIDE(PYBIND11_TYPE(MyReturnType<T1, T2>),
 PYBIND11_TYPE(Class<T3, T4>), func)
```

## PYBIND11\_MAKE\_OPAQUE



## 7.2 释放GIL

Python和C++的交互中，GIL的释放和获取是非常重要的。在Python的C++扩展中，我们通常使用`gil_scoped_release`和`gil_scoped_acquire`来管理GIL。在C++的Python扩展中，我们通常使用`PyGILState_STATE`来管理GIL。

```
class PyAnimal : public Animal {
public:
 /* Inherit the constructors */
 using Animal::Animal;

 /* Trampoline (need one for each virtual function) */
 std::string go(int n_times) {
 /* Acquire GIL before calling Python code */
 py::gil_scoped_acquire acquire;

 PYBIND11_OVERRIDE_PURE(
 std::string, /* Return type */
 Animal, /* Parent class */
 go, /* Name of function */
 n_times /* Argument(s) */
);
 }
};

PYBIND11_MODULE(example, m) {
 py::class_<Animal, PyAnimal> animal(m, "Animal");
 animal
 .def(py::init<>())
 .def("go", &Animal::go);

 py::class_<Dog>(m, "Dog", animal)
 .def(py::init<>());

 m.def("call_go", [](Animal *animal) -> std::string {
 /* Release GIL before calling into (potentially long-running) C++
 code */
 py::gil_scoped_release release;
 return call_go(animal);
 });
}
```

在C++的Python扩展中，我们通常使用`call_guard`和`call_go`来管理GIL。

```
m.def("call_go", &call_go, py::call_guard<py::gil_scoped_release>());
```

# mdbook-mermaid

## 简介Mermaid

Mermaid 是一个用于生成图表的标记语言，它使用简单的文本格式来描述图表结构，并可以生成各种类型的图表，如流程图、序列图、类图等。它支持多种图表类型，并且可以与 Markdown 文档结合使用，方便在文档中嵌入图表。

Mermaid 的特点包括：

- 使用简单的文本格式描述图表结构；
- 支持多种图表类型，如流程图、序列图、类图等；
- 可以与 Gitbook 等文档生成工具集成使用。

## 安装Mermaid

安装

```
- npm install mermaid
- npm install mermaid
- npm install mermaid
```

安装完成后，可以在 Markdown 文档中使用 Mermaid 语法生成图表。

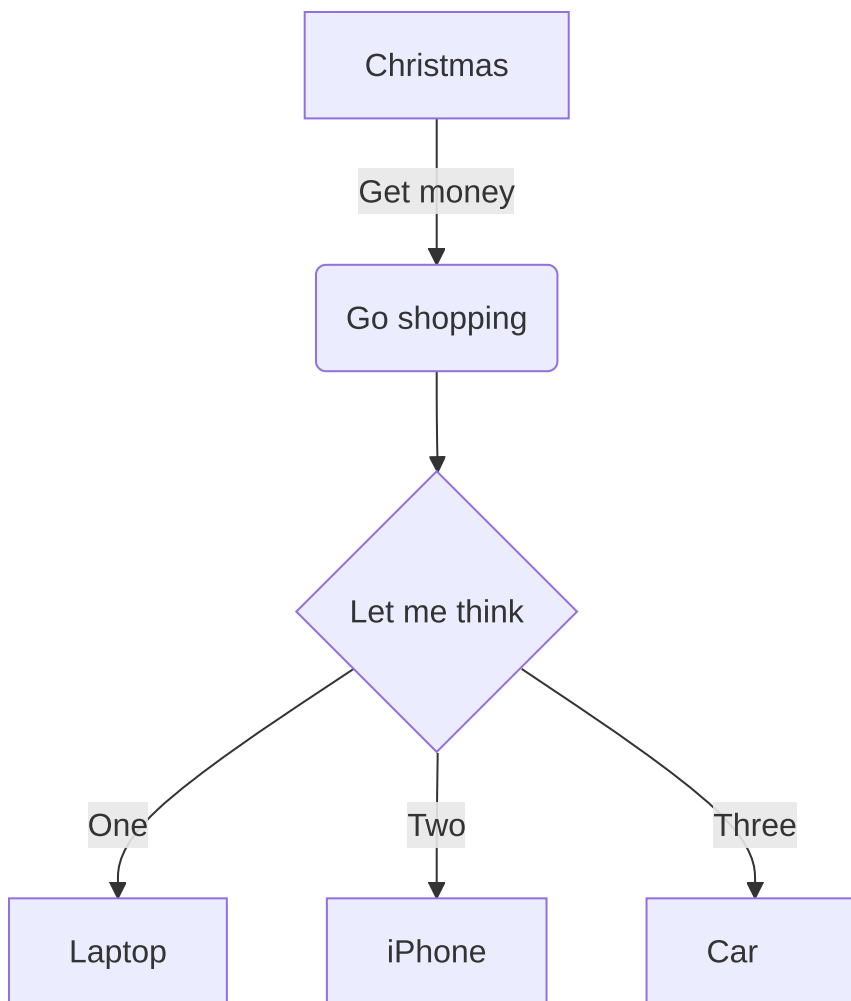
Mermaid 支持多种图表类型，如流程图、序列图、类图等。它使用简单的文本格式来描述图表结构，并可以生成各种类型的图表，如流程图、序列图、类图等。它支持多种图表类型，并且可以与 Markdown 文档结合使用，方便在文档中嵌入图表。

安装

```
M↓
```

```
graph TD
 A[Christmas] -->|Get money| B(Go shopping)
 B --> C{Let me think}
 C -->|One| D[Laptop]
 C -->|Two| E[iPhone]
 C -->|Three| F[fa:fa-car Car]
```

□□



- □□□□: <https://github.com/mermaid-js/mermaid>
- □□□□: <https://mermaidjs.github.io/mermaid-live-editor/>
- □□□□: <https://mermaid-js.github.io/mermaid/#/flowchart>
- □□□□: <https://mermaid.nodejs.cn/syntax/flowchart.html>

# Mermaid

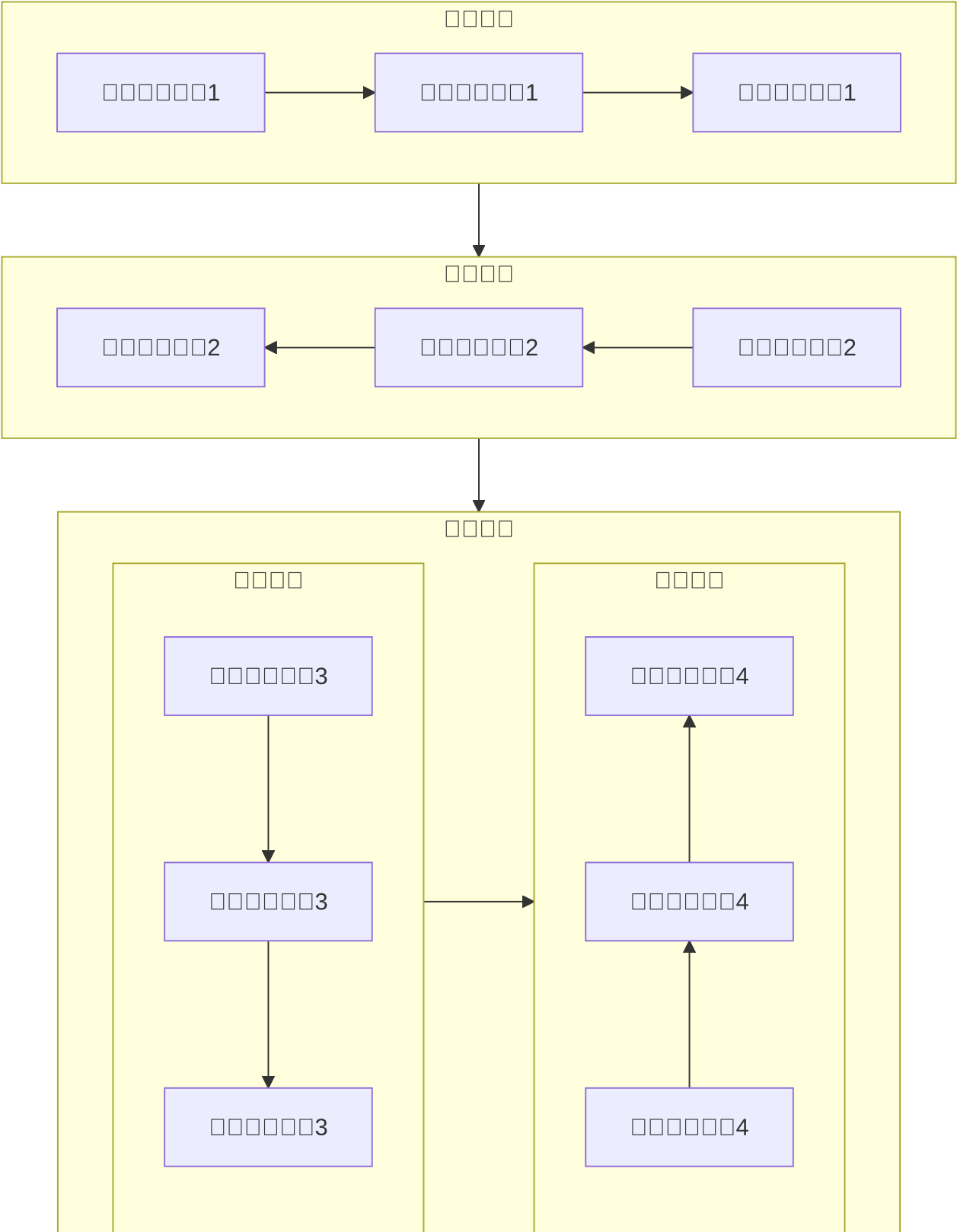
M↓

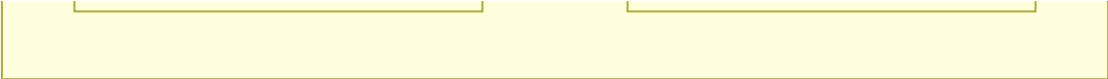
+ TB

+ BT

+ LR

+ RL





graph TD, direction TB; Start --> Stop; subgraph " " direction LR; direction TB; Start --> Stop; end

graph TD; Start --> Stop; direction TB; Start --> Stop; direction LR; Start --> Stop; direction RL; Start --> Stop;

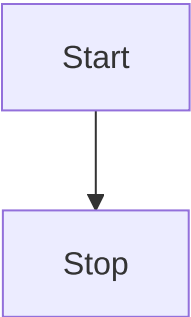
- TB

graph TD; Start --> Stop; direction TB; Start --> Stop; direction LR; Start --> Stop; direction RL; Start --> Stop;

graph TD; Start --> Stop; direction TB; Start --> Stop; direction LR; Start --> Stop; direction RL; Start --> Stop;

```
graph TB; Start --> Stop;
```

graph TD; Start --> Stop; direction TB; Start --> Stop; direction LR; Start --> Stop; direction RL; Start --> Stop;



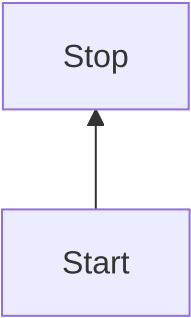
- BT

graph TD; Start --> Stop; direction TB; Start --> Stop; direction LR; Start --> Stop; direction RL; Start --> Stop;

graph TD; Start --> Stop; direction TB; Start --> Stop; direction LR; Start --> Stop; direction RL; Start --> Stop;

```
graph BT
 Start --> Stop
```

□□



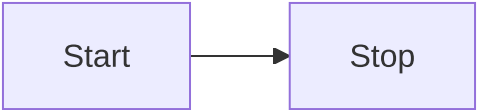
- LR

□□□□: from **L**eft to **R**ight

□□

```
graph LR
 Start --> Stop
```

□□



- RL

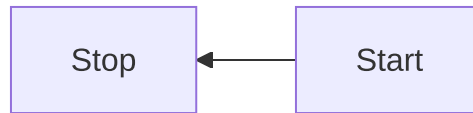
□□□□: from **R**ight to **L**eft

□□



```
graph RL
 Start --> Stop
```

11



11

111

```

- []
+ []
 - []
 - []
 - []
 - []
 - []
 - []
 - []
 - []
+ ()
 - ()
 - ()
 - ()
+ { }
 - { }
 - { }
 - { }
+ >]

```

0000000,0000000000000000,000000000000,00000000,00
 [] 0000, () 0000, {} 0000
 (000 <> 000)00.

00: 00000000,000000.

00000

00000,0000000000,000000000 id 00,0000000000000000.

id 000000000000000000000000.

00

graph TD
 id

00



00000

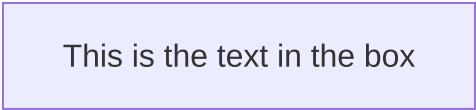
00000,000000,00000000 id 0000 <node shape> 00,0000000000000000.

id 0000000000,00000000000 <node shape> 0000,00 id 00000000000.

- 00

节点: [node description] , [] 节点描述, node description 节点描述.

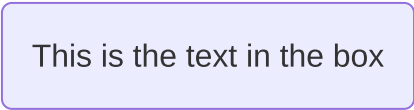
```
graph LR
 id1[This is the text in the box]
```



- 节点

节点: (node description) , () 节点描述, node description 节点描述.

```
graph LR
 id1(This is the text in the box)
```



- 节点

节点: ([node description]) , () 节点 [] 节点描述, 节点描述

`[, node description ]`.

```
graph LR
 id1([This is the text in the box])
```

This is the text in the box

- 

`[ ]`: `[(node description)]` , `[ ]` `( )` `[ ]`, `node description` `[ ]`.

```
graph LR
 id1[(Database)]
```



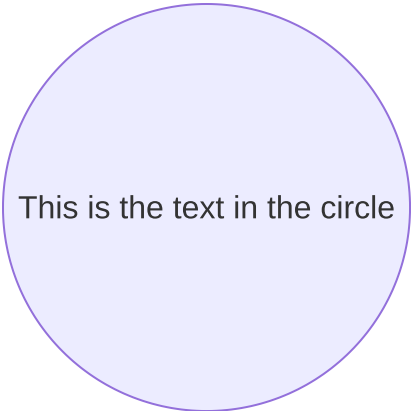
- 

`((node description))` , `( )` `[ ]` `( )` `[ ]`, `node description` `[ ]`.

□□

```
graph LR
 id1((This is the text in the circle))
```

□□



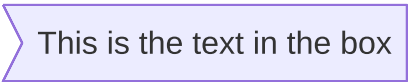
- □□□□

```
□□□□: >node description] ,□□□□□□□ > ,□□□□□□□] □□□□□□□□□□, node
description □□□□□□□□.
```

□□

```
graph LR
 id1>This is the text in the box]
```

□□



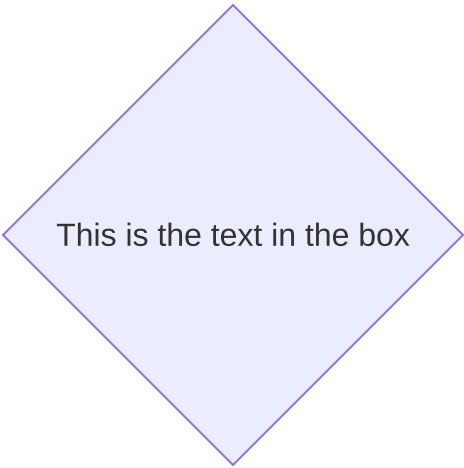
- □□

节点: {node description} , {} 节点描述, node description 节点描述.

图

```
graph LR
 id1{This is the text in the box}
```

图



- 图

节点: {{node description}} , {} 节点 {} 节点描述, node description 节点描述.

图

```
graph LR
 id1\\{\\{This is the text in the box\\}\\}
```

Gitbook 节点 {} 节点描述,节点描述,节点描述 \ 节点.

□□

This is the text in the box

- □□□□

```
□□□□: [/node description/] , [] □□□□ // □□□□□□□□□□□□, node
description □□□□□□□□.
```

□□

```
graph TD
 id1[/This is the text in the box/]
```

□□

This is the text in the box

- □□□□

```
□□□□: [\node description\] , [] □□□□ \\ □□□□□□□□□□□□, node
description □□□□□□□□.
```

□□

```
graph TD
 id1[\This is the text in the box\]
```

□□

This is the text in the box

- 

####: [/node description\] , [] / \ , node description .

```
graph TD
 A[/Christmas\]
```

Christmas

- 

####: [\node description/] , [] \ / , node description .

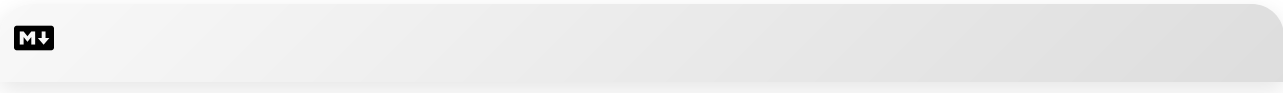
```
graph TD
 B[\Go shopping/]
```



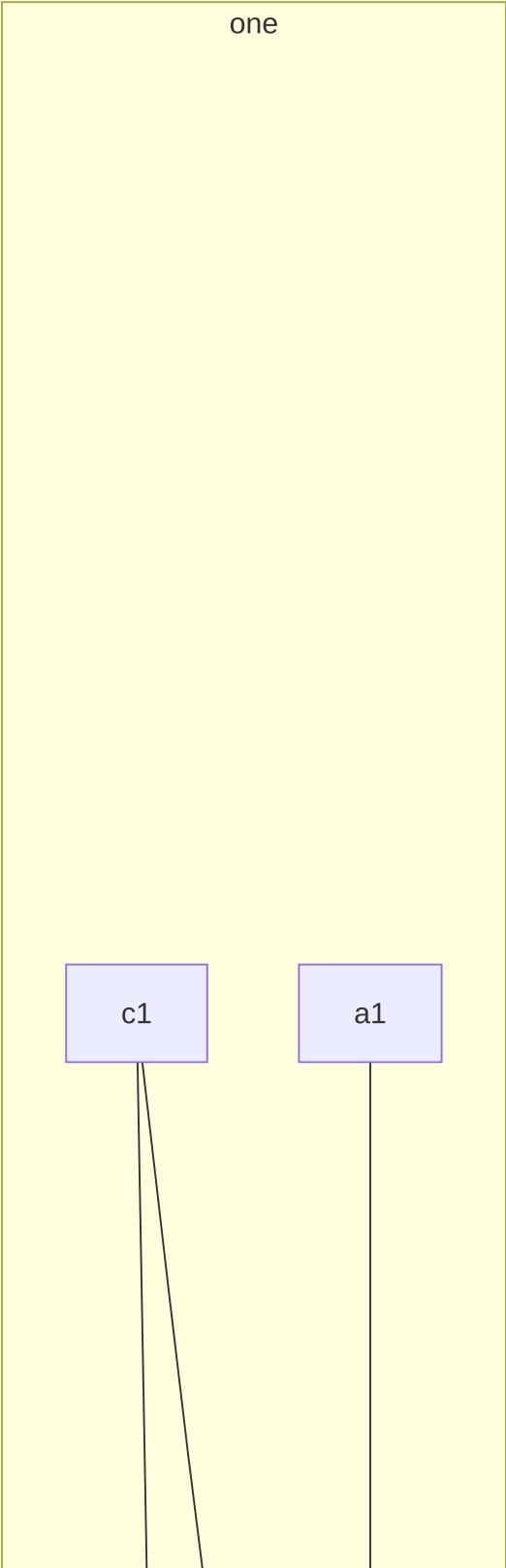
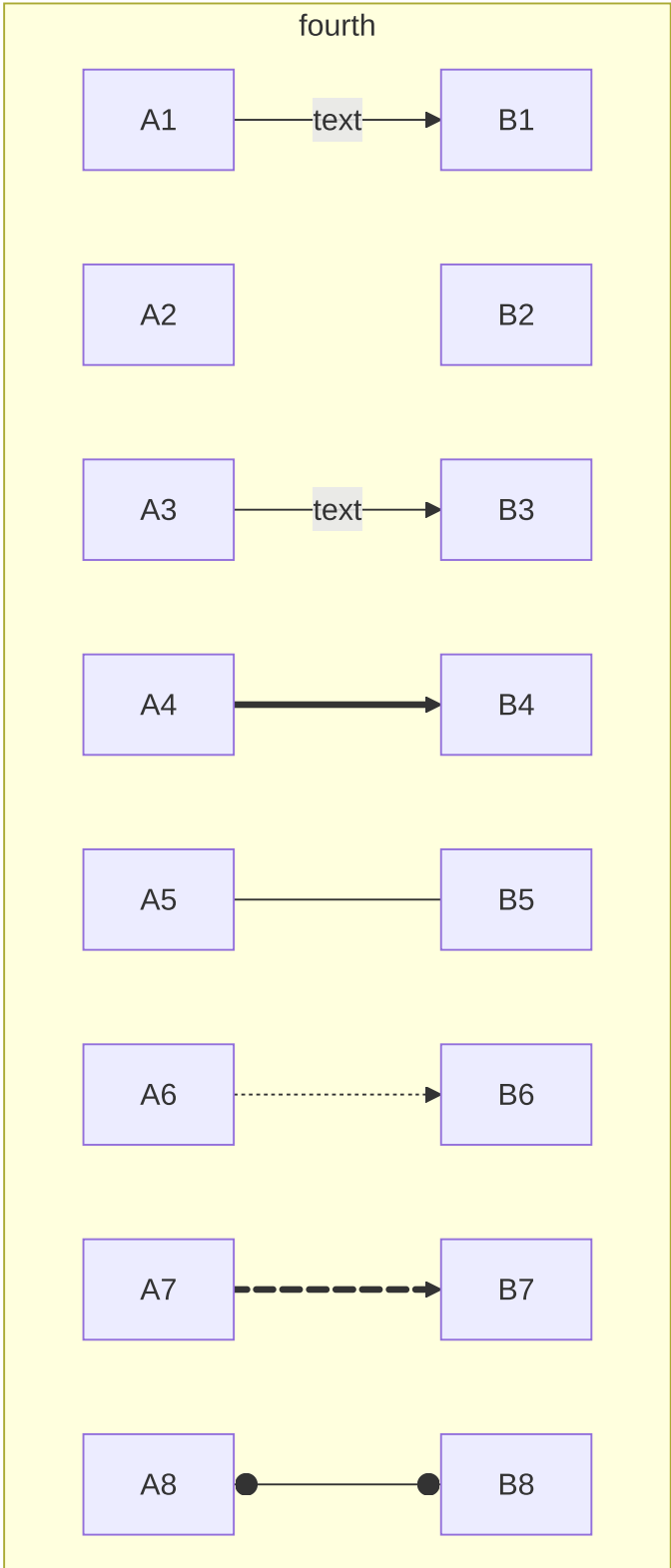
Go shopping

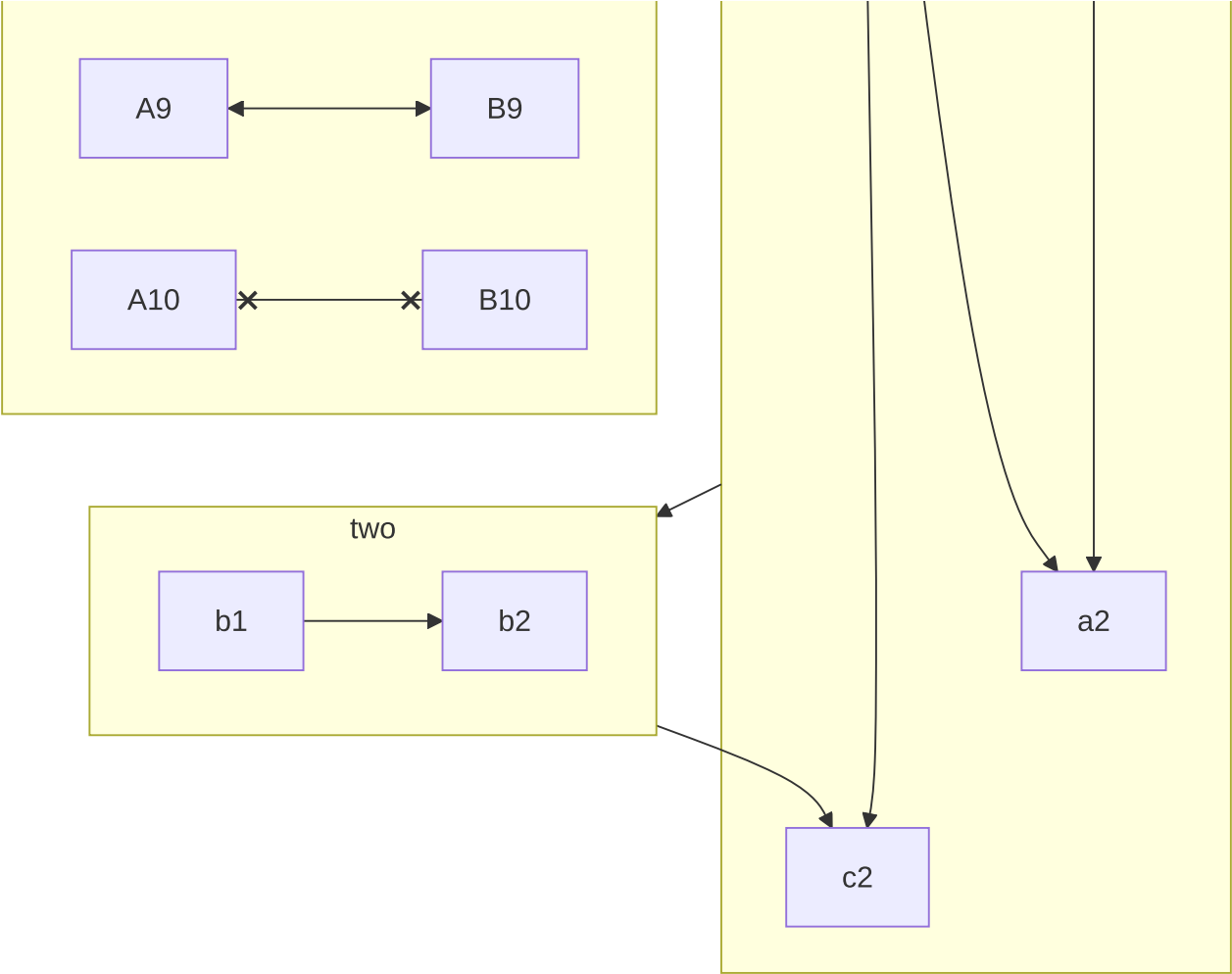
☐☐☐

☐☐☐



```
+ 00/00
- --
- -.
+ 000/000
- >
- -
+ 000/000
- 00
+ --0000
+ |0000|
- 00
+ -.0000
+ |0000|
+ 00
- ==
+ 0000
- -->
- ---
- -. ->
- .-
- 00000000
+ --0000-->
+ -->|0000|
- 00000000
+ --0000---
+ ---|0000|
- 00000000
+ -.0000-.->
+ -. ->|0000|
- 00000000
+ -.0000-.-
+ -. -|0000|
- ==>
- ===
- 0000000000(2)
+ ==0000==>
+ ==>|0000|
- 0000000000(2)
+ ==0000===
+ ===|0000|
```





00000000,00000000000000000000,0000000000000000,00 -- 0000,00000000 -. -  
0000,0000000000 > ,00000000000 - .

00: 0000000,000000000,00000000000000000000,0000000000000000.

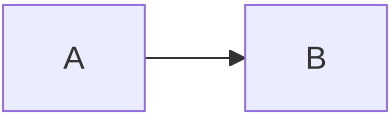
- 0000000000

0000: --> ,00 -- 0000, > 000000.

00

```
graph LR
 A-->B
```

🔗



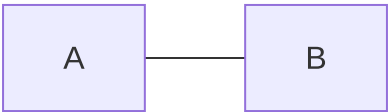
- 有向图

```
graph LR
 A --- B
```

🔗

```
graph LR
 A --- B
```

🔗



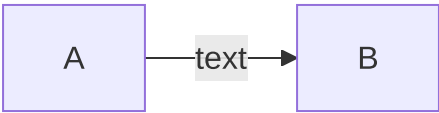
- 无向图

```
graph LR
 A -- connection line description --> B
```

🔗

```
graph LR
 A -- text --> B
```

❏

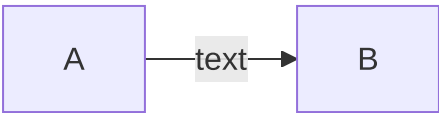


❏: |connection line description| ,❏ || ❏❏❏❏❏❏❏❏.

❏

```
graph LR
 A-->|text|B
```

❏



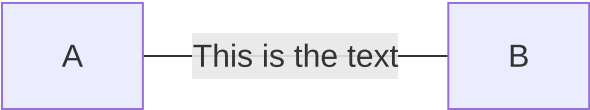
- ❏❏❏❏❏❏❏❏

❏: --connection line description ,❏❏❏❏ -- ❏❏❏❏❏❏❏❏,❏❏ --- ❏❏❏❏❏❏❏.

❏

```
graph LR
 A-- This is the text ---B
```

❏

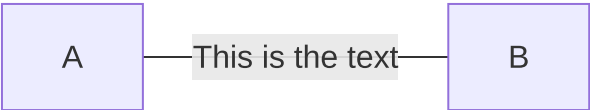


格式: |connection line description| , 逗号 || 分隔符.

图

```
graph LR
 A---|This is the text|B
```

图



- 格式

格式: -.connection line description.-> , 逗号 -. 分隔符, 逗号 .-> 图  
分隔符.

图

```
graph LR
 A-. text .-> B
```

图



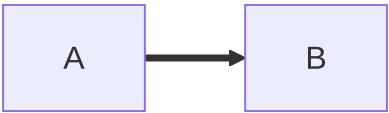
- 图的基本概念

图：==> ,图的基本概念。

图

```
graph LR
 A ==> B
```

图



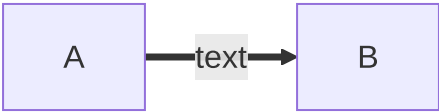
- 图的基本概念

图：==connection line description ,图 == 图的基本概念,图 ==> 图的基本概念。

图

```
graph LR
 A == text ==> B
```

图



- 图的基本概念

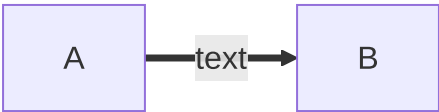


连接: |connection line description| , 连接 || 连接描述.

连接

```
graph LR
 A ==>|text| B
```

连接



连接

连接



- + -->-->
- + &
- + ""
- + %%
- + subgraph



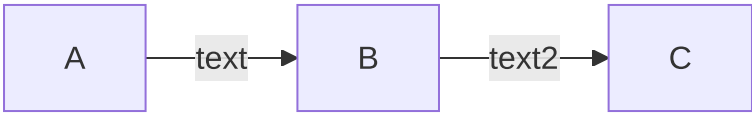
- 连接

连接

连接, A-->B-->C 连接 A-->B 连接 B-->C 连接.

```
graph LR
 A -- text --> B -- text2 --> C
```

□□



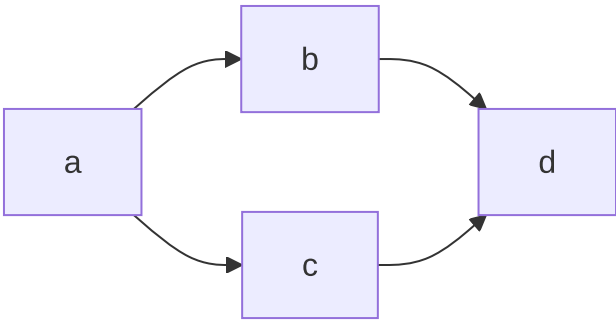
- □□□□□□□□

□□□□□□□□□, A-->B & C □□□ A-->B □ A-->C □□.

□□

```
graph LR
 a --> b & c --> d
```

□□



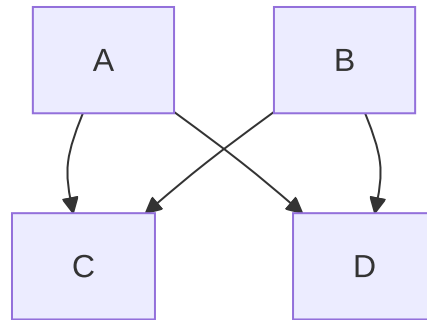
- □□□□□□□□

□□□□□□□□□□□□□, A & B --> C & D □□□ A-->C , A-->D , B-->C □ B-->D □□□□□□.

□□

```
graph TB
 A & B --> C & D
```

11



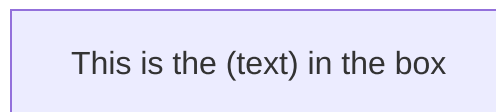
- □ □ □ □ □ □ □ □

□□□□□□□□□□□□□□□□ " " □□□□,□□□ [ ] □ ( ) □□ { } □□□□□□.

11

```
graph LR
 id1["This is the (text) in the box"]
```

11



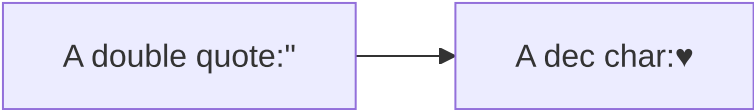
- □ □ □ □ □ □ □ □

□□ Html □□□□

11

```
graph LR
 A["A double quote:#quot;"] --> B["A dec char:#9829;"]
```

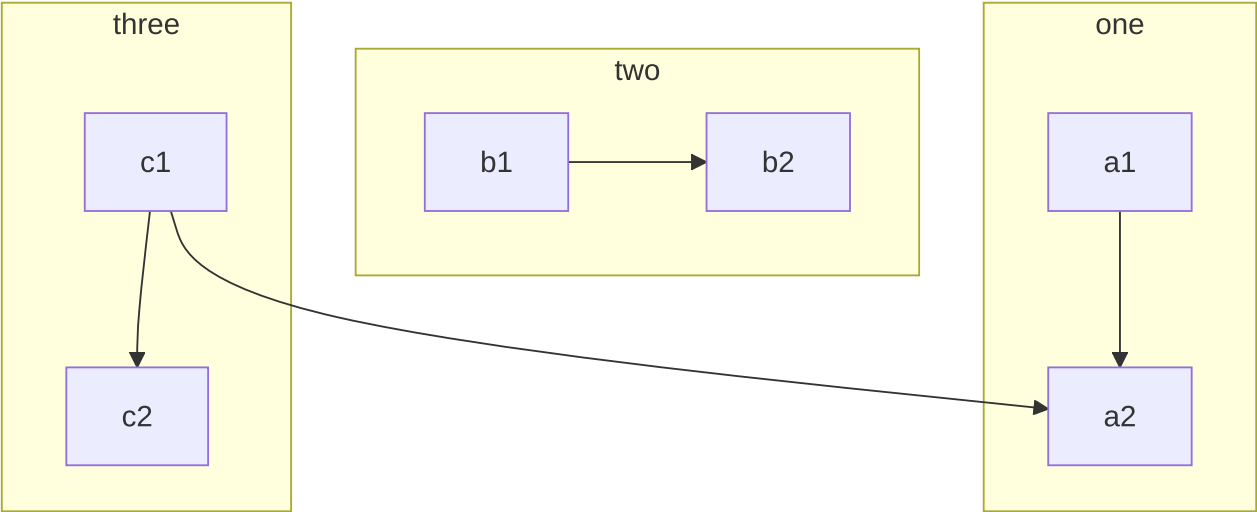
11



- 

```
subgraph title
 graph definition
end
```

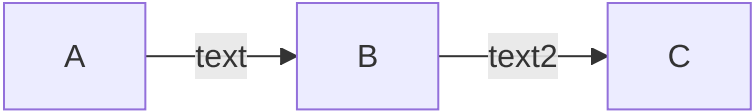
```
graph TB
 c1-->a2
 subgraph one
 a1-->a2
 end
 subgraph two
 b1-->b2
 end
 subgraph three
 c1-->c2
 end
```



- 图图图

图图 %% 图图图图图图图图.

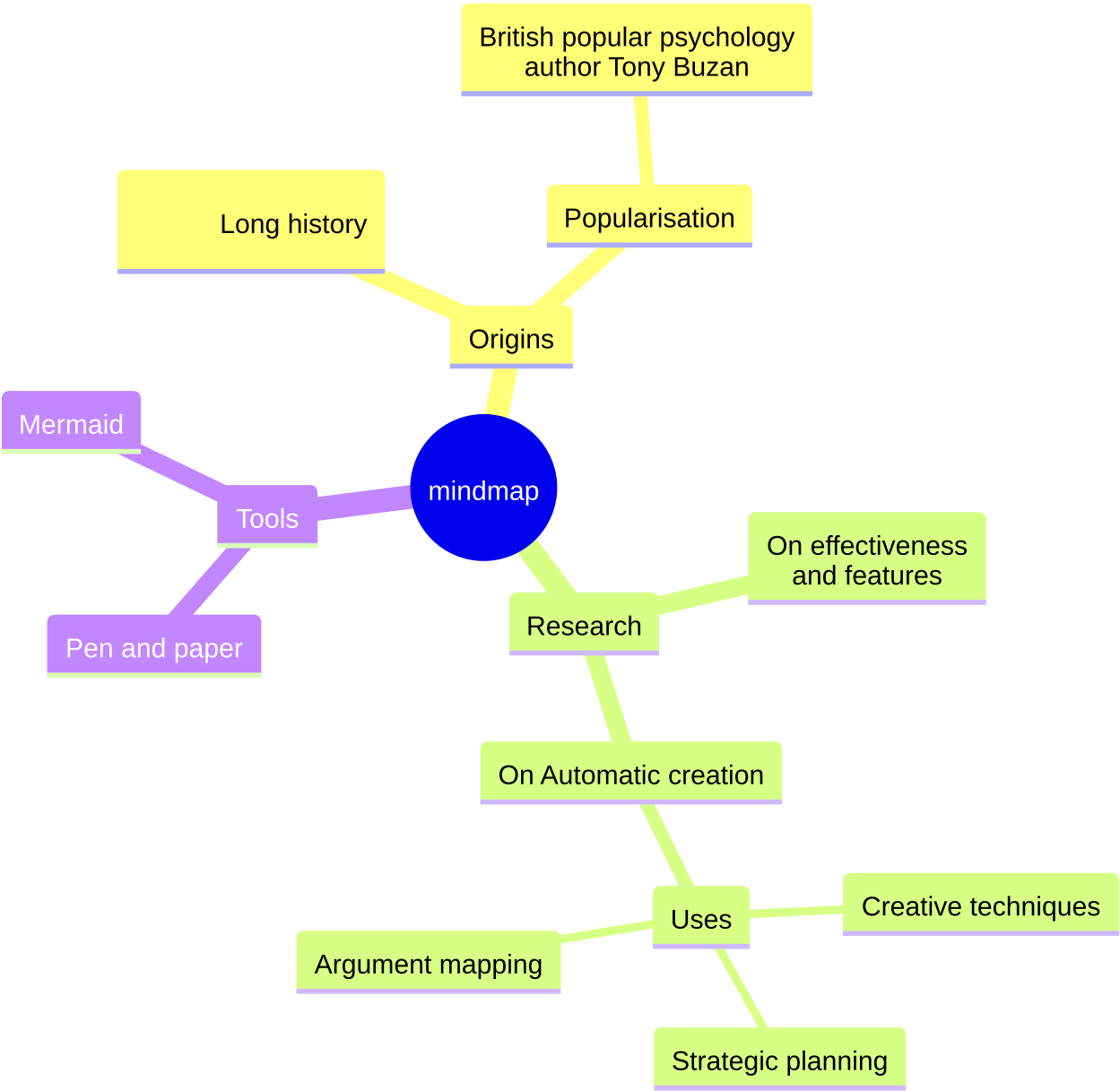
```
graph LR
%% this is a comment A -- text --> B{node}
 A -- text --> B -- text2 --> C
```



图图图图图图图图图图图图

图图

- 图图图图图图
- 图图图图图
- 图图图图



Mermaid 图生成器, 使用 Markdown 格式描述图结构, 生成 Mermaid 图代码。

图的基本概念

图是表示对象之间关系的一种模型。图由顶点和边组成。顶点表示对象，边表示对象之间的关系。

关键字	描述	说明
graph	graph	图的基本结构
subgraph	subgraph	子图
top	TB	BT, 从上到下
bottom	BT	TB, 从下到上
left	LR	RL, 从左到右
right	RL	LR, 从右到左

图的应用

图在计算机科学中有广泛的应用，如数据库、网络、人工智能等。

- 图的应用

符号	描述	说明	备注
[]	图	图的基本结构	图
()	子图	子图	图
{}	图	图的基本结构	图
<>	图	图的基本结构	图
--	图	图的基本结构	图
-.	图	图的基本结构	图
==	图	图的基本结构	图
=:	图	图的基本结构	图

符号	名称	描述	用法
>	大于	比较运算符	数值比较
-	减法	算术运算符	数值运算
xxx	变量名	标识符	变量声明
--	双减	比较运算符	数值比较
-.	负数	算术运算符	数值运算
==	等于	比较运算符	数值比较
=:	赋值	赋值运算符	变量赋值

- 运算符

符号	名称	描述	用法
[[]]	数组	数组操作	数组访问
[()]	函数	函数调用	函数调用
[{}]	字典	字典操作	字典访问
(( ))	圆括号	表达式	表达式
([])	列表	列表操作	列表访问
({})	字典	字典操作	字典访问
xxxx	变量名	标识符	变量声明
{[]}	字典	字典操作	字典访问
{() }	字典	字典操作	字典访问
-->	右箭头	比较运算符	数值比较
---	三减	比较运算符	数值比较
-.>	负数右箭头	比较运算符	数值比较

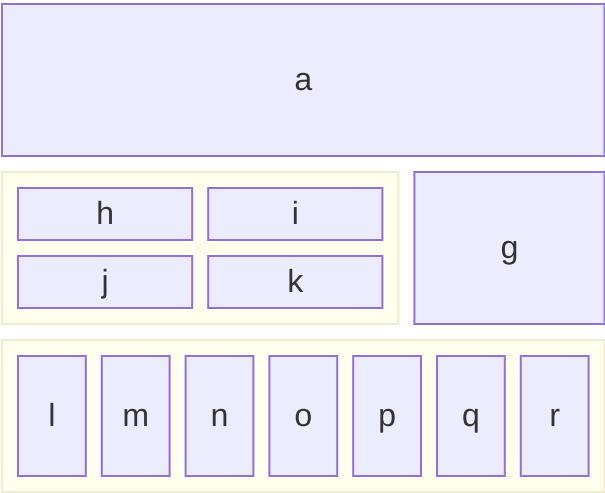


␣␣␣	␣␣	␣␣	␣␣
<code>-.-&gt;</code>	␣␣␣␣␣	␣␣␣␣␣	␣␣
<code>.-&gt;</code>	␣␣␣␣␣	␣␣␣␣␣	␣␣
<code>-.-</code>	␣␣␣␣␣	␣␣␣␣␣	␣␣
<code>.-</code>	␣␣␣␣␣	␣␣␣␣␣	␣␣
<code>==&gt;</code>	␣␣␣␣␣␣␣	␣␣␣␣␣	␣␣
<code>===</code>	␣␣␣␣␣␣␣	␣␣␣␣␣	␣␣
<code>:=&gt;</code>	␣␣␣␣␣␣␣	␣␣␣␣␣	␣␣ ␣
<code>:=:&gt;</code>	␣␣␣␣␣␣␣	␣␣␣␣␣	␣␣ ␣
<code>:=</code>	␣␣␣␣␣␣␣	␣␣␣␣␣	␣␣ ␣
<code>␣␣␣</code>	␣␣␣␣␣␣␣␣␣	␣␣␣␣␣ ␣	␣␣
<code>--connection line description-&gt;</code>	␣␣␣␣␣␣␣␣␣␣␣␣ ␣	␣␣␣␣␣ ␣	␣␣
<code>-.connection line description.-&gt;</code>	␣␣␣␣␣␣␣␣␣␣␣␣ ␣	␣␣␣␣␣ ␣	␣␣
<code>--connection line description--</code>	␣␣␣␣␣␣␣␣␣␣␣␣ ␣	␣␣␣␣␣ ␣	␣␣
<code>-.connection line description.-</code>	␣␣␣␣␣␣␣␣␣␣␣␣ ␣	␣␣␣␣␣ ␣	␣␣
<code>==connection line description==&gt;</code>	␣␣␣␣␣␣␣␣␣␣␣␣ ␣␣␣	␣␣␣␣␣ ␣	␣␣
<code>:=connection line description:=&gt;</code>	␣␣␣␣␣␣␣␣␣␣␣␣ ␣␣␣	␣␣␣␣␣ ␣	␣␣ ␣
<code>==connection line description===</code>	␣␣␣␣␣␣␣␣␣␣␣␣ ␣␣␣	␣␣␣␣␣ ␣	␣␣

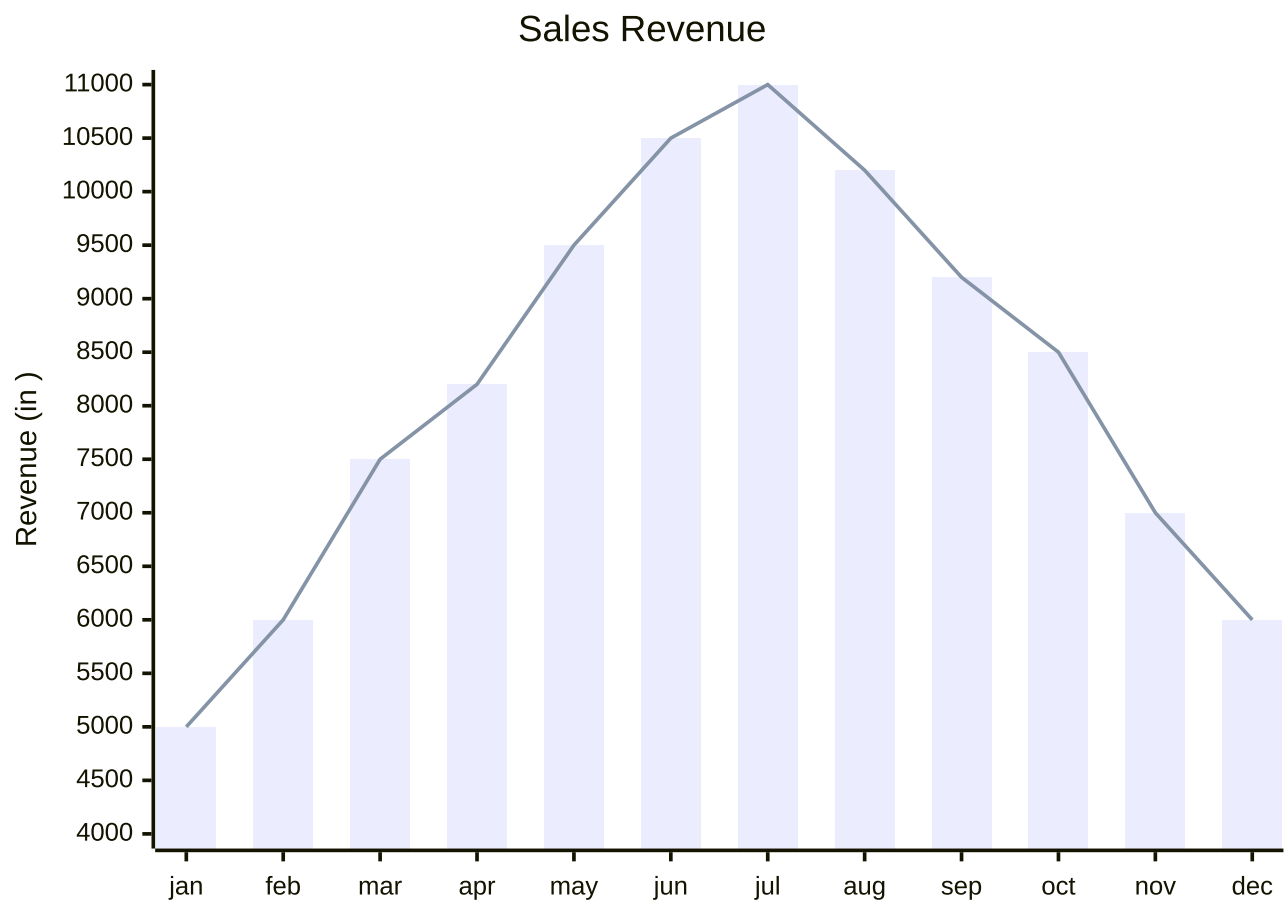


□□□

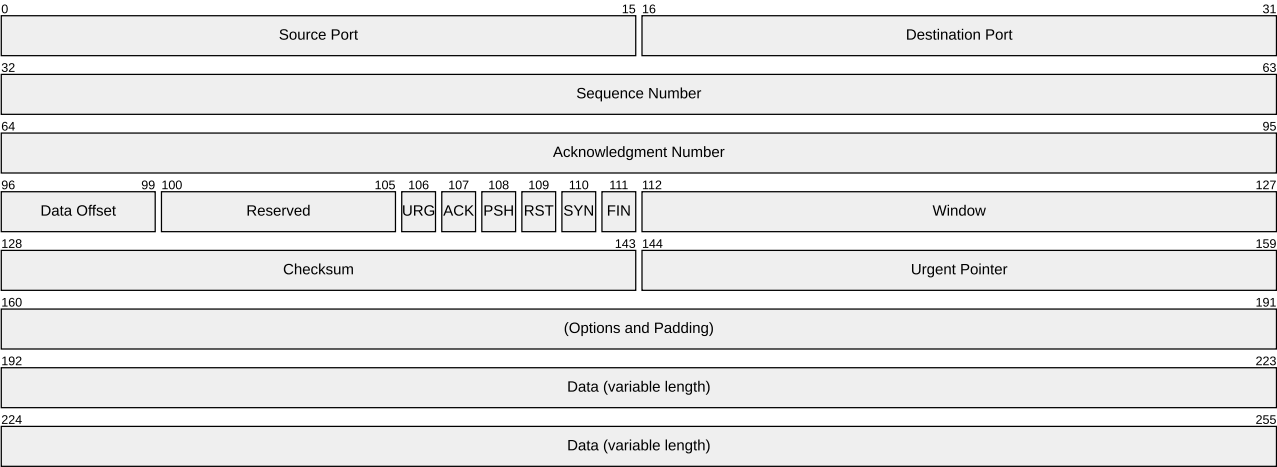
□□



XY



□□□



TCP Packet

# mdbook-admonish

The following admonishments are implemented by the [mdbook-admonish](#) plugin and are automatically themed to match Catppuccin.

## Directives

All supported directives are listed below.

`note`

### Note

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`abstract`, `summary`, `tldr`

### Abstract

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`info`, `todo`

 **Info**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

tip, hint, important

 **Tip**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

success, check, done

 **Success**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

question, help, faq

 **Question**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

warning, caution, attention

 **Warning**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

failure, fail, missing

 **Failure**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

danger, error

 **Danger**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

bug

 **Bug**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

example



### Example

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

quote, cite

### ” Quote

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

## Bienvenue sur notre site de développement 3D !

Bienvenue sur notre site dédié au développement 3D. Ici, vous trouverez des ressources, des tutoriels et des informations utiles pour vous lancer dans le monde passionnant de la 3D.

### Info

A beautifully styled message.

### Un example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

 **Une note**

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

 **Un warning**

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

 **Collapsing note** 

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

 **Le javascript c'est yolo préférez Typescript**

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

 **Referencing and dereferencing**

The opposite of *referencing* by using `&` is *dereferencing*, which is accomplished with the *dereference operator*, `*`.

## Bug

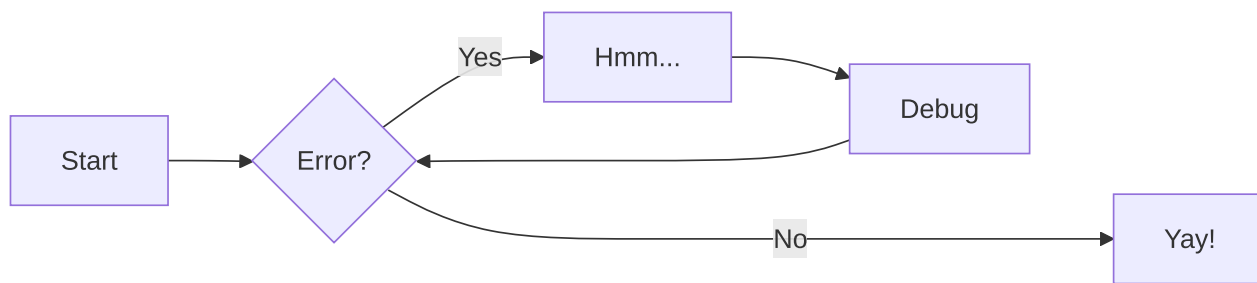
This syntax won't work in Python 3:



```
print "Hello, world!"
```

## À propos de nous

Nous sommes une équipe passionnée par la 3D et nous avons pour mission de partager nos connaissances avec la communauté. Vous trouverez ici des articles, des exemples de code et des démonstrations pour vous aider à démarrer votre voyage dans le développement 3D.



## Pour commencer

Si vous êtes nouveau dans le domaine de la 3D, ne vous inquiétez pas ! Notre page ["Getting Started"](#) vous guidera à travers les étapes essentielles pour démarrer rapidement.

## Restons en contact

N'hésitez pas à nous suivre sur les réseaux sociaux pour rester à jour avec nos dernières publications et annonces. Si vous avez des questions ou des commentaires, n'hésitez pas à nous contacter !

## Mizux



# mdbook\_mathjax

## Section 1

## Section 2

## Chapter 2

## Section 1

## Section 2

## Chapter 3

## Section 1

## Section 2

## Chapter 4

## Section 1

## Section 2

# Chapter 5

## Section 1

## Section 2



# Chapter 6

## Section 1

### Note

Highlights information that users should take into account, even when skimming.

### Tip

Optional information to help a user be more successful.

### Important

Crucial information necessary for users to succeed.

### Warning

Critical content demanding immediate user attention due to potential risks.

### Caution

Negative potential consequences of an action.

## Section 2

Here is an inline example,  $\pi(\theta)$ ,  
an equation,



$$\nabla f(x) \in \mathbb{R}^n,$$

and a regular \$ symbol.

Define  $f(x)$ :

$$f(x) = x^2$$

$$x \in \mathbb{R}$$

```
graph TD
 A[Anyone] -->|Can help | B(Go to github.com/yuzutech/kroki)
 B --> C{ How to contribute? }
 C --> D[Reporting bugs]
 C --> E[Sharing ideas]
 C --> F[Advocating]
```

□□□□

- □□□□□□□□Python: \\ □□□□ \\,□□□□□□□

```
def ex_gcd(a, b):
 if b == 0:
 return 1, 0, a
 else:
 x, y, g = ex_gcd(b, a % b)
 return y, x - (a // b) * y, g
```

```
N = 5
for n in range(1, N + 1):
 for m in range(1, N + 1):
 print(ex_gcd(n, m))
```

- □□□□□□□□

1

$x_1$

$\vdots$

$x_1^{n-1}$

1

$x_2$

$\vdots$

$x_2^{n-2}$

$\dots$

$\dots$

$\ddots$

$\dots$

1

$x_n$

$\vdots$

$x_n^{n-1}$

$$= \prod_{1 \leq j \leq i \leq n} (x_i - x_j)$$

- 



- 

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

$$i\hbar \frac{\partial}{\partial t} \Psi(x,t) = \left[ -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x,t) \right] \Psi(x,t)$$

- 

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} = \mu_0 \left( \mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \end{array} \right.$$

Gauss’s law

Gauss’s law for magnetism

Maxwell–Faraday equation

Ampère’s circuital law

- 

$x$	$y$	$S$
0	0	0
0	1	1
1	0	1
1	0	0

-

$$\mathbf{1}_{\mathbb{Q}}(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

- 同余式

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } a \equiv 0 \pmod{p} \\ 1 & \text{if } a \not\equiv 0 \pmod{p} \wedge \exists x : x^2 \equiv a \pmod{p} \\ -1 & \text{if } a \not\equiv 0 \pmod{p} \wedge \nexists x : x^2 \equiv a \pmod{p} \end{cases}$$

- 微分形式

$$\oint_{\Gamma} Pdx + Qdy + Rdz = \iiint_{\Sigma} \left( \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dydz \tag{1}$$

$$+ \left( \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dzdx \tag{2}$$

$$+ \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy \tag{3}$$

- 特殊函数

$$\Gamma(z) = \int_0^\infty t^{z-1} \mathrm{e}^{-t} dt$$

$$\zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s}$$

- 逻辑

$\neg(P \vee Q) \iff \neg P \wedge \neg Q$

(propositional logic)

$\neg(P \wedge Q) \iff \neg P \vee \neg Q$

(4)

$(A \cup B)^c = A^c \cap B^c$

(set theory)

$(A \cap B)^c = A^c \cup B^c$

(5)

$\overline{A \cup B} = \overline{A} \cap \overline{B}$

(probability theory)

$\overline{A \cap B} = \overline{A} \cup \overline{B}$

(6)

- 全概率公式

$$P(A_i|B) = \frac{P(B|A_i) \cdot P(A_i)}{\sum_{j=1}^n P(B|A_j) \cdot P(A_j)}$$

- 泰勒级数

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \tag{7}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \tag{8}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \tag{9}$$

$$\ln(x+1) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{(n+1)!} \tag{10}$$

- 自然常数e的表达式

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$$

- 二项式定理

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

# mdbook-katex

HTML:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\Rightarrow x + iy = re^{i\theta}$$

Markdown (requires `mdbook-katex`):

$$\oint_C f(x, y) \, dA$$

Inspect element and use `Sources` tab (under `Debugger` on Firefox) to check that all CSS and fonts are properly loaded from GitHub pages instead of external CDN.

▼ Proof that  $e^{ix} = \cos x + i \sin x$

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \implies e^{ix} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} \\ \cos x &= \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{(2m)!} = \sum_{m=0}^{\infty} \frac{(ix)^{2m}}{(2m)!} \\ \sin x &= \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+1}}{(2s+1)!} = \sum_{s=0}^{\infty} \frac{(ix)^{2s+1}}{i(2s+1)!} \\ \cos x + i \sin x &= \sum_{l=0}^{\infty} \frac{(ix)^{2l}}{(2l)!} + \sum_{s=0}^{\infty} \frac{(ix)^{2s+1}}{(2s+1)!} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} \\ &= e^{ix} \end{aligned}$$

Fourier Transform:

$$f(t) = \int_{-\infty}^{\infty} F(\omega) i^{4t\omega} d\omega$$
$$F(\omega) = \int_{-\infty}^{\infty} f(t) i^{-4t\omega} dt$$

Pauli Matrices:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$
$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

# mdbook-whichlang



```
#include <stdio.h>

int main(void) {
 printf("Hello World\n");
}
```



```
#include <iostream>

int main()
{
 std::cout << "Hello World" << std::endl;
}
```



```
console.log("Hello World");
```



```
console.log("Hello World");
```



```
Console.log("Hello World")
```



```
h1 {
 color: blue;
}
```



```
set syntax=ruby
```



init.lua

```
print("Hello World")
```



```
print("Hello World")
```



```
echo "Hello World"
```



hello.zig

```
const std = @import("std");

pub fn main() !void {
 const stdout = std.io.getStdOut().writer();
 try stdout.print("Hello, {s}!\n", .{"world"});
}
```





```
package main

import "core:fmt"

main :: proc() {
 fmt.println("Hellope!")
}
```

WA

hello.wat

```
(module
 (import "wasi_unstable" "fd_write"
 (func $fd_write (param i32 i32 i32 i32) (result i32))
)

 (memory 1)
 (export "memory" (memory 0))

 (data (i32.const 0) "\08\00\00\00\0c\00\00\00Hello World\n")

 (func $main (export "_start")
 i32.const 1
 i32.const 0
 i32.const 1
 i32.const 20
 call $fd_write
 drop
)
)
```

# mdbook-langtabs

## Some code

let's try with a simple example:



```
LogChannel(n"DEBUG", "hello world");
```



```
print("Hello, World!")
```



```
fn main() { println!("hello world"); }
```



```
print("Hello World")
```



```
#include <iostream>

int main() {
 std::cout << "Hello World!";
 return 0;
}
```



```
some:
 interesting:
 - property
```



```
{
 "some": { "interesting": ["property"] }
}
```

*xml:*

```
<some>
 <interesting>
 <property />
 </interesting>
</some>
```

contrary to code blocks, `inline` and `fenced` are left untouched.

it should also work when deeply nested:

1. outer:

1. inner:



```
GameInstance
 .GetStatusEffectSystem(this.GetGame())
 .ApplyStatusEffect(
 this.GetEntityID(),
 t"BaseStatusEffect.SplinterAddicted",
 this.GetRecordID(),
 this.GetEntityID());
```

# Hello World

Here's a simple "Hello World" example in different languages:



```
fn main() {
 println!("Hello, World!");
}
```



```
def main():
 print("Hello, World!")

if __name__ == "__main__":
 main()
```



```
function main() {
 console.log("Hello, World!");
}

main();
```



```
package main

import "fmt"

func main() {
 fmt.Println("Hello, World!")
}
```



```
public class HelloWorld {
 public static void main(String[] args) {
 System.out.println("Hello, World!");
 }
}
```

## Simple Function Example

Here's how you might define a function to calculate factorial in different languages:



```
fn factorial(n: u64) -> u64 {
 match n {
 0 | 1 => 1,
 _ => n * factorial(n - 1)
 }
}

fn main() {
 println!("5! = {}", factorial(5)); // Outputs: 5! = 120
}
```



```
def factorial(n):
 if n <= 1:
 return 1
 return n * factorial(n - 1)

print(f"5! = {factorial(5)}") # Outputs: 5! = 120
```



```
function factorial(n) {
 if (n <= 1) return 1;
 return n * factorial(n - 1);
}

console.log(`5! = ${factorial(5)}`); // Outputs: 5! = 120
```



```
package main

import "fmt"

func factorial(n uint64) uint64 {
 if n <= 1 {
 return 1
 }
 return n * factorial(n-1)
}

func main() {
 fmt.Printf("5! = %d\n", factorial(5)) // Outputs: 5! = 120
}
```



```
public class FactorialExample {
 static long factorial(int n) {
 if (n <= 1) return 1;
 return n * factorial(n - 1);
 }

 public static void main(String[] args) {
 System.out.println("5! = " + factorial(5)); // Outputs: 5! = 120
 }
}
```

# Data Structures Example

Here's how you might implement a simple stack in different languages:



```
struct Stack<T> {
 items: Vec<T>,
}

impl<T> Stack<T> {
 fn new() -> Self {
 Stack { items: Vec::new() }
 }

 fn push(&mut self, item: T) {
 self.items.push(item);
 }

 fn pop(&mut self) -> Option<T> {
 self.items.pop()
 }

 fn is_empty(&self) -> bool {
 self.items.is_empty()
 }
}

fn main() {
 let mut stack = Stack::new();
 stack.push(1);
 stack.push(2);
 stack.push(3);

 while let Some(item) = stack.pop() {
 println!("Popped: {}", item);
 }
}
```



```
class Stack:
 def __init__(self):
 self.items = []

 def push(self, item):
 self.items.append(item)

 def pop(self):
 if not self.is_empty():
 return self.items.pop()
 return None

 def is_empty(self):
 return len(self.items) == 0

Usage
stack = Stack()
stack.push(1)
stack.push(2)
stack.push(3)

while not stack.is_empty():
 print(f"Popped: {stack.pop()}")
```

JS



```
class Stack {
 constructor() {
 this.items = [];
 }

 push(item) {
 this.items.push(item);
 }

 pop() {
 if (!this.isEmpty()) {
 return this.items.pop();
 }
 return null;
 }

 isEmpty() {
 return this.items.length === 0;
 }
}

// Usage
const stack = new Stack();
stack.push(1);
stack.push(2);
stack.push(3);

while (!stack.isEmpty()) {
 console.log(`Popped: ${stack.pop()}`);
}
```

[GO](#)

```
package main

import "fmt"

type Stack struct {
 items []int
}

func (s *Stack) Push(item int) {
 s.items = append(s.items, item)
}

func (s *Stack) Pop() (int, bool) {
 if s.IsEmpty() {
 return 0, false
 }

 index := len(s.items) - 1
 item := s.items[index]
 s.items = s.items[:index]
 return item, true
}

func (s *Stack) IsEmpty() bool {
 return len(s.items) == 0
}

func main() {
 stack := Stack{}
 stack.Push(1)
 stack.Push(2)
 stack.Push(3)

 for !stack.IsEmpty() {
 item, _ := stack.Pop()
 fmt.Printf("Popped: %d\n", item)
 }
}
```



```
import java.util.ArrayList;

public class StackExample {
 static class Stack<T> {
 private ArrayList<T> items = new ArrayList<>();

 public void push(T item) {
 items.add(item);
 }

 public T pop() {
 if (isEmpty()) {
 return null;
 }
 return items.remove(items.size() - 1);
 }

 public boolean isEmpty() {
 return items.isEmpty();
 }
 }

 public static void main(String[] args) {
 Stack<Integer> stack = new Stack<>();
 stack.push(1);
 stack.push(2);
 stack.push(3);

 while (!stack.isEmpty()) {
 System.out.println("Popped: " + stack.pop());
 }
 }
}
```

# kroki

## kroki-graphviz

```
digraph G {
 a -> b [dir=both color="red:blue"]
 c -> d [dir=none color="green:red;0.25:blue"]
}
```

# kroki-plantuml

```
@startuml
clock "Clock_0" as C0 with period 50
clock "Clock_1" as C1 with period 50 pulse 15 offset 10
binary "Binary" as B
concise "Concise" as C
robust "Robust" as R
analog "Analog" as A

@0
C is Idle
R is Idle
A is 0

@100
B is high
C is Waiting
R is Processing
A is 3

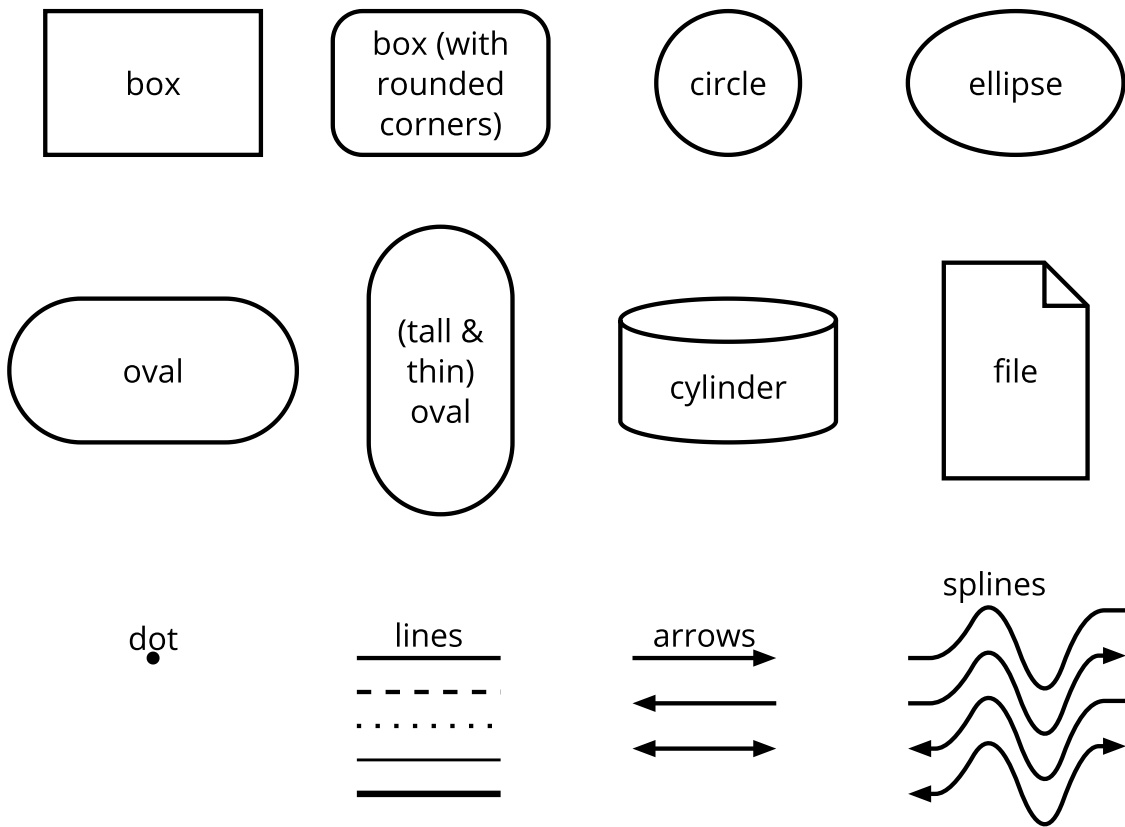
@300
R is Waiting
A is 1
@enduml
```

# kroki-symbolator

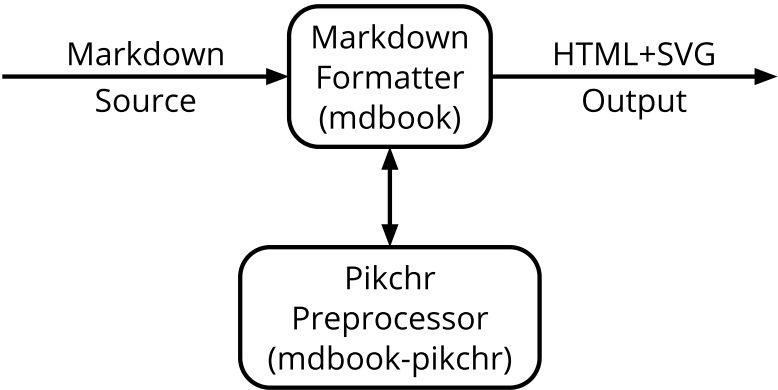
```
module demo_device #(
 //# {{}}
 parameter SIZE = 8,
 parameter RESET_ACTIVE_LEVEL = 1
) (
 //# {{clocks|Clocking}}
 input wire clock,
 //# {{control|Control signals}}
 input wire reset,
 input wire enable,
 //# {{data|Data ports}}
 input wire [SIZE-1:0] data_in,
 output wire [SIZE-1:0] data_out
);
 // ...
endmodule
```

# pikchr-demo1

## Examples Of Pikchr Objects



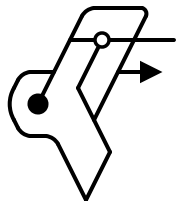
pikchr-demo2





# svgbob-demo

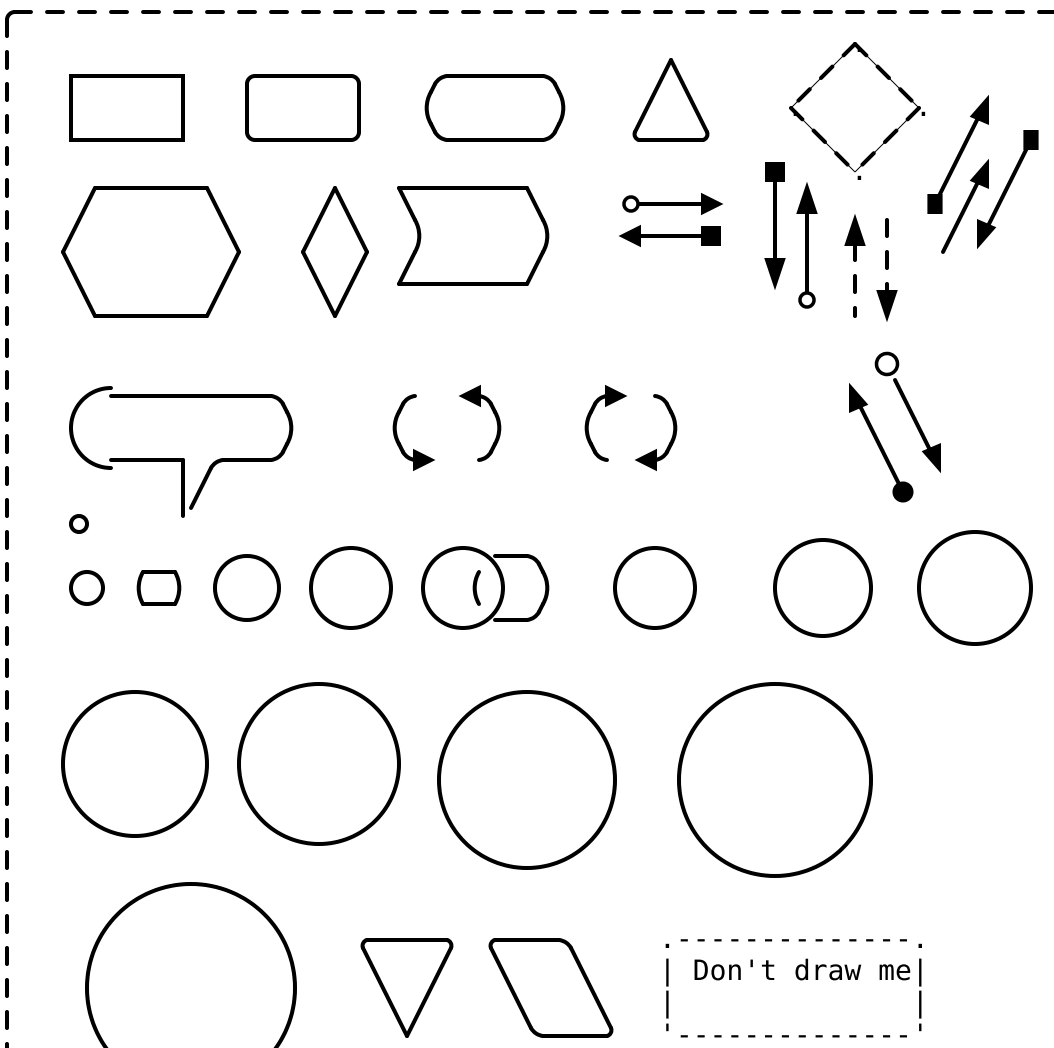
Svgbob is a diagramming model  
which uses a set of typing characters  
to approximate the intended shape.



It uses a combination of characters  
which are readily available on your keyboards.

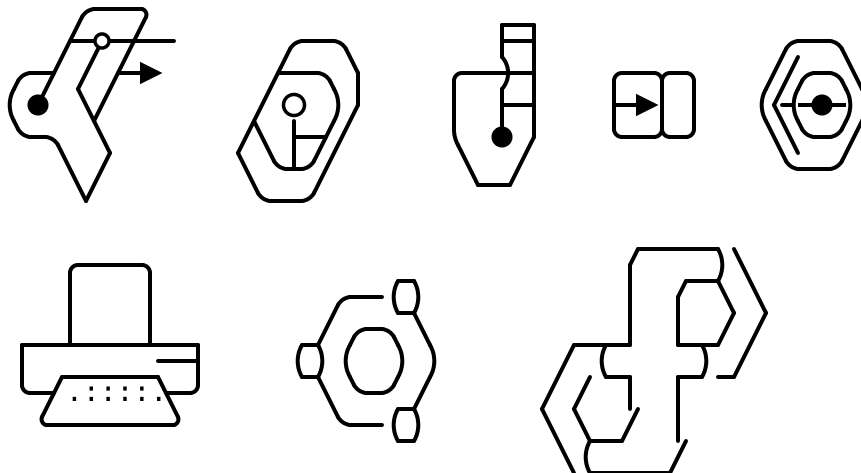
What can it do?

#### ➡ Basic shapes

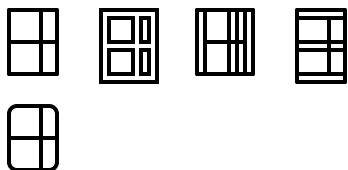




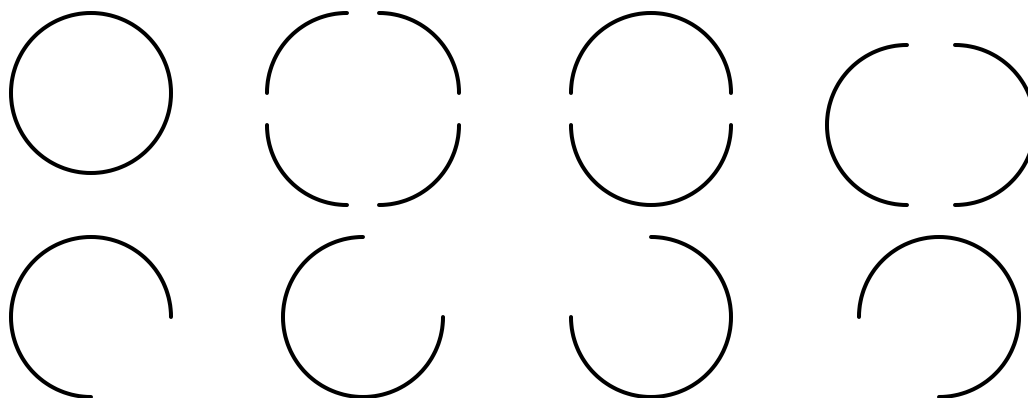
### ➡ Quick logo scribbles



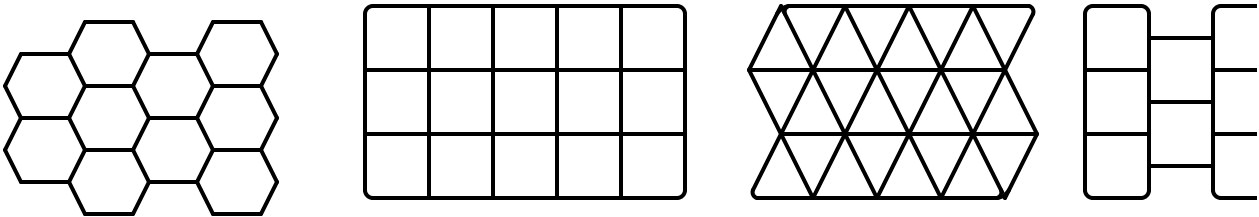
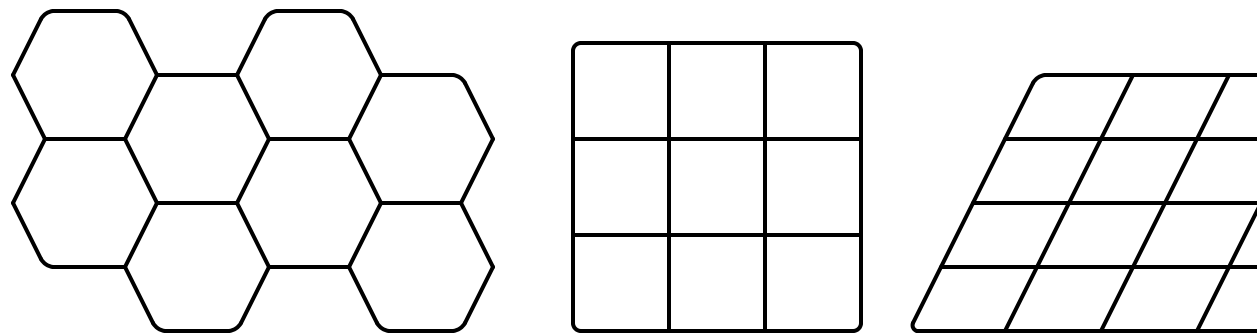
### ➡ Even unicode box drawing characters are supported



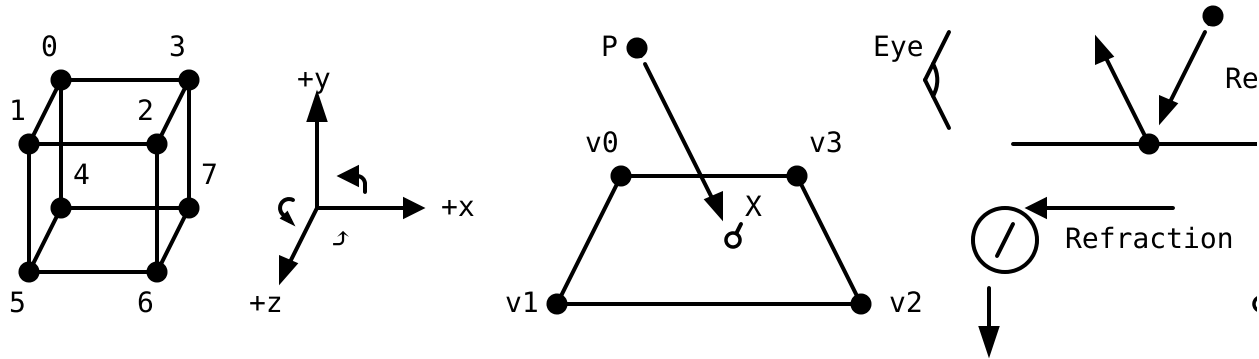
### ➡ Circle, quarter arcs, half circles, 3/4 quarter arcs



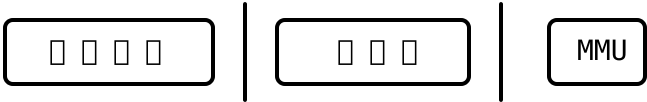
### ➡ Grids



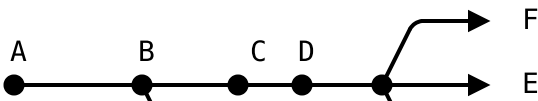
➡ Graphics Diagram

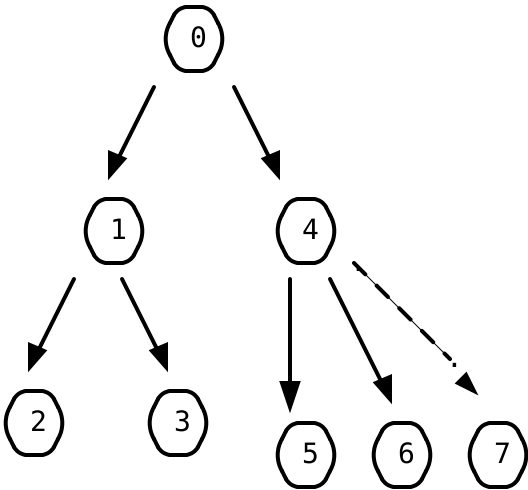
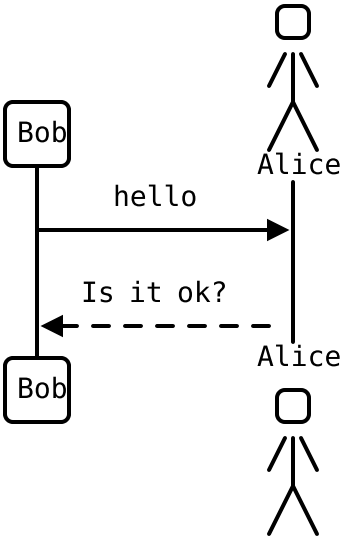
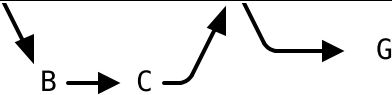


➡ CJK characters

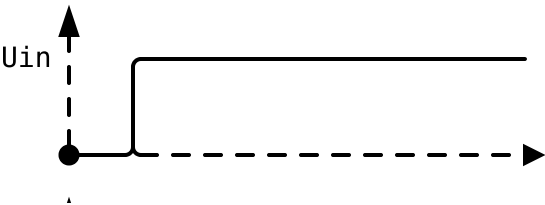


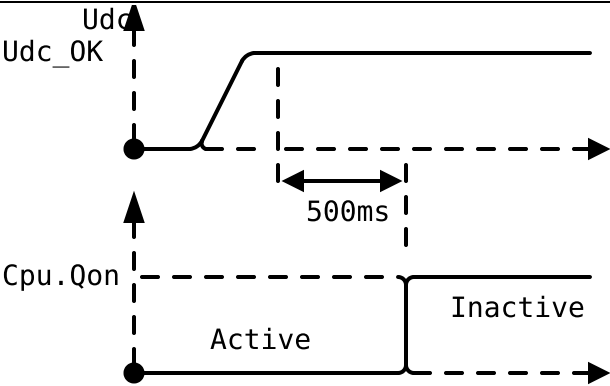
➡ Sequence Diagrams



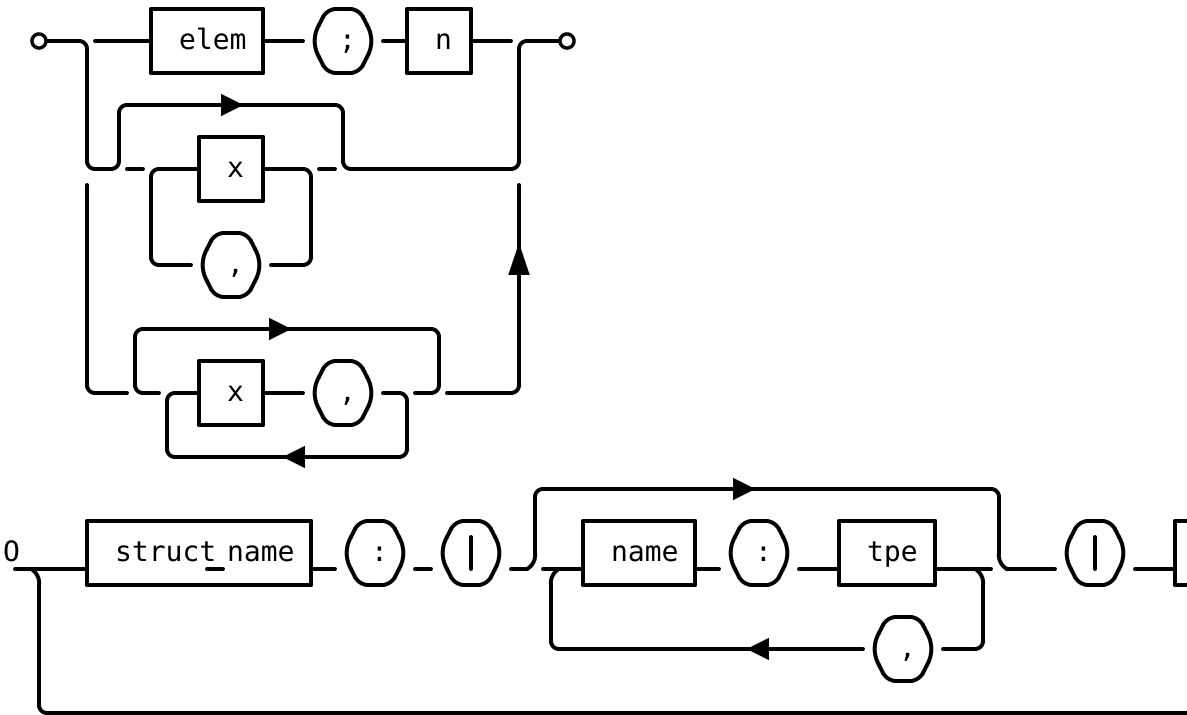


➡ Plot diagrams

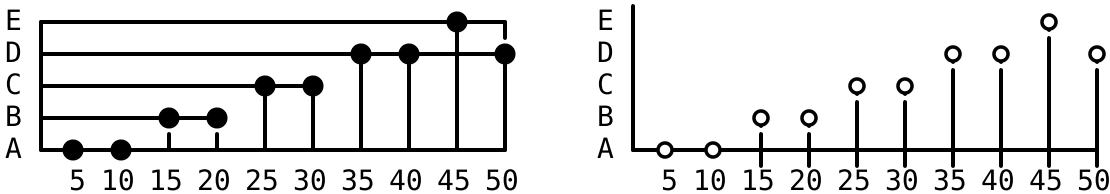


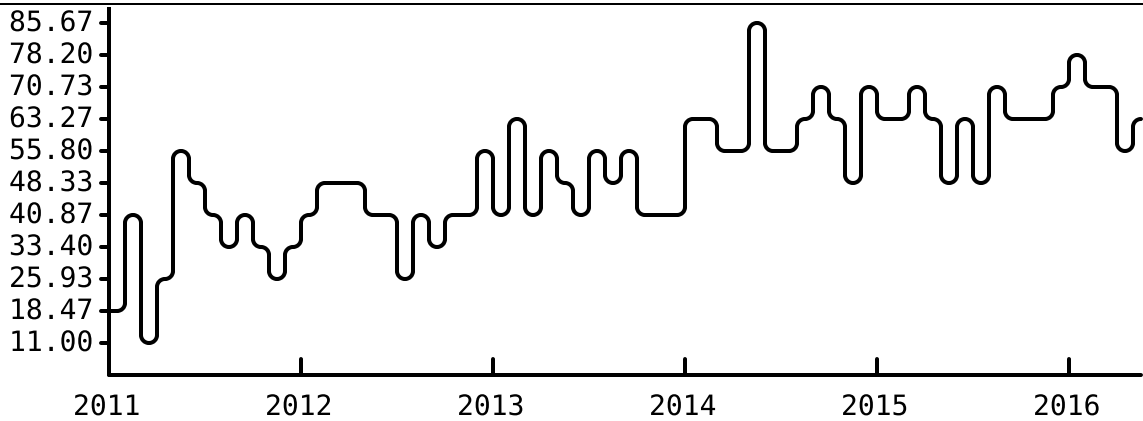


➤ Railroad diagrams

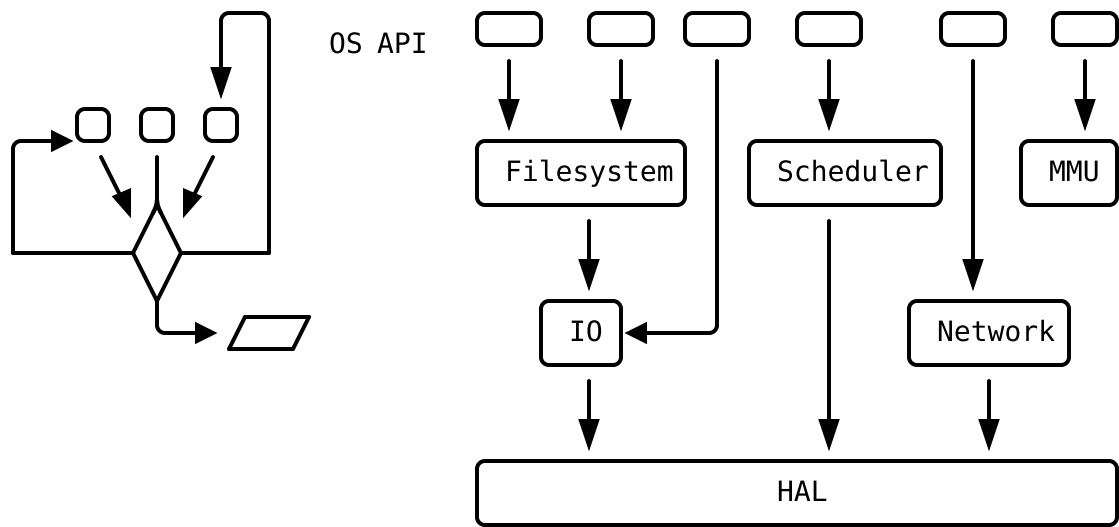


➤ Statistical charts

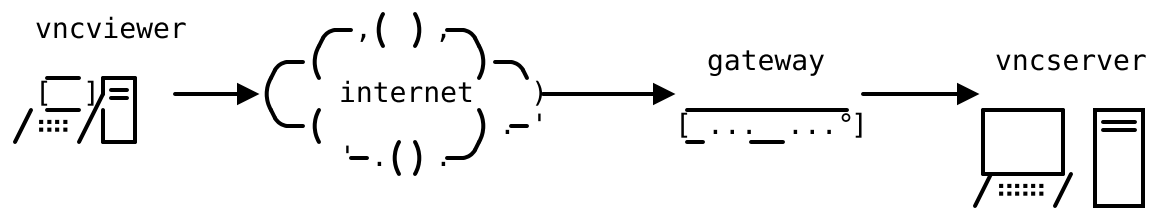


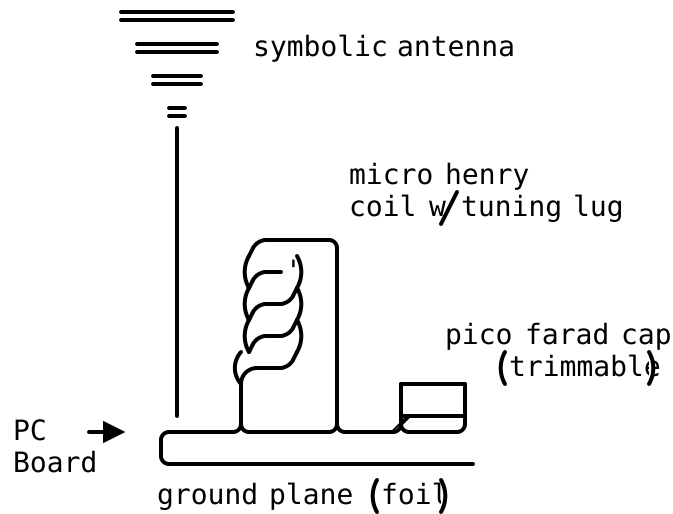
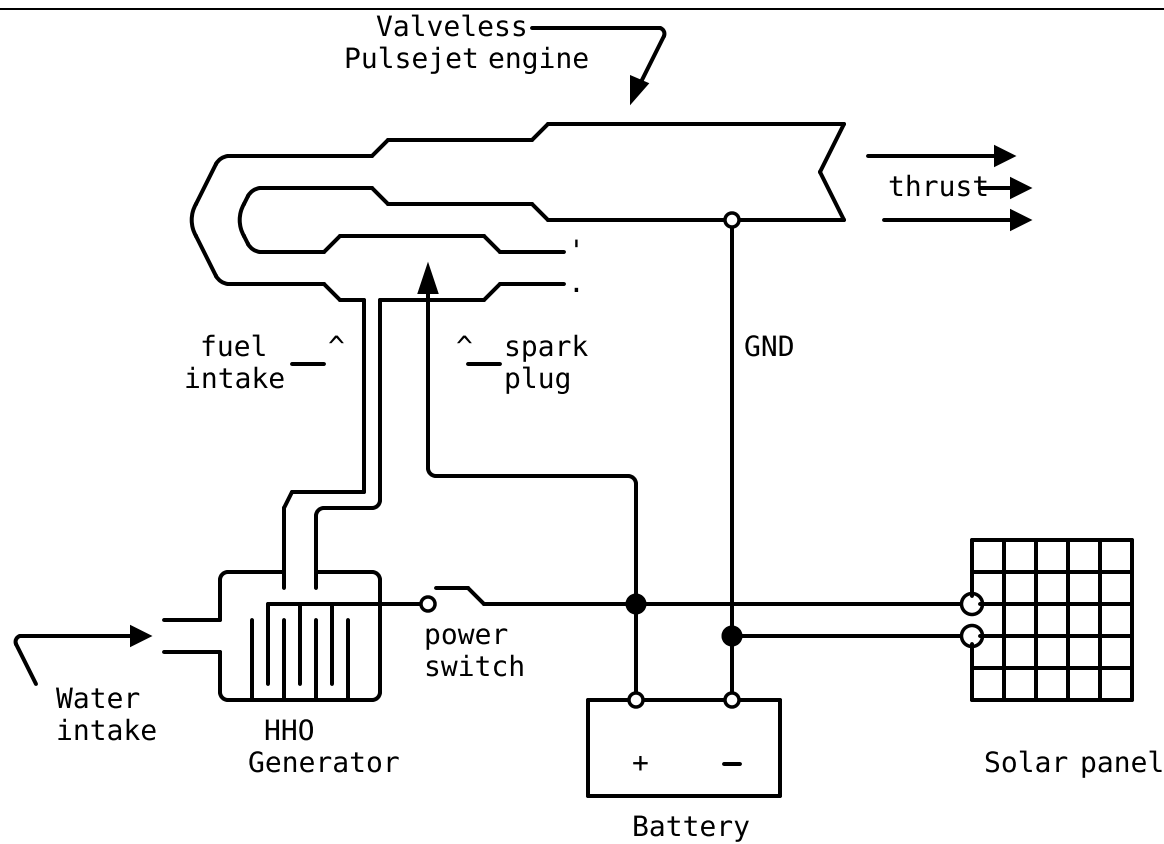


➡ Flow charts



➡ Block diagrams

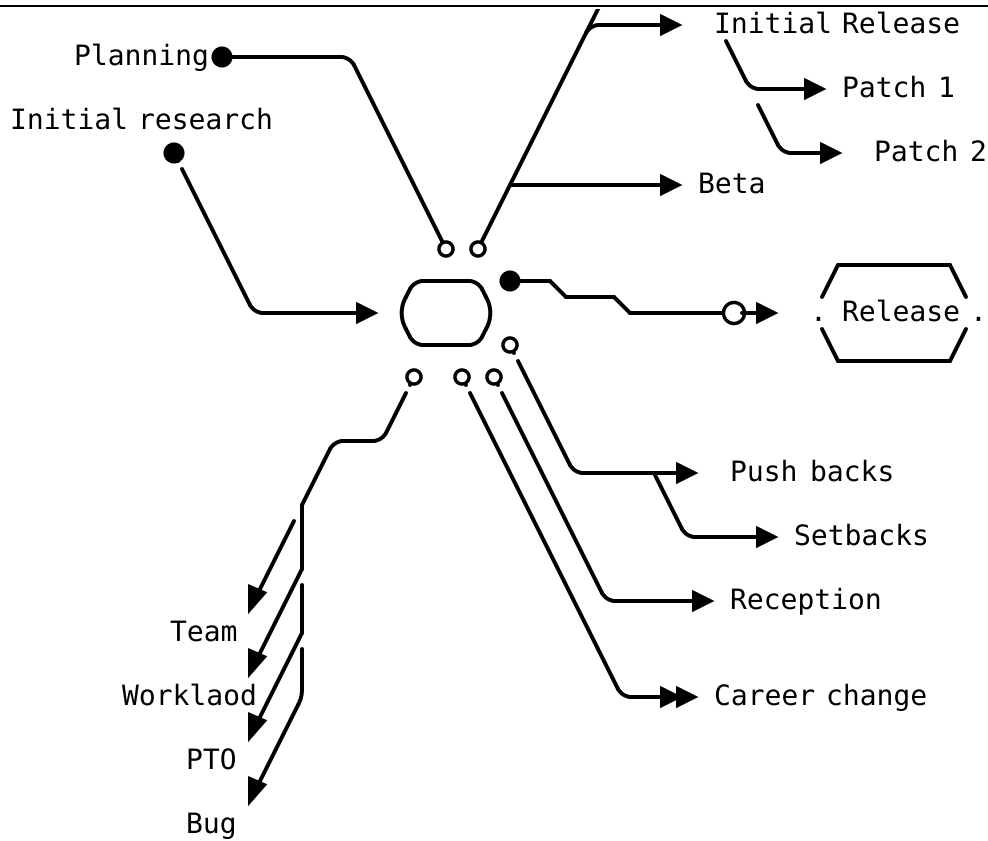




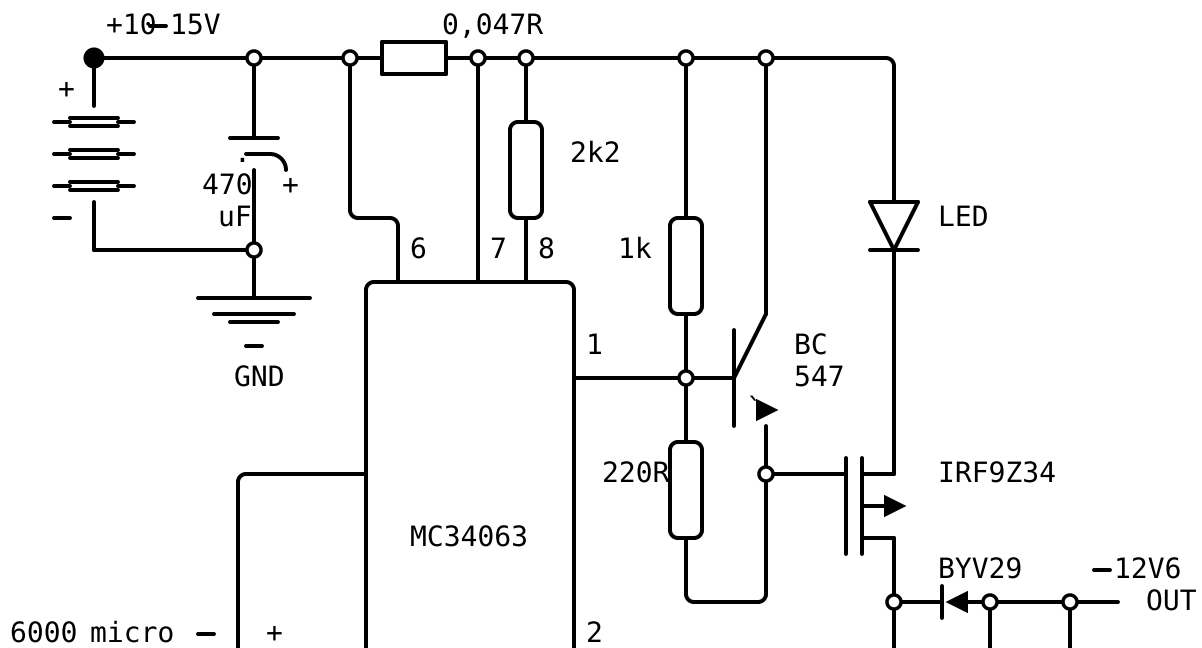
➡ Mindmaps

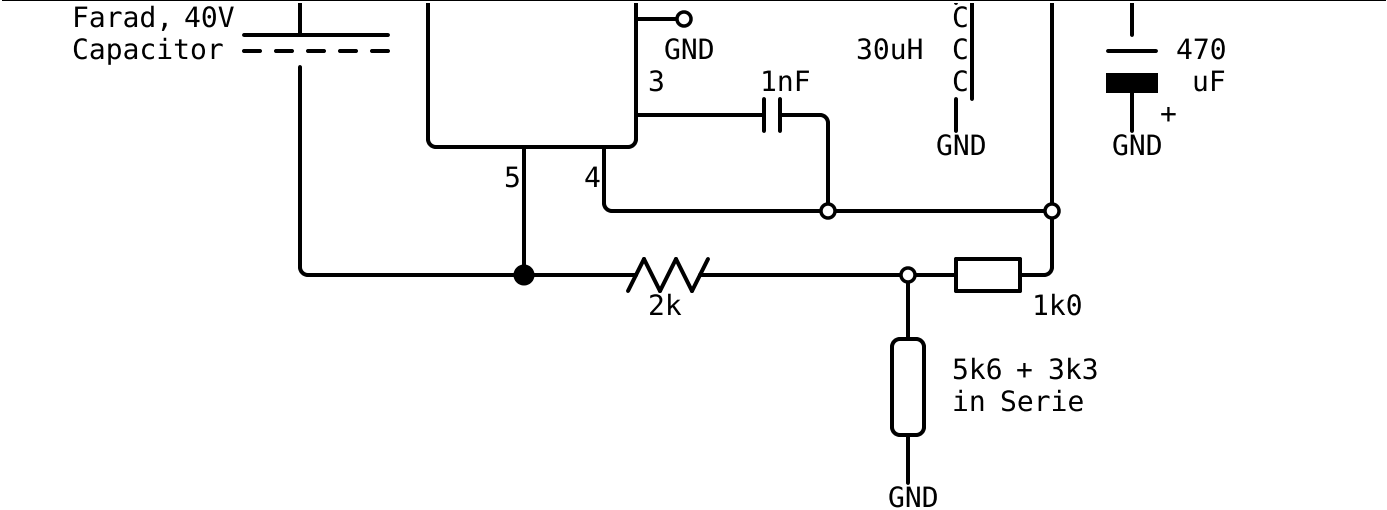
➡ Alpha



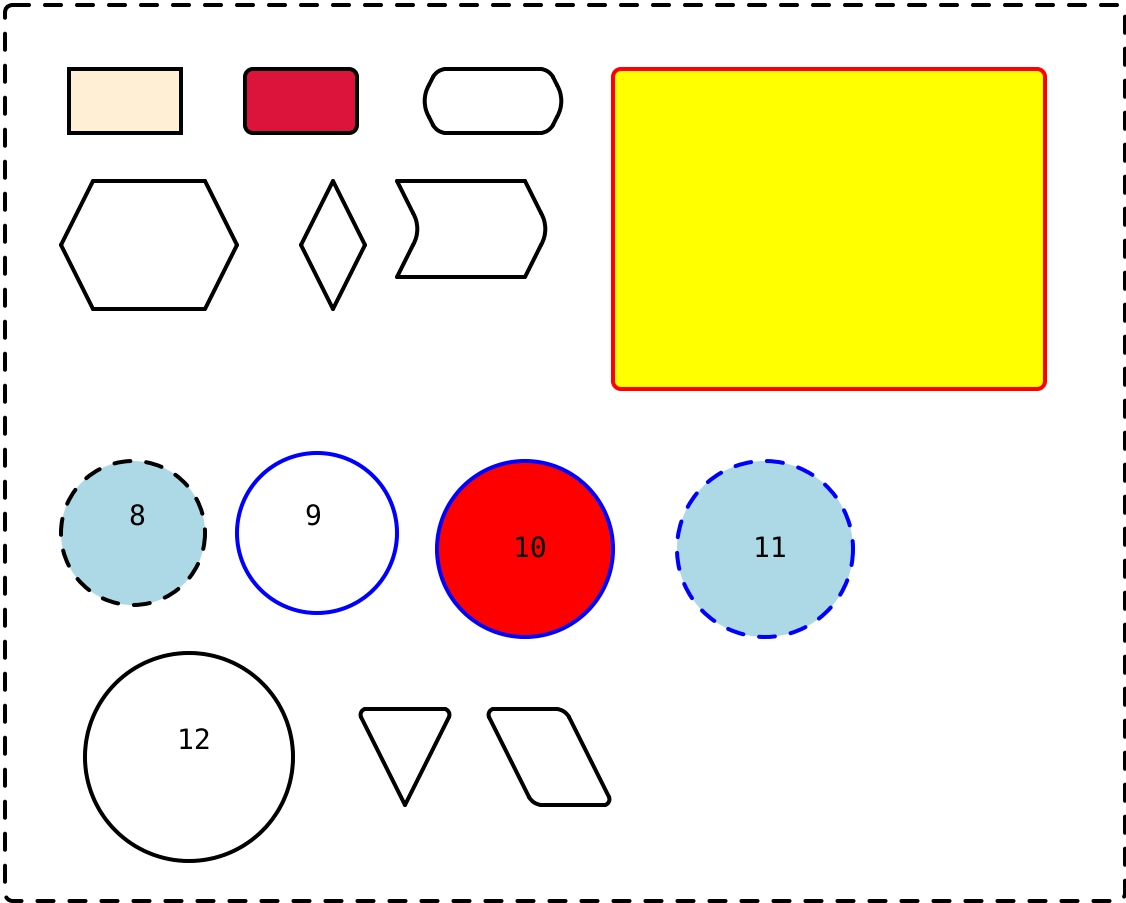


➡ It can do complex stuff such as circuit diagrams





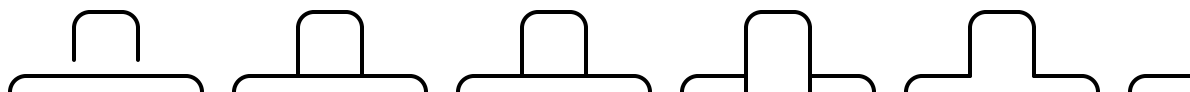
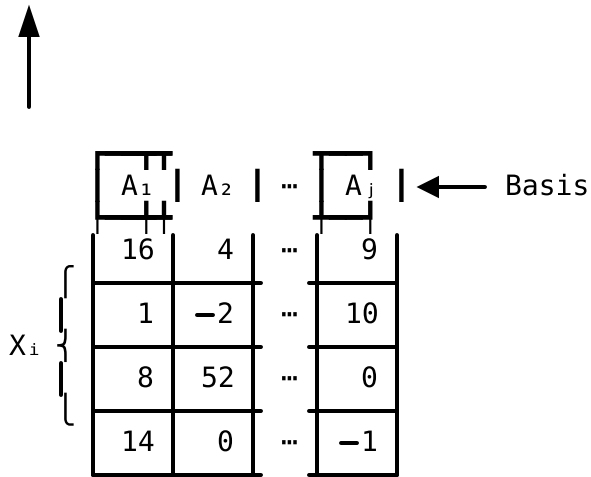
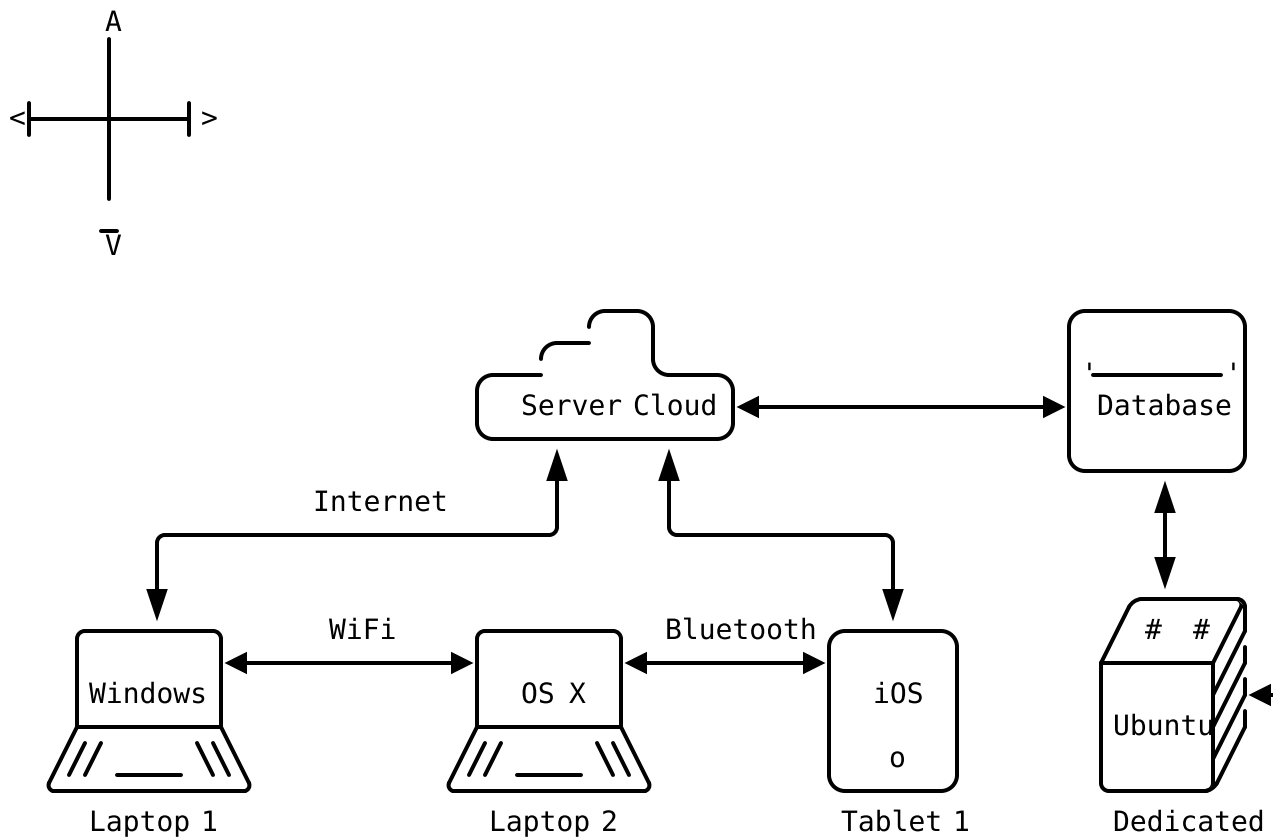
➔ Latest addition: Styling of tagged shapes

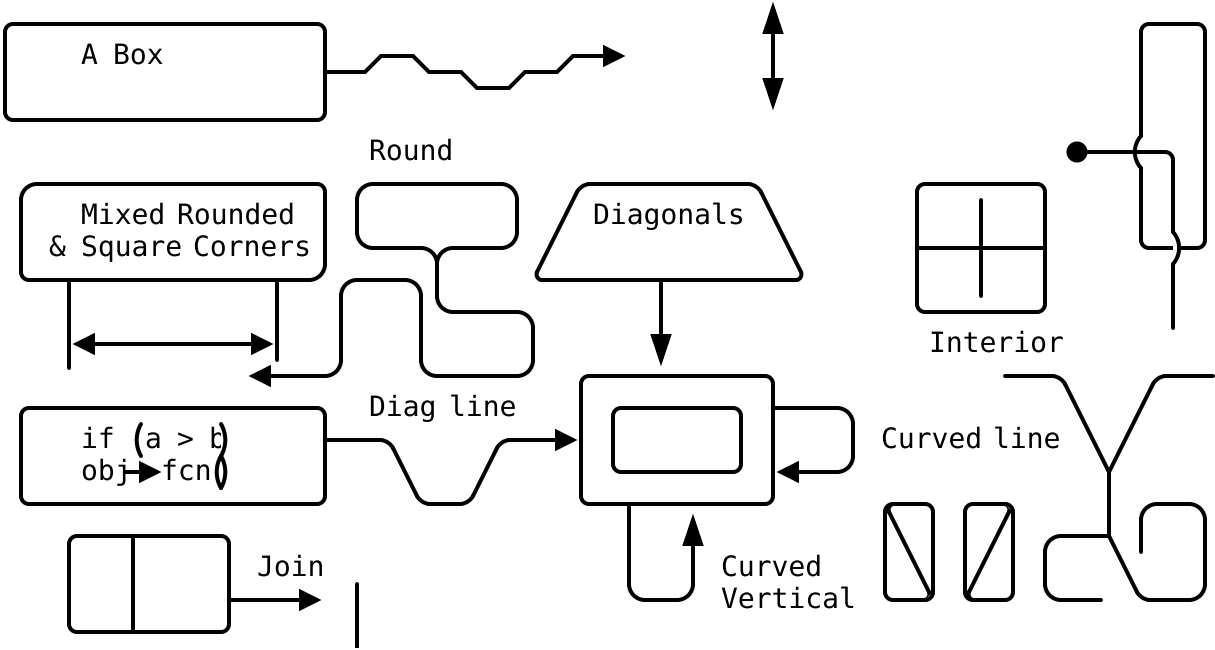
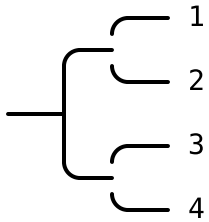
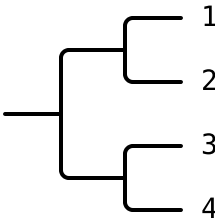
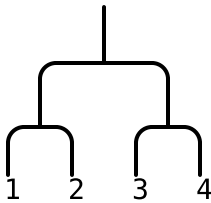
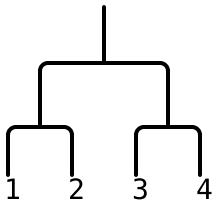
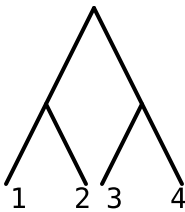
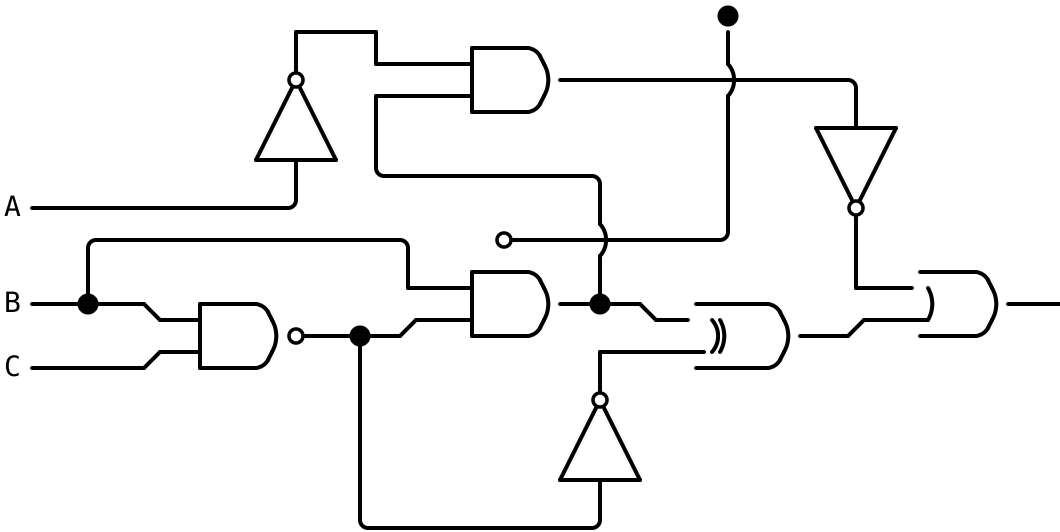
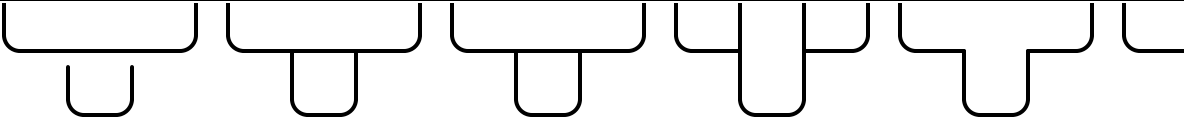


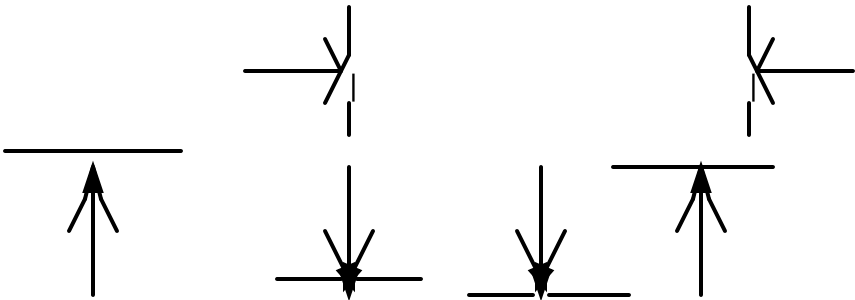
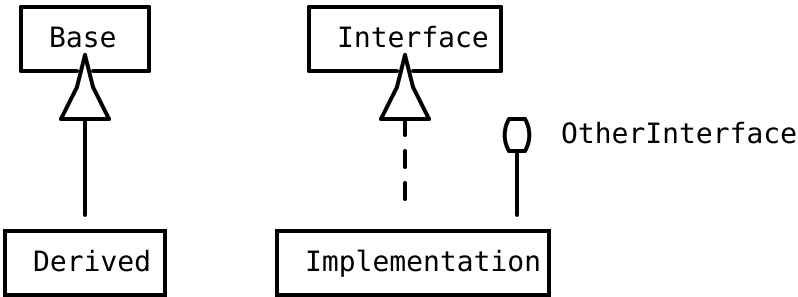
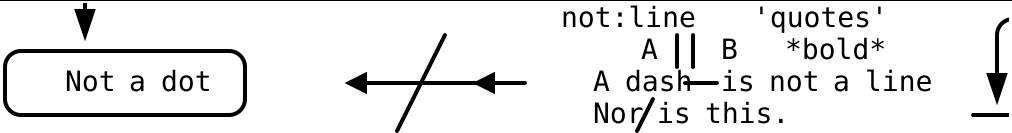
### Advantages:

- ➊ Plain text format  
Ultimately portable, backward compatible and future proof.
- ➋ Degrades gracefully  
Even when not using a graphical renderer, it would still looks good as text based diagrams. Paste the text in your source code.
- ➌ Easiest to use. Anyone knows how to edit text.

## svgbob-otherdemo

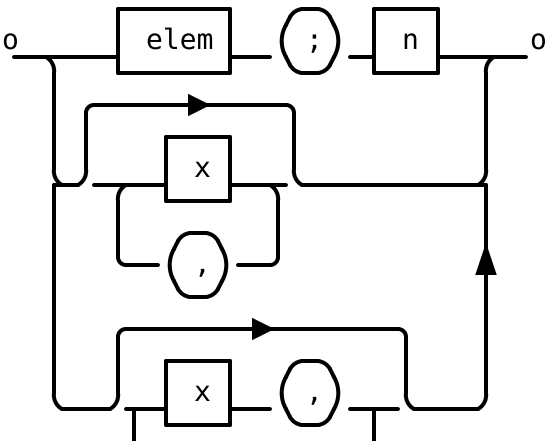


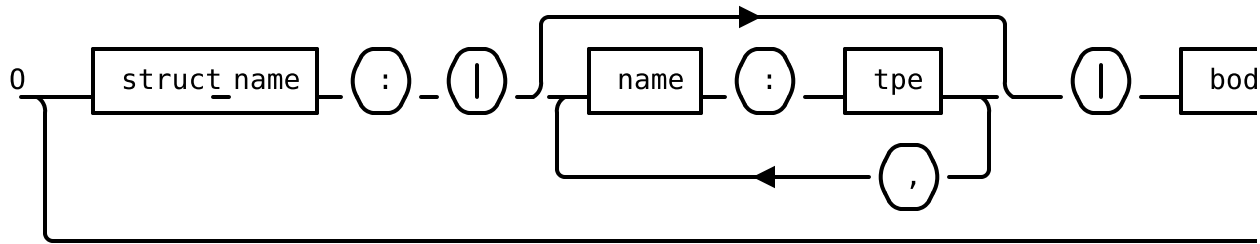




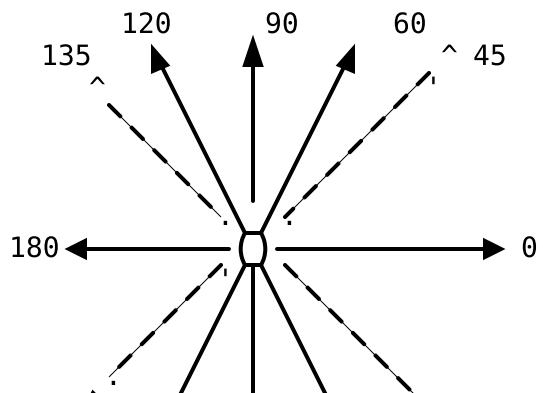
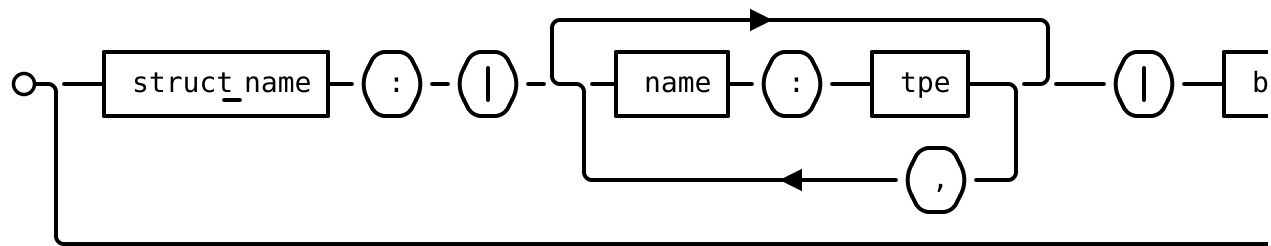
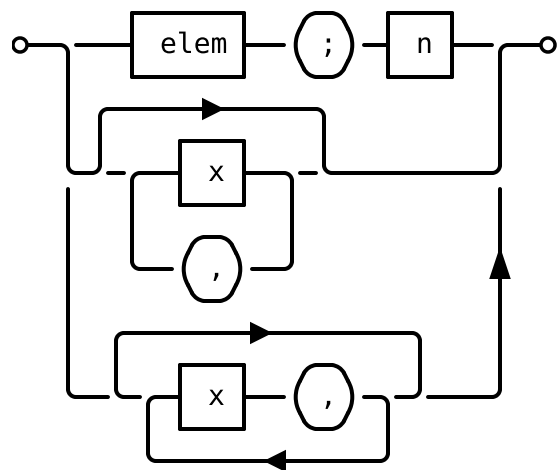
Poor man's rail road

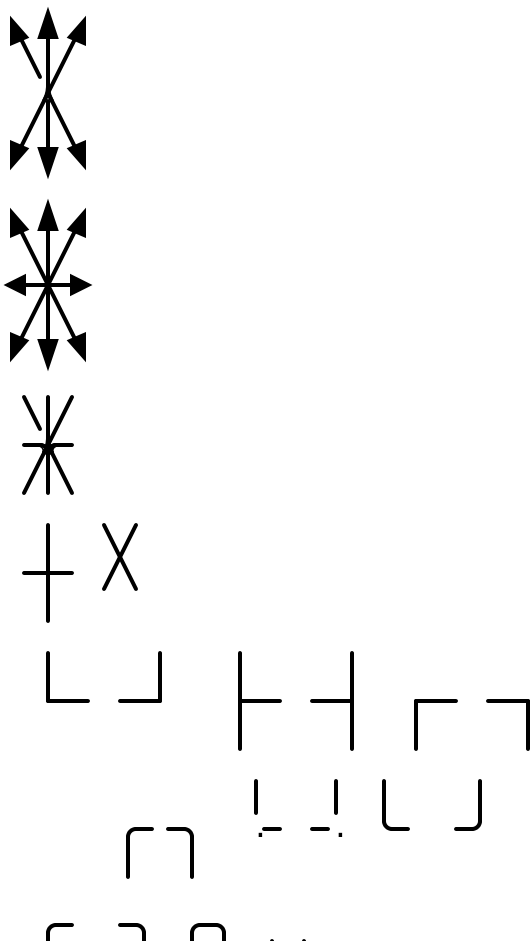
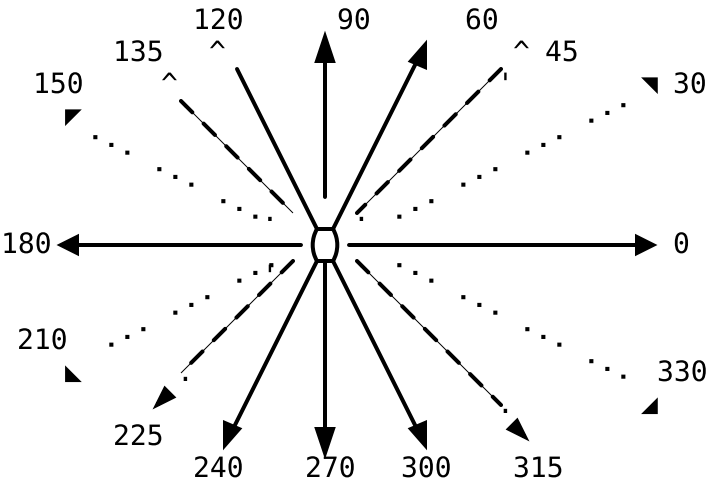
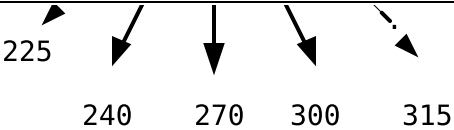
Style #1





Style #2







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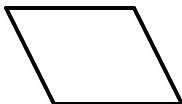
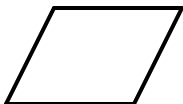
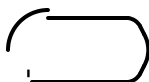
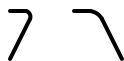
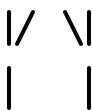
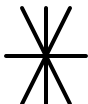
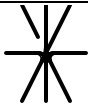
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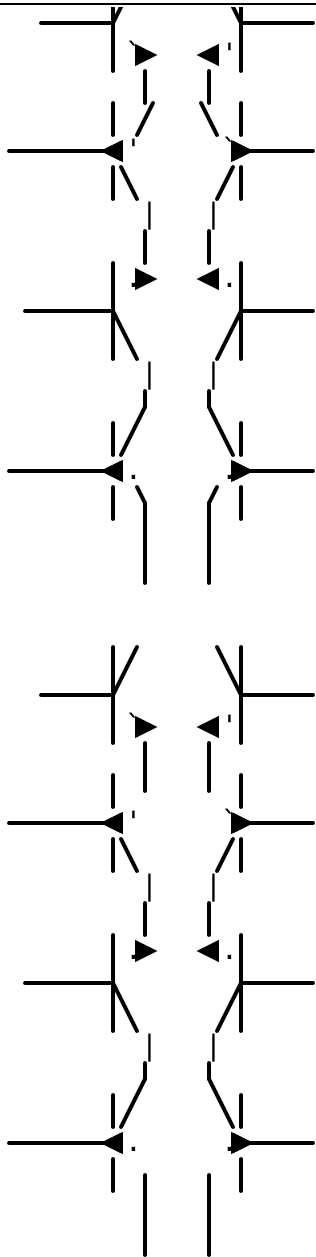
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...





# ECharts

## echarts-multi-function

linear

quadraticIn

quadraticOut

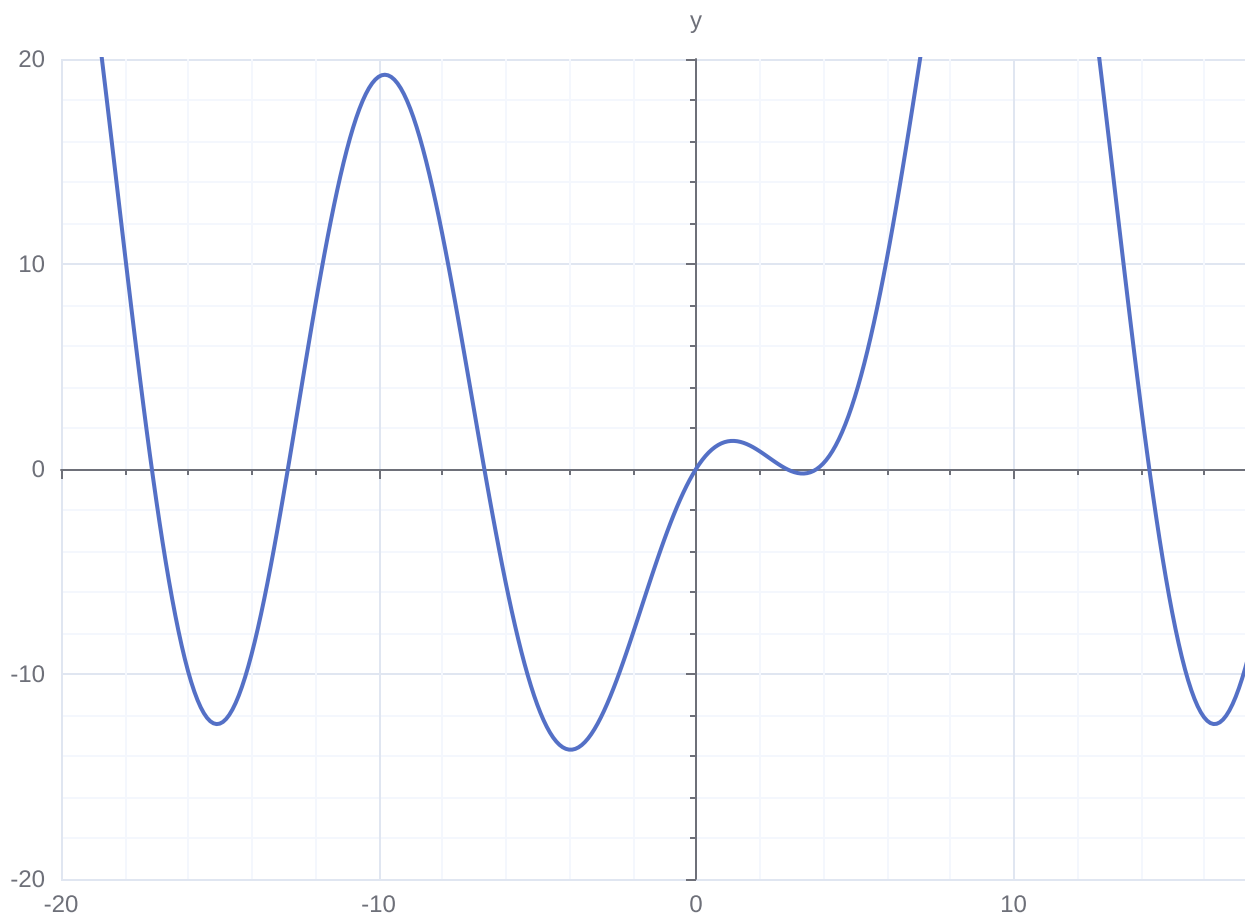
quadraticInOut

cubicIn

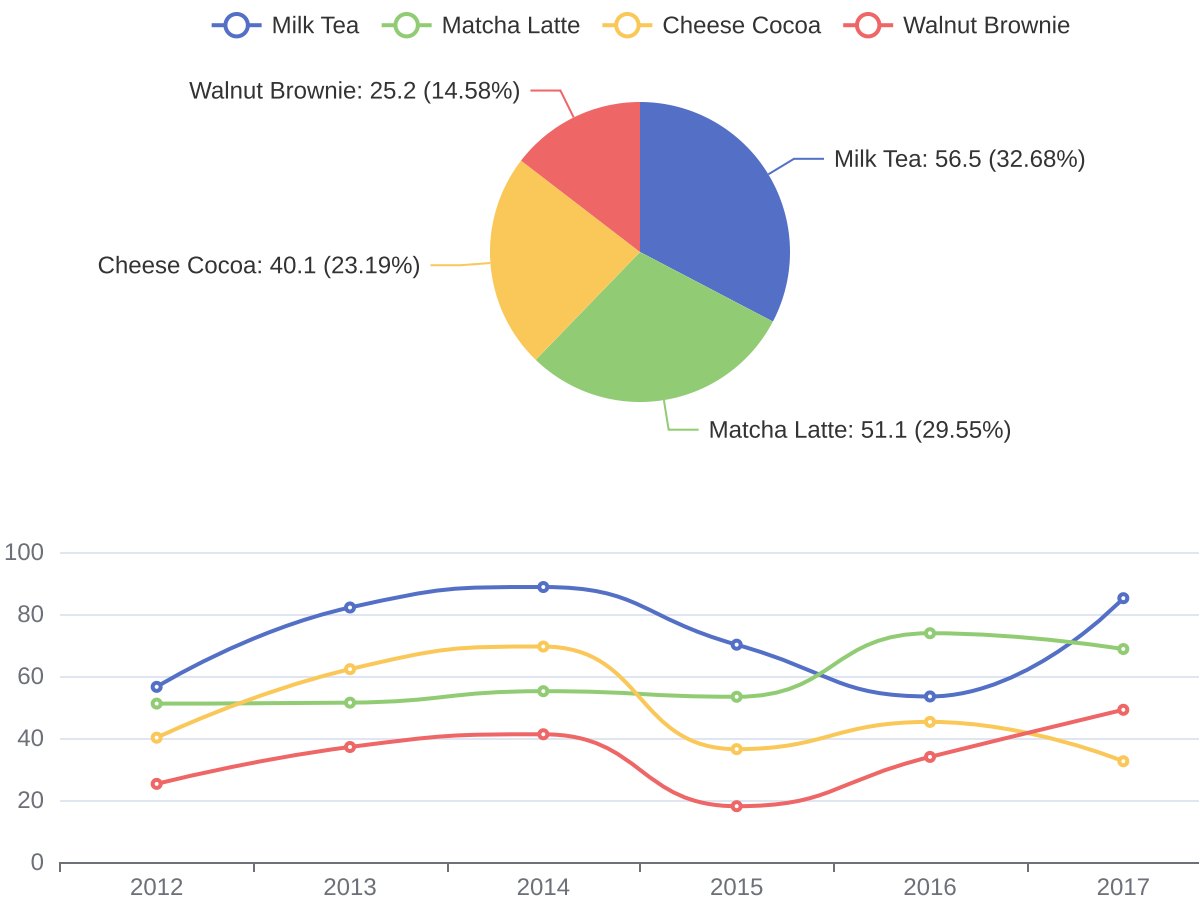


Different Easing Functions

echarts-function

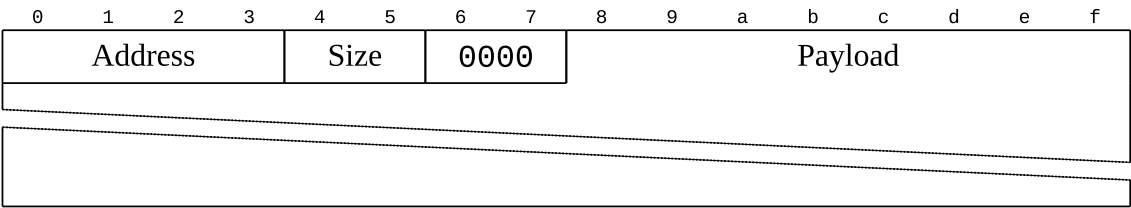


echarts-dataset

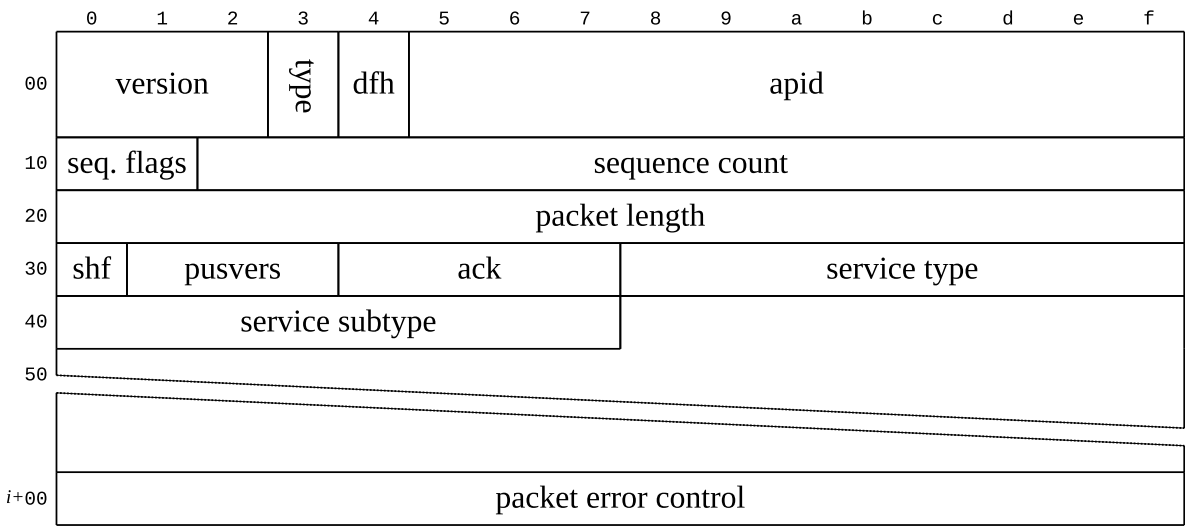


# Bytefield

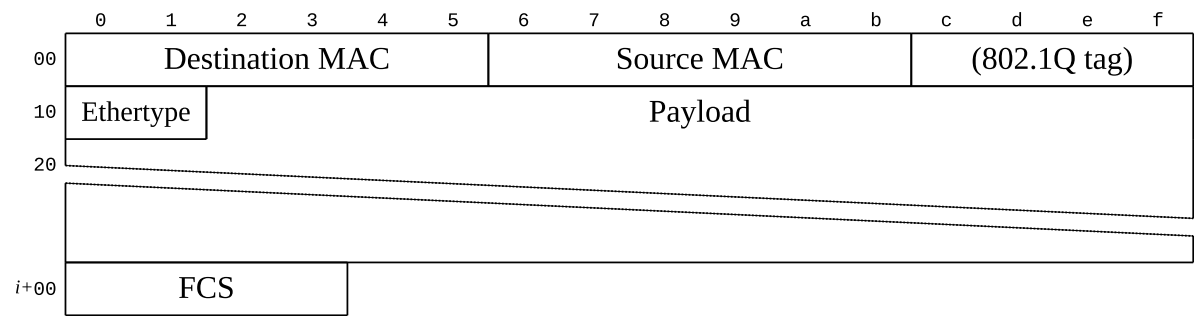
## Basic.



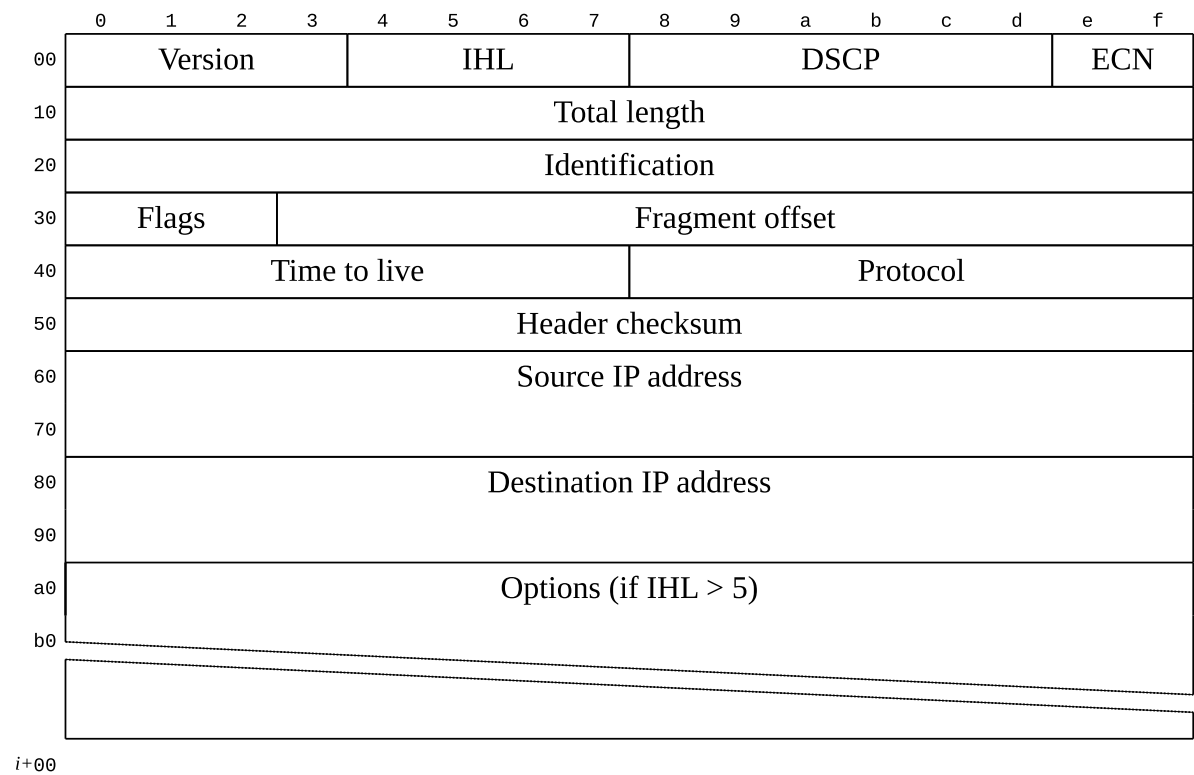
## ccsds.edn



# ethernet.edn

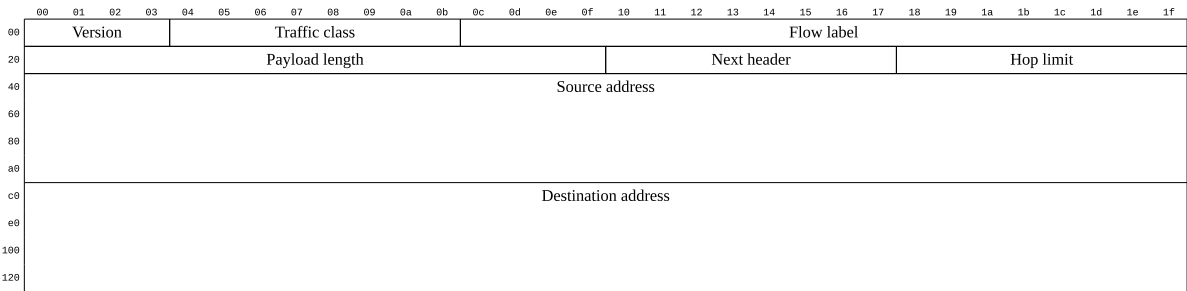


# ipv4.edn

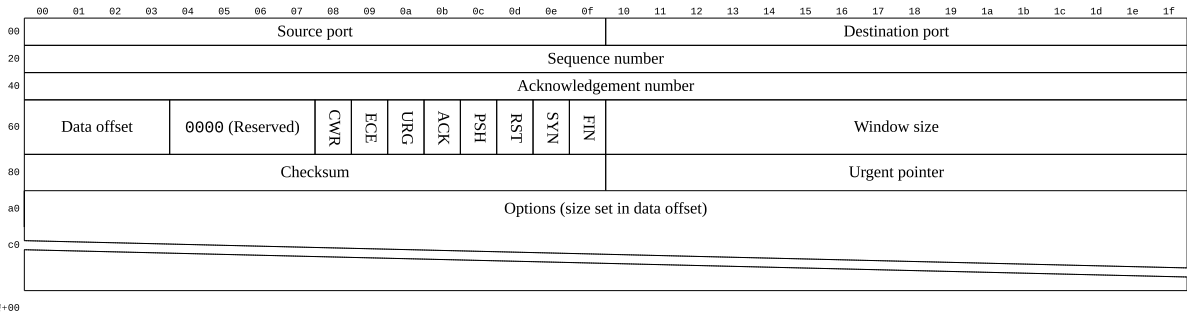




# ipv6.edn



# tcp.edn



udp.edn

00	01	02	03	04	05	06	07	08	09	0a	0b	0c	0d	0e	0f	10	11	12	13	14	15	16	17	18	19	1a	1b	1c	1d	1e	1f
Source port																Destination port															
Length																Checksum															

test.edn

	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f	
	start						TxID					type		args			tags
00	11	872349ae				11	TxID				10	4702		0f	09	14	
10	0000000c (12)				06	06	06	03	06	06	06	06	03	00	00	00	
20	11	00002104				11	00000000				11	length <sub>1</sub>				14	
30	length <sub>1</sub>				Cue and loop point bytes												
40																	
i+00	11	00000036				11	num <sub>hot</sub>				11	num <sub>cue</sub>					
i+10	11	length <sub>2</sub>				14	length <sub>2</sub>				Unknown bytes						
i+20																	
i+00																	

# Latex Pictures



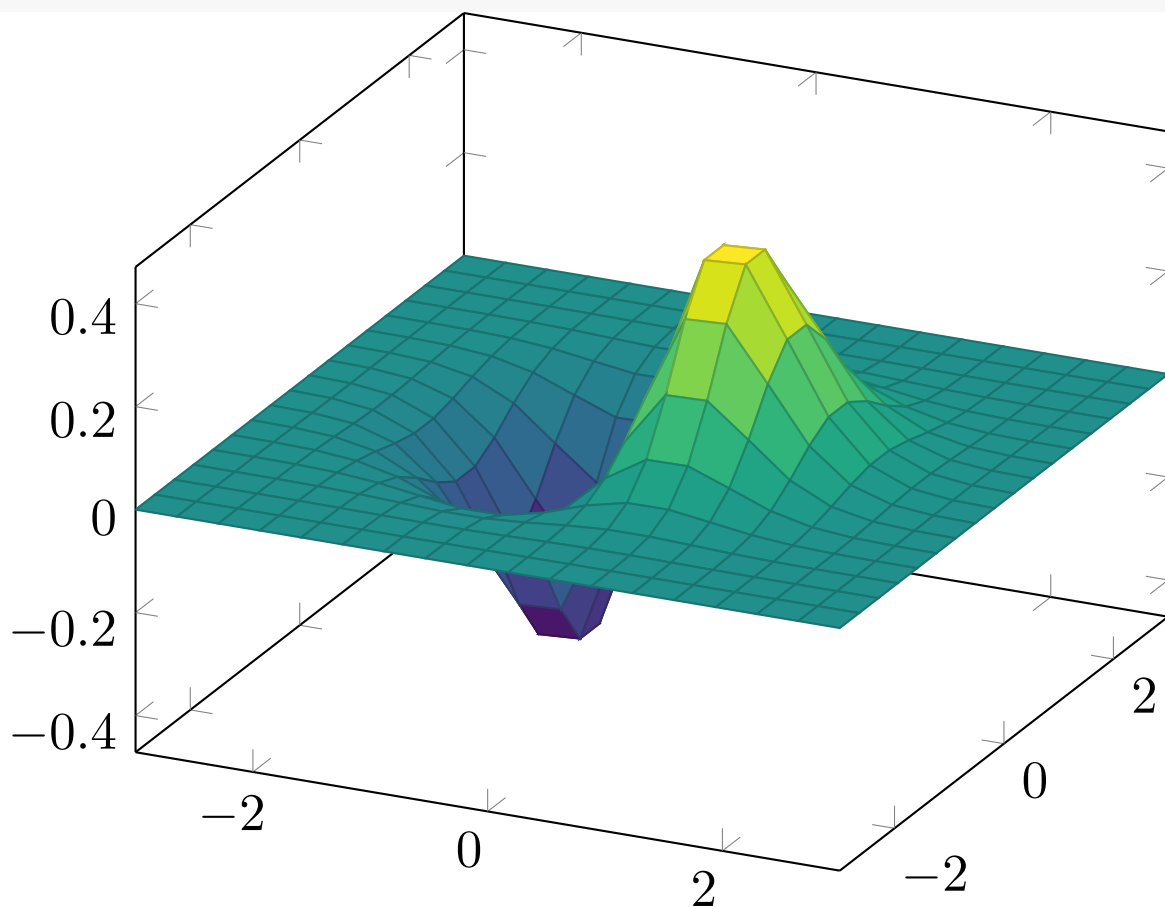
```
node-tikzjax
chemfig
tikz-cd
circuitikz
pgfplots
array
amsmath
amstext
amsfonts
amssymb
tikz-3dplot
```

## tikz demo1

▼ demo1\_0.tex



```
% demo1
\documentclass[tikz]{standalone}
\usepackage{pgfplots}
\pgfplotsset{compat=1.16}
\begin{document}
\begin{tikzpicture}[scale=2]
\begin{axis}[colormap/viridis]
\addplot3[
 surf,
 samples=18,
 domain=-3:3
]
{exp(-x^2-y^2)*x};
\end{axis}
\end{tikzpicture}
\end{document}
```

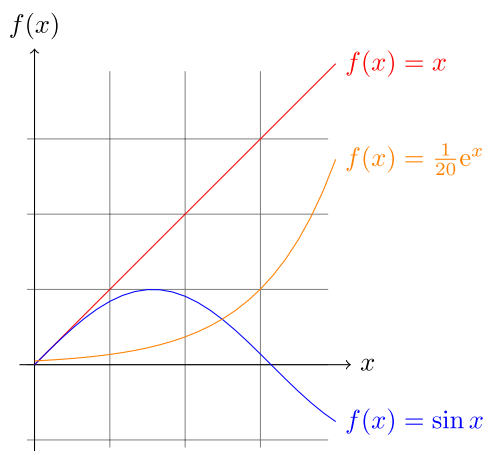


## tikz demo2

### ▼ demo2\_1.tex



```
% demo2
\documentclass[tikz]{standalone}
\begin{document}
\begin{tikzpicture}[domain=0:4]
\draw[very thin,color=gray] (-0.1,-1.1) grid (3.9,3.9);
\draw[->] (-0.2,0) -- (4.2,0) node[right] {x};
\draw[->] (0,-1.2) -- (0,4.2) node[above] {$f(x)$};
\draw[color=red] plot (\x,\x) node[right] {$f(x) = x$};
\draw[color=blue] plot (\x,{sin(\x r)}) node[right] {$f(x) = \sin$
x$};
\draw[color=orange] plot (\x,{0.05*exp(\x)}) node[right] {$f(x) =
\frac{1}{20} \mathrm{e}^x$};
\end{tikzpicture}
\end{document}
```

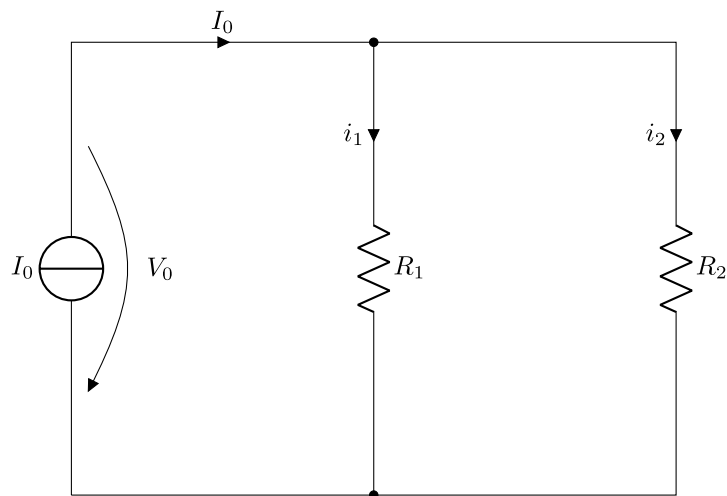


## tikz demo3

### ▼ demo3\_2.tex



```
% demo3
\documentclass[margin=10pt]{standalone}
\usepackage[siunitx]{circuitikz}
\begin{document}
\begin{tikzpicture}[scale=2]
\draw (0,0)
to[isource, l=I_0, v=V_0] (0,3)
to[short, -*, i=I_0] (2,3)
to[R=R_1, i>_=i_1] (2,0) -- (0,0);
\draw (2,3) -- (4,3)
to[R=R_2, i>_=i_2]
(4,0) to[short, -*] (2,0);
\end{tikzpicture}
\end{document}
```



## tikz demo4

▼ demo4\_3.tex



```

% demo4
% Poincaré Diagram: Classification of Phase Portraits in the $(\det A, \text{Tr } A)$ -plane
% Author: Gernot Salzer, 22 Jan 2017
\documentclass[tikz,border=10pt]{standalone}
\usetikzlibrary{decorations.markings}
\tikzset
{
 every pin/.style = {pin edge = {<-}}, % pins are arrows from label to point
 > = stealth, % arrow tips look like stealth bombers
 flow/.style = % everything marked as "flow" will be decorated with an arrow
 {
 decoration = {markings, mark=at position #1 with {\arrow{>}}},
 postaction = {decorate}
 },
 flow/.default = 0.5, % default position of the arrow is in the middle
 main/.style = {line width=1pt} % thick lines for main graph
}
% \newtemplate[Scaling, default 0.18]{\NameOfTemplate}{Caption}{Code}
%
% Typesets Code and stores it in the box \NameOfTemplate.
% This way we avoid nested tikzpictures when inserting the templates into the
% main picture, since nesting is not guaranteed to work.
\newcommand\newtemplate[4][0.18]%
{
 \newsavebox#2%
 \savebox#2%
 {
 \begin{tabular}{@{}c@{}}
 \begin{tikzpicture}[scale=#1]
 #4
 \end{tikzpicture}
 \end{tabular}
 \templatecaption{#3}
 }
 \%
}
\newcommand\template[1]{\usebox{#1}} % use the Code stored in box #1
\newcommand\templatecaption[1]{\sffamily\scriptsize#1} % typeset caption
\newcommand\Tr{\mathop{\mathrm{Tr}}}

```

```

\newtemplate\sink{sink}%
{\foreach \sx in {+,-} % for right/left half do:
 {\draw[flow] (\sx4,0) -- (0,0); % draw half of horizontal axis
 \draw[flow] (0,\sx4) -- (0,0); % draw half of vertical axis
 \foreach \sy in {+,-} % for upper/lower quadrant do:
 \foreach \a/\b in {2/1,3/0.44} % draw two half-parabolas
 \draw[flow,domain=\sx\a:0] plot (\x, {\sy\b*\x*\x});
 }
}

\newtemplate\source{source}%
{\foreach \sx in {+,-} % for right/left half do:
 {\draw[flow] (0,0) -- (\sx4,0); % draw half of horizontal axis
 \draw[flow] (0,0) -- (0,\sx4); % draw half of vertical axis
 \foreach \sy in {+,-} % for upper/lower quadrant do:
 \foreach \a/\b in {2/1,3/0.44} % draw two half-parabolas
 \draw[flow,domain=0:\sx\a] plot (\x, {\sy\b*\x*\x});
 }
}

\newtemplate\stablefp{line of stable fixed points}%
{\draw (-4,0) -- (4,0); % draw horizontal axis
\foreach \sy in {+,-} % for upper/lower half do:
 {\draw[flow] (0,\sy4) -- (0,0); % draw half of vertical axis
 \foreach \x in {-3,-2,-1,1,2,3} % draw six vertical half-lines
 \draw[flow] (\x,\sy3) -- (\x,0);
 }
}

\newtemplate\unstablefp{line of unstable fixed points}%
{\draw (-4,0) -- (4,0); % draw horizontal axis
\foreach \sy in {+,-} % for upper/lower half do:
 {\draw[flow] (0,0) -- (0,\sy4); % draw half of vertical axis
 \foreach \x in {-3,-2,-1,1,2,3} % draw six vertical half-lines
 \draw[flow] (\x,0) -- (\x,\sy3);
 }
}

\newtemplate\spiralsink{spiral sink}%
{\draw (-4,0) -- (4,0); % draw horizontal axis
\draw (0,-4) -- (0,4); % draw vertical axis
\draw [samples=100,smooth,domain=27:7] % draw spiral
 plot ({\x r}:{0.005*\x*\x}); % Using "flow" here gives
"Dimension
 \def\x{26} % too large", so we draw
a tiny
 \draw[->] ({\x r}:{0.005*\x*\x}) -- +(0.01,-0.01);% tangent for the
arrow.
}

```



```

\newtemplate\spiralsource{spiral source}%
{\draw (-4,0) -- (4,0); % draw horizontal axis
 \draw (0,-4) -- (0,4); % draw vertical axis
 \draw [samples=100,smooth,domain=10:28] % draw spiral
 plot ({-\x r}:{0.005*\x*\x}); % Using "flow" here gives
"Dimension
 \def\x{27.5} % too large", so we draw
a tiny
 \draw[<-] ({-\x r}:{0.005*\x*\x}) -- +(0.01,-0.008);% tangent for the
arrow.
}
\newtemplate[0.15]\centre{center}% British spelling since \center is in
use
{\draw (-4,0) -- (4,0); % draw horizontal axis
 \draw (0,-4) -- (0,4); % draw vertical axis
 \foreach \r in {1,2,3} % draw three circles
 \draw[flow=0.63] (\r,0) arc (0:-360:\r cm);
}
\newtemplate\saddle{saddle}%
{\foreach \sx in {+,-} % for right/left half do:
 {\draw[flow] (\sx4,0) -- (0,0); % draw half of horizontal axis
 \draw[flow] (0,0) -- (0,\sx4); % draw half of vertical axis
 \foreach \sy in {+,-} % for upper/lower quadrant do:
 \foreach \a/\b/\c/\d in {2.8/0.3/0.7/0.6, 3.9/0.4/1.3/1.1}
 \draw[flow] (\sx\a,\sy\b) % draw two bent lines
 .. controls (\sx\c,\sy\d) and (\sx\d,\sy\c)
 .. (\sx\b,\sy\a);
 }
}
\newtemplate\degensink{degenerate sink}%
{\draw (0,-4) -- (0,4); % draw vertical axis
 \foreach \s in {+,-} % for upper/lower half do:
 {\draw[flow] (\s4,0) -- (0,0); % draw half of horizontal axis
 \foreach \a/\b/\c/\d in {3.5/4/1.5/1, 2.5/2/1/0.8}
 \draw[flow] (\s-3.5,\s\a) % draw two bent lines
 .. controls (\s\b,\s\c) and (\s\b,\s\d)
 .. (0,0);
 }
}
\newtemplate\degensource{degenerate source}%
{\draw (0,-4) -- (0,4); % draw vertical axis
 \foreach \s in {+,-} % for upper/lower half do:
 {\draw[flow] (0,0) -- (\s4,0); % draw half of horizontal axis
 \foreach \a/\b/\c/\d in {3.5/4/1.5/1, 2.5/2/1/0.8}
 \draw[flow] (0,0) % draw two bent lines

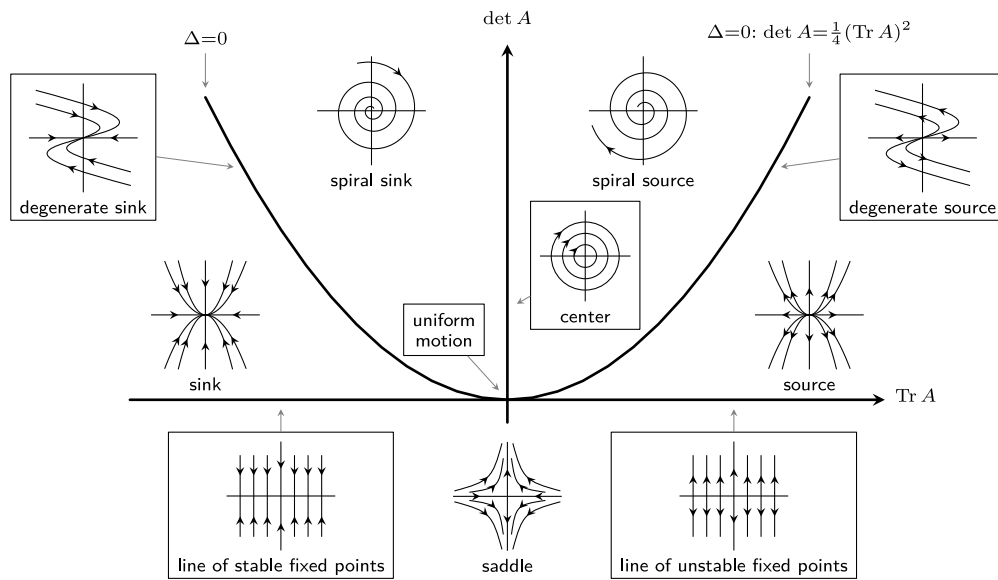
```

```

 .. controls (\s\b,\s\d) and (\s\b,\s\c)
 .. (\s-3.5,\s\ a);
 }
}
\begin{document}
\begin{tikzpicture}[line cap=round,line join=round]
 % MAIN DIAGRAM
 \draw [main,->] (0,-0.3) -- (0,4.7) %
vertical axis
 node [label={[above]\scriptstyle\det A$}] {}
 node [label={[above,yshift=0.8cm]%
 {\sffamily\large Poincar\'e Diagram: Classification of Phase
 Portraits in the $(\det A,\Tr A)$-plane}}] {};
 \draw [main,->] (-5,0) -- (5,0) %
horizontal axis
 node [label={[right,yshift=-0.5ex]\scriptstyle\Tr A$}] {};
 \draw [main, domain=-4:4] plot (\x, {0.25*\x*\x}); % main
graph
 \node at (-4,4) [pin={[above]\scriptstyle\Delta=0$}] {};
 \node at (4,4) [pin={[above,align=left]%
 {\scriptstyle\Delta=0:\;\det A=\frac{1}{4}(\Tr A)^2$}}] {};
 % TEMPLATES describing areas
 \node at (0 , -1.4) {\template\saddle};
 \node at (-4 , 1) {\template\sink};
 \node at (4 , 1) {\template\source};
 \node at (-1.8, 3.7) {\template\spiralsink};
 \node at (1.8, 3.7) {\template\spiralsource};
 % TEMPLATES labeling lines and points
 \node at (0 , 1.2) [pin={[draw,right,xshift=0.3cm]%
 \template\centre}] {};
 \node at (-3 , 0) [pin={[draw,below,yshift=-1cm]%
 \template\stablefp}] {};
 \node at (3 , 0) [pin={[draw,below,yshift=-1cm]%
 \template\unstablefp}] {};
 \node at (-3.5,{0.25*3.5*3.5}) [pin=
 {[draw,left,xshift=-1.15cm,yshift=-0.3cm]%
 \template\degensink}] {};
 \node at (3.5,{0.25*3.5*3.5}) [pin=
 {[draw,right,xshift=0.9cm,yshift=-0.3cm]%
 \template\degensource}] {};
 \node at (0 , 0) [pin={[draw,above left,align=center,xshift=-0.3cm]%
 \templatecaption{uniform}\[-1ex]\templatecaption{motion}}] {};
\end{tikzpicture}
\end{document}

```

## Poincaré Diagram: Classification of Phase Portraits in the $(\det A, \text{Tr } A)$ -plane



## tikz demo5

▼ demo5\_4.tex



```

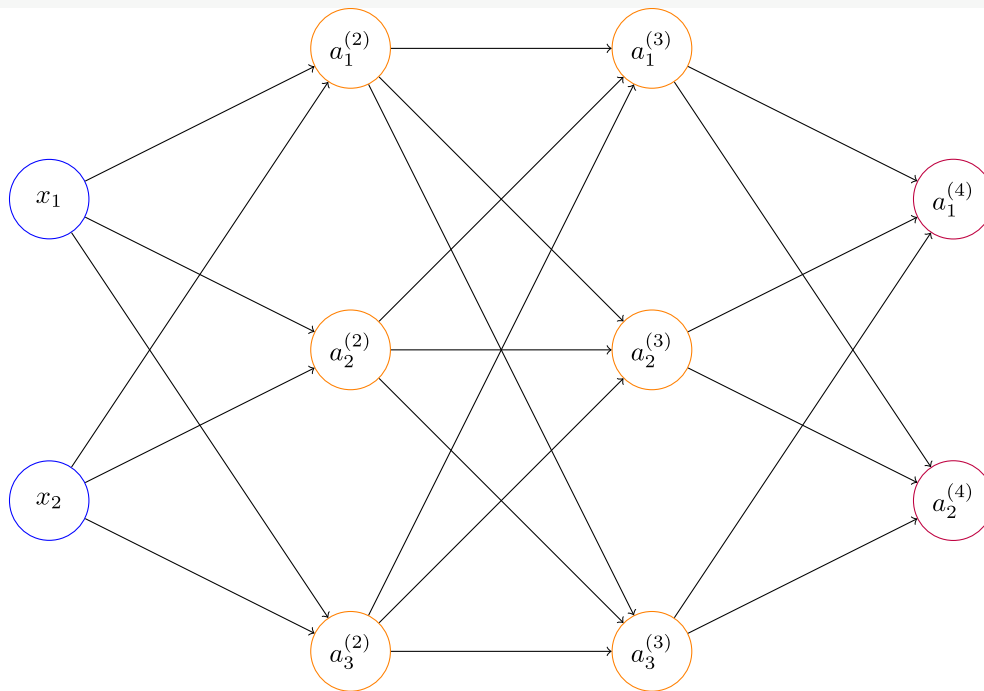
% demo5
\documentclass[tikz]{standalone}
\begin{document}
\begin{tikzpicture}[scale=2]
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=blue] (1) at(0,2){ x_1 };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=blue] (2) at(0,0){ x_2 };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (3) at(2,-1){ $a_3^{(2)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (4) at(2,1){ $a_2^{(2)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (5) at(2,3){ $a_1^{(2)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (6) at(4,-1){ $a_3^{(3)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (7) at(4,1){ $a_2^{(3)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (8) at(4,3){ $a_1^{(3)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=purple] (9) at(6,2){ $a_1^{(4)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=purple] (10) at(6,0){ $a_2^{(4)}$ };
\draw[->] (1) --(3);
\draw[->] (1) --(4);
\draw[->] (1) --(5);
\draw[->] (2) --(3);
\draw[->] (2) --(4);
\draw[->] (2) --(5);
\draw[->] (3) --(6);
\draw[->] (3) --(7);
\draw[->] (3) --(8);

```

```

\draw[->] (4) --(6);
\draw[->] (4) --(7);
\draw[->] (4) --(8);
\draw[->] (5) --(6);
\draw[->] (5) --(7);
\draw[->] (5) --(8);
\draw[->] (6) --(9);
\draw[->] (6) --(10);
\draw[->] (7) --(9);
\draw[->] (7) --(10);
\draw[->] (8) --(9);
\draw[->] (8) --(10);
\end{tikzpicture}
\end{document}

```



## tikz demo6

▼ demo6\_5.tex



```

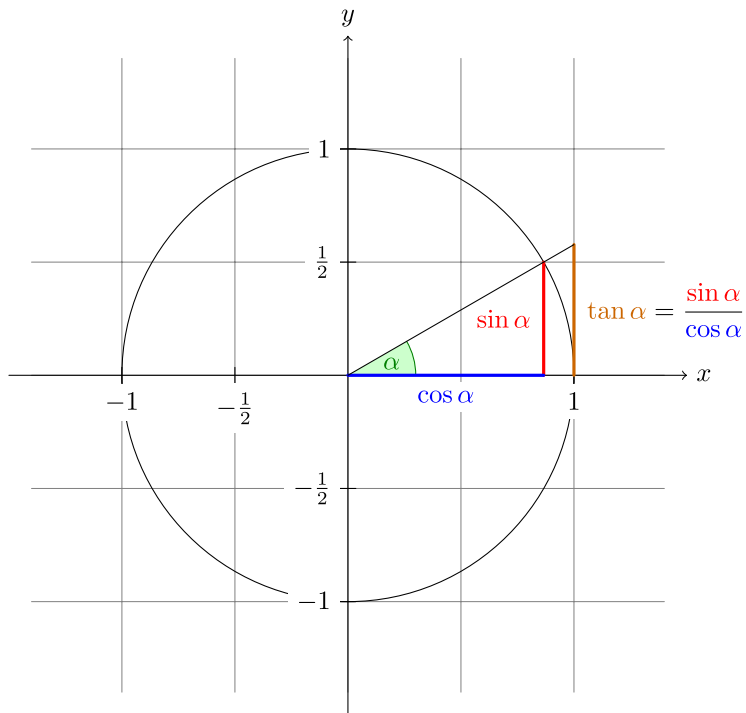
% demo6
% Author: Till Tantau
% Source: The PGF/TikZ manual
\documentclass[tikz]{standalone}
\begin{document}
\pagestyle{empty}
\begin{tikzpicture}[scale=3,cap=round]
% Local definitions
\def\costhirty{0.8660256}
% Colors
\colorlet{anglecolor}{green!50!black}
\colorlet{sincolor}{red}
\colorlet{tancolor}{orange!80!black}
\colorlet{coscolor}{blue}
% Styles
\tikzstyle{axes}=[]
\tikzstyle{important line}=[very thick]
\tikzstyle{information text}=[rounded corners,fill=red!10,inner sep=1ex]
% The graphic
\draw[style=help lines,step=0.5cm] (-1.4,-1.4) grid (1.4,1.4);
\draw (0,0) circle (1cm);
\begin{scope}[style=axes]
\draw[->] (-1.5,0) -- (1.5,0) node[right] {x};
\draw[->] (0,-1.5) -- (0,1.5) node[above] {y};
\foreach \x/\xtext in {-1, -.5/-\frac{1}{2}, 1}
\draw[xshift=\x cm] (0pt,1pt) -- (0pt,-1pt) node[below,fill=white]
{$\xtext$};
\foreach \y/\ytext in {-1, -.5/-\frac{1}{2}, .5/\frac{1}{2}, 1}
\draw[yshift=\y cm] (1pt,0pt) -- (-1pt,0pt) node[left,fill=white]
{$\ytext$};
\end{scope}
\filldraw[fill=green!20,draw=anglecolor] (0,0) -- (3mm,0pt) arc(0:30:3mm);
\draw (15:2mm) node[anglecolor] {α};
\draw[style=important line,sincolor]
(30:1cm) -- node[left=1pt,fill=white] {$\sin \alpha$} +(0,-.5);
\draw[style=important line,coscolor]
(0,0) -- node[below=2pt,fill=white] {$\cos \alpha$} (\costhirty,0);
\draw[style=important line,tancolor] (1,0) --
node [right=1pt,fill=white]
{
$\displaystyle \tan \alpha \color{black}=$
\frac{\color{sincolor}\sin \alpha}{\color{coscolor}\cos \alpha}$
} (intersection of 0,0--30:1cm and 1,0--1,1) coordinate (t);

```

```

\draw (0,0) -- (t);
\draw[xshift=1.85cm] node [right,text width=6cm,style=information text]
{
 The {\color{anglecolor} angle α } is 30° in the
 example ($\pi/6$ in radians). The {\color{sincolor}sine} of
 α , which is the height of the red line, is
 \[
 {\color{sincolor} \sin \alpha} = 1/2.
 \]
 By the Theorem of Pythagoras we have ${\color{coscolor} \cos^2 \alpha}$
+
 ${\color{sincolor} \sin^2 \alpha} = 1$. Thus the length of the blue
 line, which is the {\color{coscolor}cosine} of α , must be
 \[
 {\color{coscolor} \cos \alpha} = \sqrt{1 - 1/4} = \textstyle
 \frac{1}{2} \sqrt{3}.
 \]
 This shows that {\color{tancolor} $\tan \alpha$ }, which is the
 height of the orange line, is
 \[
 {\color{tancolor} \tan \alpha} = \frac{{\color{sincolor} \sin
 \alpha}}{{\color{coscolor} \cos \alpha}} = 1/\sqrt{3}.
 \]
};
\end{tikzpicture}
\end{document}

```



The **angle**  $\alpha$  is  $30^\circ$  in the example ( $\pi/6$  in radians). The **sine of**  $\alpha$ , which is the height of the red line, is

$$\sin \alpha = 1/2.$$

By the Theorem of Pythagoras we have  $\cos^2 \alpha + \sin^2 \alpha = 1$ . Thus the length of the blue line, which is the **cosine of**  $\alpha$ , must be

$$\cos \alpha = \sqrt{1 - 1/4} = \frac{1}{2}\sqrt{3}.$$

This shows that **tan**  $\alpha$ , which is the height of the orange line, is

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = 1/\sqrt{3}.$$

## tikz demo7

▼ demo7\_6.tex

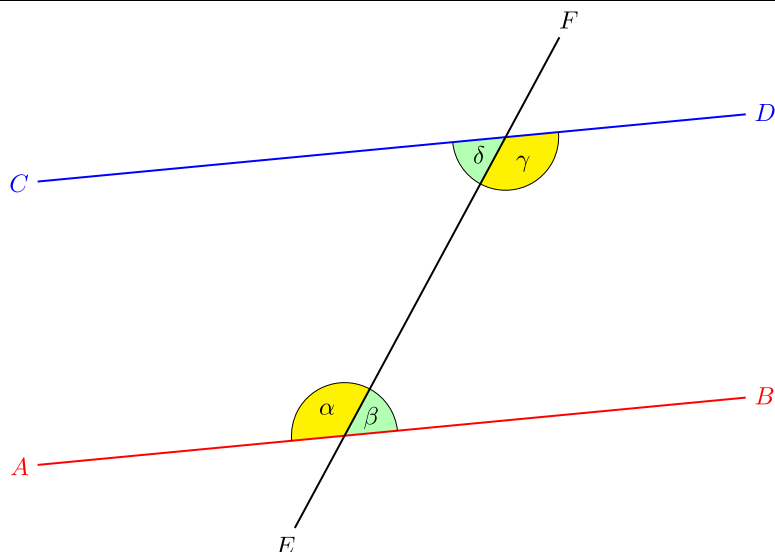




```

% demo7
\documentclass[tikz]{standalone}
\usetikzlibrary {angles,calc,quotes}
\begin{document}
\begin{tikzpicture}[angle radius=.75cm, scale=2]
\node (A) at (-2,0) [red,left] {A};
\node (B) at (3,.5) [red,right] {B};
\node (C) at (-2,2) [blue,left] {C};
\node (D) at (3,2.5) [blue,right] {D};
\node (E) at (60:-5mm) [below] {E};
\node (F) at (60:3.5cm) [above] {F};
\coordinate (X) at (intersection cs:first line={(A)--(B)}, second line=
{(E)--(F)});
\coordinate (Y) at (intersection cs:first line={(C)--(D)}, second line=
{(E)--(F)});
\path
 (A) edge [red, thick] (B)
 (C) edge [blue, thick] (D)
 (E) edge [thick] (F)
 pic ["α", draw, fill=yellow] {angle = F--X--A}
 pic ["β", draw, fill=green!30] {angle = B--X--F}
 pic ["γ", draw, fill=yellow] {angle = E--Y--D}
 pic ["δ", draw, fill=green!30] {angle = C--Y--E};
\node at ($ (D)!.5!(B) $) [right=1cm,text width=6cm,rounded
corners,fill=red!20,inner sep=1ex]
{
 When we assume that $\color{red}AB$ and $\color{blue}CD$ are
 parallel, i.\,e., $\color{red}AB \parallel \color{blue}CD$,
 then $\alpha = \gamma$ and $\beta = \delta$.
};
\end{tikzpicture}
\end{document}

```



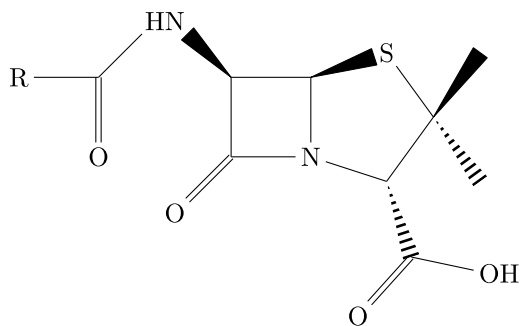
When we assume that  $AB$  and  $CD$  are parallel, i. e.,  $AB \parallel CD$ , then  $\alpha = \gamma$  and  $\beta = \delta$ .

## tikz demo8

### ▼ demo8\_7.tex

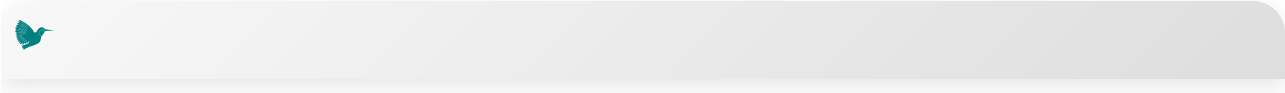


```
% demo8
\documentclass[tikz]{standalone}
\usepackage{chemfig}
\begin{document}
\chemfig{[: -90]HN(-[: -45](-[: -45]R)=[: +45]O)>[: +45]*4(-(=O)-N*5(-(<[: -60]O)-[: +60]OH)-(<[: +0])(<[: -108])-S>--)}
\end{document}
```



# tikz demo9

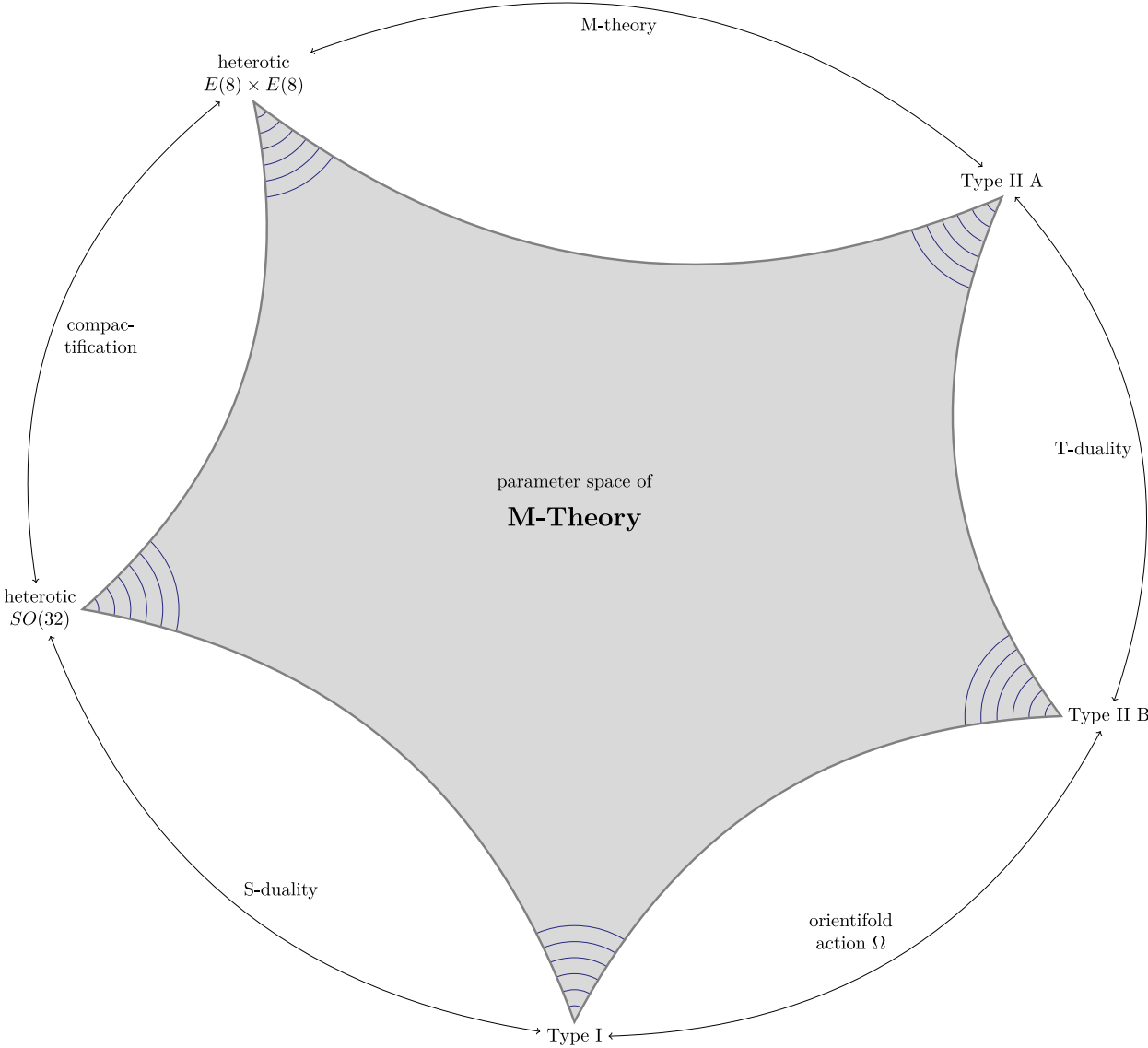
▼ samples\_8.tex



```

\documentclass[tikz]{standalone}
\definecolor{DarkBlue}{rgb}{0,0,0.5} % 深蓝色
\begin{document}
\begin{tikzpicture}[scale=2]
 \node (so32) [align=center] at (-5,-1) {heterotic\\$SO(32)$};
 \node (e8e8) [align=center] at (-3,4) {heterotic\\$E(8) \times E(8)$};
 \node (tiia) [align=center] at (4,3) {Type II A};
 \node (tiib) [align=center] at (5,-2) {Type II B};
 \node (ti) [align=center] at (0,-5) {Type I};
 \draw[bend left,<->] (so32) to node [below right,align=center] {compact-\\tification} (e8e8);
 \draw[bend left,<->] (e8e8) to node [below left] {M-theory} (tiia);
 \draw[bend left,<->] (tiia) to node [below left] {T-duality} (tiib);
 \draw[bend left,<->] (tiib) to node [above left,align=center]
{orientifold\\action Ω} (ti);
 \draw[bend left,<->] (ti) to node [above right] {S-duality} (so32);
 \begin{scope}
 % 深蓝色
 \clip (so32.east) to [bend right] (e8e8.south)
 to [bend right] (tiia.south)
 to [bend right] (tiib.west)
 to [bend right] (ti.north)
 to [bend right] cycle; % 深蓝色
 \foreach \c in {so32.east,e8e8.south,tiia.south,tiib.west,ti.north}{%
 \foreach \r in {1,...,6}{%
 \draw[DarkBlue] (\c) circle (\r*0.15cm); % 深蓝色
 }
 }
 \end{scope}
 % 深蓝色
 \draw[bend right,very thick,gray,fill,fill opacity=0.3]
 (so32.east) to [bend right] (e8e8.south)
 to [bend right] (tiia.south)
 to [bend right] (tiib.west)
 to [bend right] (ti.north)
 to [bend right] cycle; % 深蓝色
 \node (mth) [align=center] at (0,0) {parameter space of\\[2ex]{\Large
\textbf{M-Theory}}}};
\end{tikzpicture}
\end{document}

```



TexNote.



```
latex c/c++/rust
\# ==> \\\#
\$ ==> \\\$
\% ==> \\%\
\& ==> \\&
_ ==> _\
\{ ==> \\{\
\} ==> \\}\
\~{} ==> \\~\{\}
\^{} ==> \\^\{\}
\ ==> \\
\< ==> \\<
\> ==> \\>
```

# Latex Document.

## Latex demo

test in  
margin  
here

### L<sup>A</sup>T<sub>E</sub>X.js Showcase

made with ♥ by Michael Brade

2017–2021

#### Abstract

This document will show most of the features of L<sup>A</sup>T<sub>E</sub>X.js while at the same time being a gentle introduction to L<sup>A</sup>T<sub>E</sub>X. In the appendix, the API as well as the format of custom macro definitions in JavaScript will be explained.

### 1 Characters

It is possible to input any UTF-8 character either directly or by character code using one of the following:

- `\symbol{"00A9}`: ©
- `\char"A9`: ©
- `^^A9` or `~~~~00A9`: © or ©

Special characters, like those:

`$ & % # _ { } ~ ^ \`

have to be escaped. More than 200 symbols are accessible through macros. For instance: 30 °C is 86 °F. See also Section [11](#).

### 2 Spaces and Comments

Spaces and comments, of course, work just as they do in L<sup>A</sup>T<sub>E</sub>X. This is an example: Supercalifragilisticexpialidocious It does not matter whether you enter one or several spaces after a word, it will always be typeset as one space—unless you force several spaces, like now. New T<sub>E</sub>Xusers may miss whitespaces after a command. Experienced T<sub>E</sub>X users are T<sub>E</sub>Xperts, and know how to use whitespaces. Longer com-

ments can be embedded in the `comment` environment: This is another example for embedding comments in your document.

### 3 Dashes and Hyphens

$\text{\LaTeX}$  knows four kinds of dashes. Access three of them with different numbers of consecutive dashes. The fourth sign is actually not a dash at all—it is the mathematical minus sign:

daughter-in-law, X-rated  
pages 13–67  
yes—or no?  
0, 1 and −1

The names for these dashes are: ‘-’ hyphen, ‘—’ en-dash, ‘—’ em-dash, and ‘−’ minus sign.  $\text{\LaTeX}$ .js outputs the actual true unicode character for those instead of using the `hyphen`-`minus`.

### 4 Text and Paragraphs, Ligatures

An empty line starts a new paragraph, and so does `\par`.

Like this. A new line usually starts automatically when the previous one is full. However, using `\newline` or `\\`, one can force

to start a new line. Ligatures are supported as well, for instance:

fi, fl, ff, ffi, ffl ... instead of fi, fl, ffl ...

Use `\/` to prevent a ligature.

#### 4.1 Multicolumns

The multi-column layout, using the `multicols` environment, allows easy definition of multiple columns of text—just like in newspapers. The first and mandatory argument specifies the number of columns the text should be divided into. It is often convenient to spread some text over all columns, just before the multicolumn

output. In  $\text{\LaTeX}$ , this was needed to prevent any page break in between. To achieve this, the `multicols` environment has an optional second argument which can be used for this purpose. For instance, this text you are reading now was started with the argument `\subsection{Multicolumns}`.

### 5 Boxes

$\text{\LaTeX}$ .js supports most of the standard  $\text{\LaTeX}$  boxes.



```
\mbox{text}
```

We already know one of them: it's called `\mbox`. It simply packs up a series of boxes into another one, and can be used to prevent `LATEX.js` from breaking two words. As boxes can be put inside boxes, these horizontal box packers give you ultimate flexibility.

```
\makebox[width][pos]{text}
```

```
\framebox[width][pos]{text}
```

*width* defines the width of the resulting box as seen from the outside. The *pos* parameter takes a one letter value: `center`, `flushleft`, or `flushright`. `spread` is not really working in HTML. The command `\framebox` works exactly the same as `\makebox`, but it draws a box around the text. The following example shows you some things you could do with the `\makebox` and `\framebox` commands.

```
c e n t r a l
```

```
Guess I'm framed now!
```

Bummer, I am too wide

```
never mind, I can source it this?
```

```
\parbox[pos][height][inner-pos]{width}{text}
```

The `\parbox` command produces a box the contents of which are created in paragraph mode. However, only small pieces of text should be used, paragraph-making environments shouldn't be used inside a `\parbox` argument. For larger pieces of text, including ones containing a paragraph-making environment, you should use a `\minipage` environment. By default `LATEX` will position vertically a parbox so its center lines up with the center of the surrounding text line. When the optional position argument is present and equal either to `'t'` or `'b'`, this allows you respectively to align either the top or bottom line in the parbox with the baseline of the surrounding text. You may also specify `'m'` for position to get the default behaviour. The optional *height* argument overrides the natural height of the box. The *inner-pos* argument controls the placement of the text inside the box, as follows; if it is not specified, *pos* is used. `t` text is placed at the top of the box. `c` text is centered in the box. `b` text is placed at the bottom of the box. `s` is not supported in HTML.

The following examples demonstrate simple positioning:

• simple alignment: Some text

parbox default alignment, parbox test text

some text

parbox top alignment, text parbox test text

some text

parbox bottom alignment, text parbox test text

some text.

• alignment with a given height: Some text

parbox default alignment, parbox test text

some text

parbox top alignment, text parbox test text

some text

parbox bottom alignment, text parbox test text

some text.

The following examples demonstrate all explicit *pos/inner-pos* combinations:

□□□

166 / 246

BSN

- center alignment: Some text

parbox default alignment, parbox test text

some text

parbox top alignment, text parbox test text

some text

parbox bottom alignment, text parbox test text

some text.

- top alignment: Some text

parbox default alignment, parbox test text

some text

parbox top alignment, text parbox test text

some text

parbox bottom alignment, text parbox test text

some text.

- bottom alignment: Some text

parbox default alignment, parbox test text

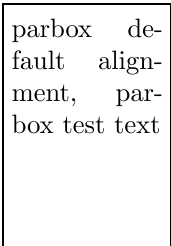
some text

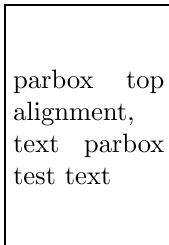
parbox top alignment, text parbox test text

some text

parbox bottom alignment, text parbox test text

some text.

- top/bottom in one line: Some text  some text.



### 5.1 Low-level box-interface

L<sup>A</sup>T<sub>E</sub>X.js supports the following low-level T<sub>E</sub>X commands: `\llap{text}`, `\rlap{text}`, and `\smash{text}`, as well as `\hphantom{text}`, `\vphantom{text}`, and `\phantom{text}`. A phantom looks like this: yes, now the phantom is ~~gone~~ `\glap` could be used to put something in the margin. However, there are better alternatives for that. See `\marginpar`.

## 6 Spacing

The following horizontal spaces are supported:

Negative thin space: ||  
 No space (natural): ||  
 Thin space: || or ||  
 Normal space: ||  
 Normal space: ||  
 Non-break space: ||  
 en-space: | |  
 em-space: | |  
 2x em-space: | |  
 3cm horizontal space: | |

## 7 Environments

### 7.1 Lists: Itemize, Enumerate, and Description

The `itemize` environment is suitable for simple lists, the `enumerate` environment for enumerated lists, and the `description` environment for descriptions.

1. You can nest the list environments to your taste:
  - But it might start to look silly.

- With a dash.

2. Therefore remember:

**Stupid** things will not become smart because they are in a list.

**Smart** things, though, can be presented beautifully in a list.

important Technical note: Viewing this in Chrome, however, will show too much vertical space at the end of a nested environment (see above). On top of that, margin collapsing for inline-block boxes is not allowed. Maybe using `dl` elements is too complicated for this and a simple nested `div` should be used instead.

Lists can be deeply nested:

- list text, level one
  - list text, level two
    - \* list text, level three And a new paragraph can be started, too.
      - list text, level four And a new paragraph can be started, too. This is the maximum level.
      - list text, level four
    - \* list text, level three
  - list text, level two
- list text, level one
- list text, level one

## 7.2 Flushleft, Flushright, and Center

The `flushleft` environment:

This text is left-aligned.  $\text{\LaTeX}$  is not trying to make each line the same length.

The `flushright` environment:

This text is right-aligned.  $\text{\LaTeX}$  is not trying to make each line the same length.

And the `center` environment:

At the centre  
of the earth

## 7.3 Quote, Quotation, and Verse

The `quote` environment is useful for quotes, important phrases and examples. A typographical rule of thumb for the

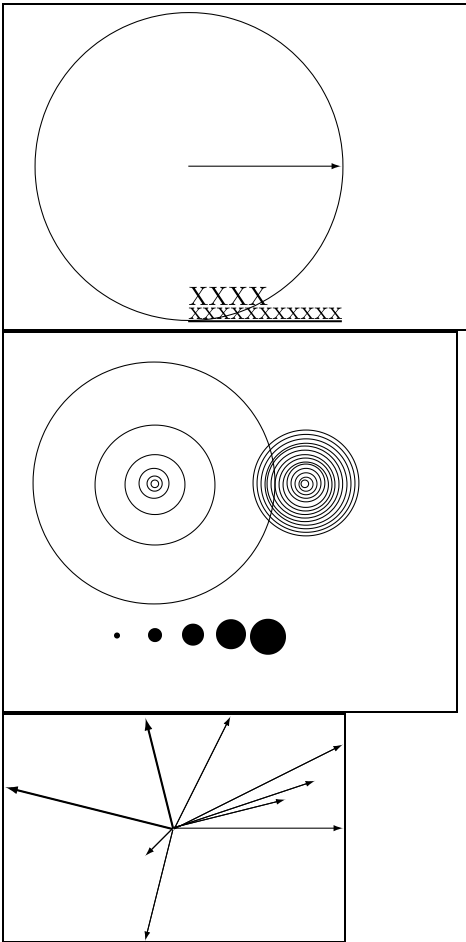
line length is:

On average, no line should be longer than 66 characters.

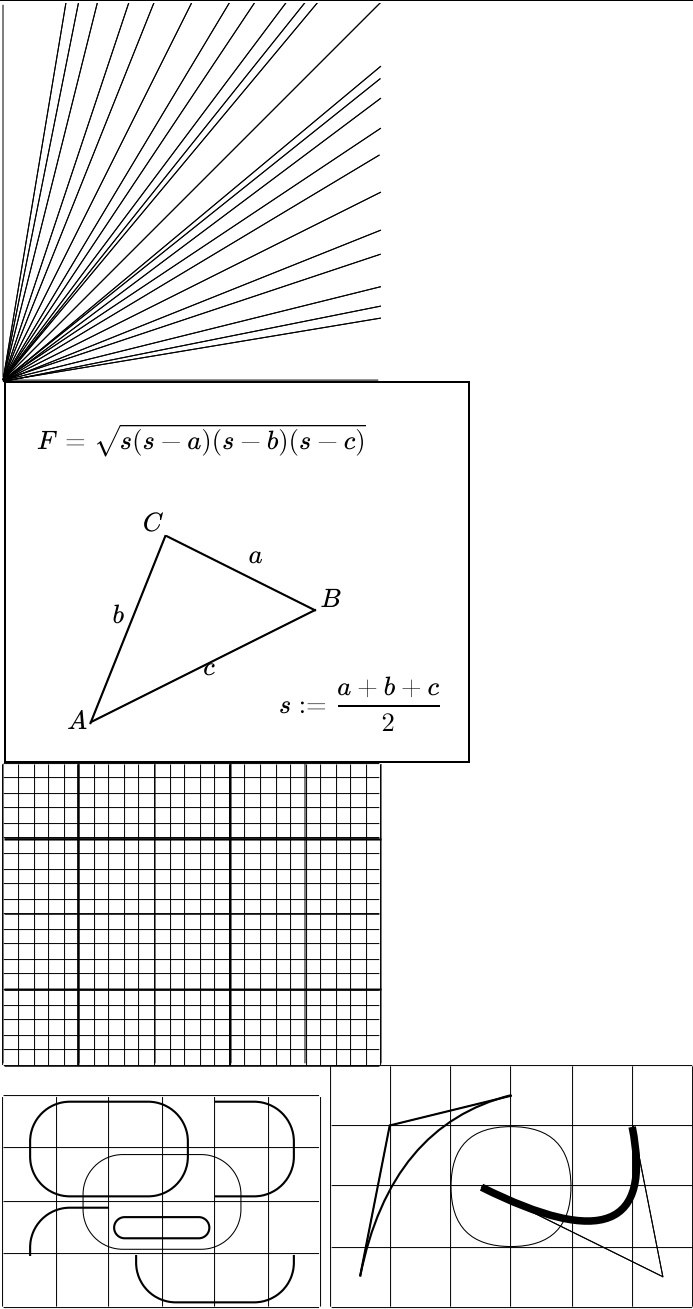
There are two similar environments: the `quotation` and the `verse` environments. The `quotation` environment is useful for longer quotes going over several paragraphs, because it indents the first line of each paragraph. The `verse` environment is useful for poems where the line breaks are important. The lines are separated by issuing a `\\` at the end of a line and an empty line after each verse.

Humpty Dumpty sat on a wall:  
Humpty Dumpty had a great fall.  
All the King's horses and all the King's men  
Couldn't put Humpty together again. —J.W.  
Elliott

7.4 Picture



.....



8   Labels and References

Labels can be attached to parts, chapters, sections, items of enumerations, footnotes, tables and figures. For instance: item [2](#) was important, and regarding fonts, read Section [12.1](#). And below, we can reference item [1\(a\)iB](#) and [1b](#).

- 1. list text, level one
  - (a) list text, level two

- i. list text, level three And a new paragraph can be started, too.

A. list text, level four And a new paragraph can be started, too. This is the maximum level.

B. list text, level four
- ii. list text, level three
- (b) list text, level two
2. list text, level one
3. list text, level one

9 Mathematical Formulae

Math is typeset using KaTeX. Inline math:  $f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi x} d\xi$  as well as display math is supported:

$$f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ 3n + 1, & \text{if } n \text{ is odd} \end{cases}$$

10 Groups

Today is Tuesday, June 10, 2025. Actually, what about some groups? They are nice.

11 Symbols

lowercase greek letters:  $\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\lambda\mu\nu\xi\omicron\pi\rho\sigma\tau\upsilon\varphi\chi\psi\omega$ uppercase greek letters:  $\text{A}\text{B}\text{C}\text{D}\text{E}\text{F}\text{G}\text{H}\text{I}\text{J}\text{K}\text{L}\text{M}\text{N}\text{O}\text{P}\text{Q}\text{R}\text{S}\text{T}\text{U}\text{V}\text{W}\text{X}\text{Y}\text{Z}$ current-cies:  $\text{€}\text{¢}\text{£}\text{¥}\text{¢}\text{d}\text{f}\text{¢}\text{N}\text{P}\text{W}\text{Y}$ old-style numerals: 0123456789math:  $\%$ 00  $\%$ 00  $\%$ 000<sup>131</sup>244/ $\div\times-\pm\sqrt{-}\ast$ <sup>123</sup>arrows:  $\leftarrow\rightarrow\downarrow$ misc:  $\checkmark\ast$ <sup>a0</sup>00d|||©©©©©<sup>SM</sup>™№℞♣%non-ASCII: ÆæĬĵĵĈĉPpBbDdOoDdŁłHhJjDj

12 Fonts

Usually, L<sup>A</sup>T<sub>E</sub>X.js chooses the right font—just like L<sup>A</sup>T<sub>E</sub>X. In some cases, one might like to change fonts and sizes by hand. To do this, use the standard commands. The actual size of each font is a design issue and depends on the document class (in this case on the CSS file). The small and bold Romans ruled all of great big *Italy*. You can also emphasize text if it is set in italics, in a sans-serif font, or in typewriter style. The environment form of the font commands is available, too:

*This whole paragraph is emphasized, for instance.*

12.1 An advice



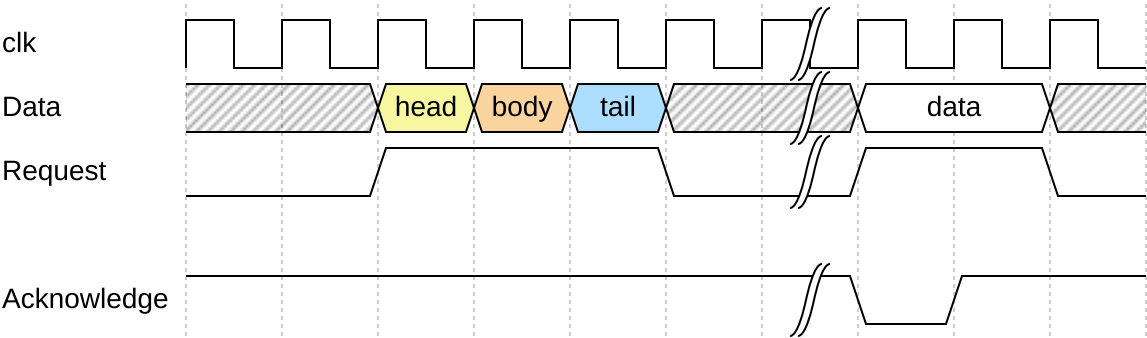
**Remember!** *The MORE fonts **you** use in a document, *the* more  
READABLE and *beautiful* it *become***S**.*

## A Source

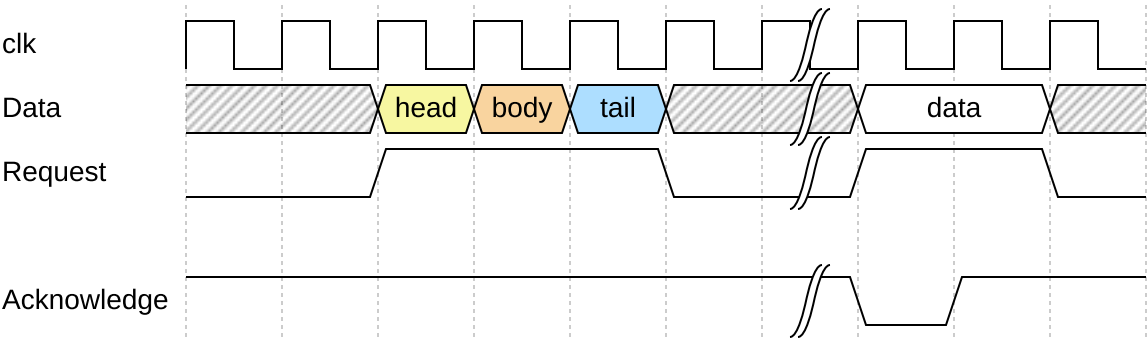
The source of LATEX.js is here on GitHub: <https://github.com/michael-brade/LaTeX.js>

# WaveDrom

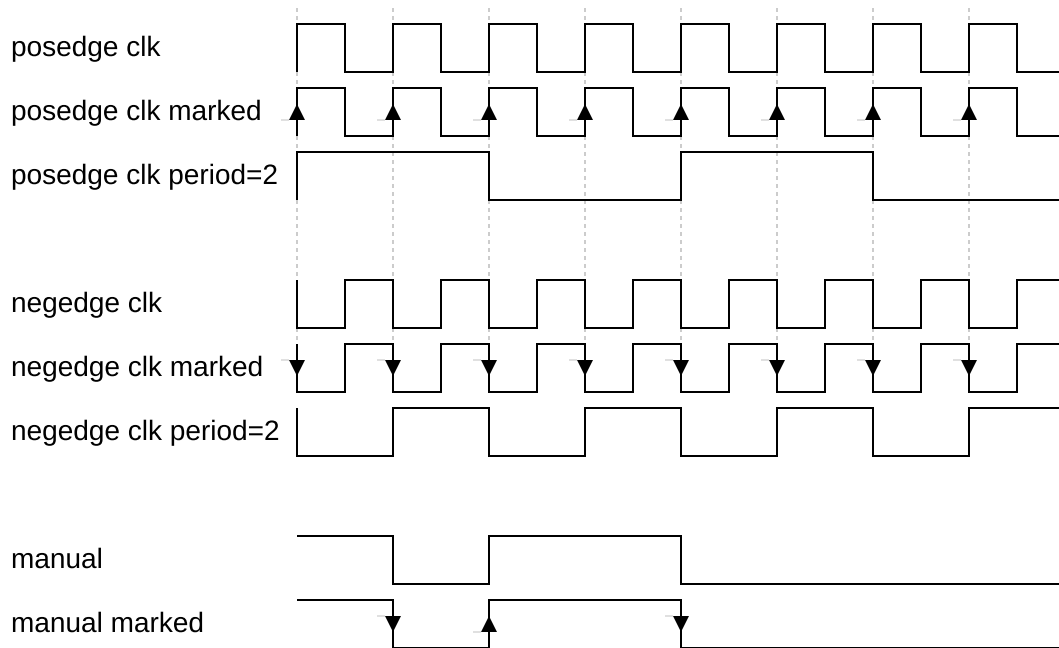
## basic1



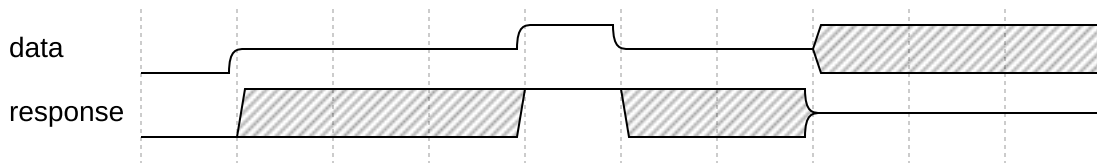
## basic2



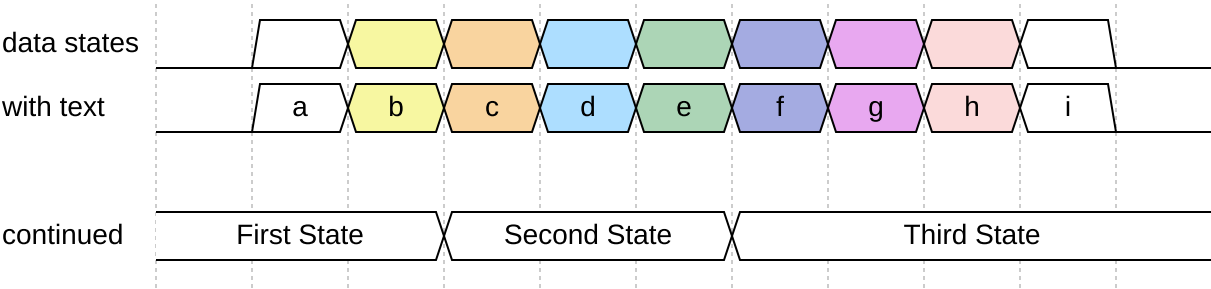
# clock



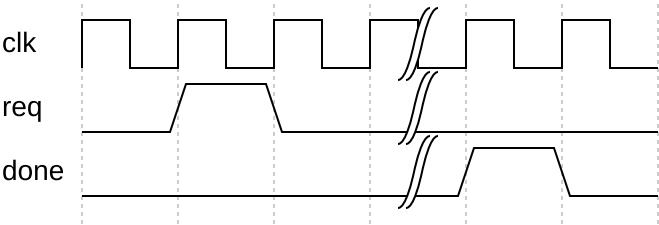
## High Impedance & Undefined



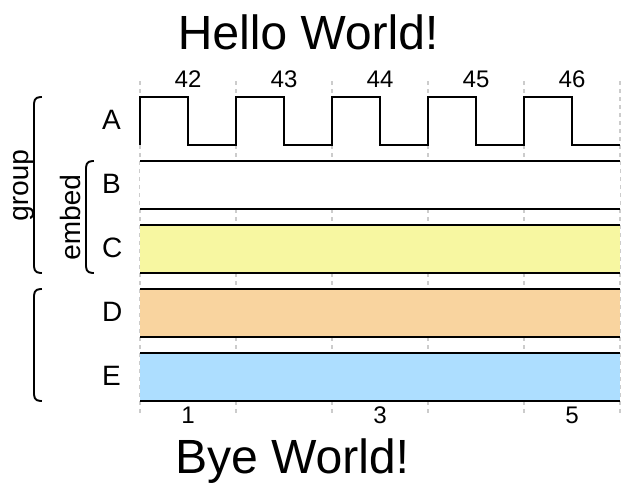
# Data types



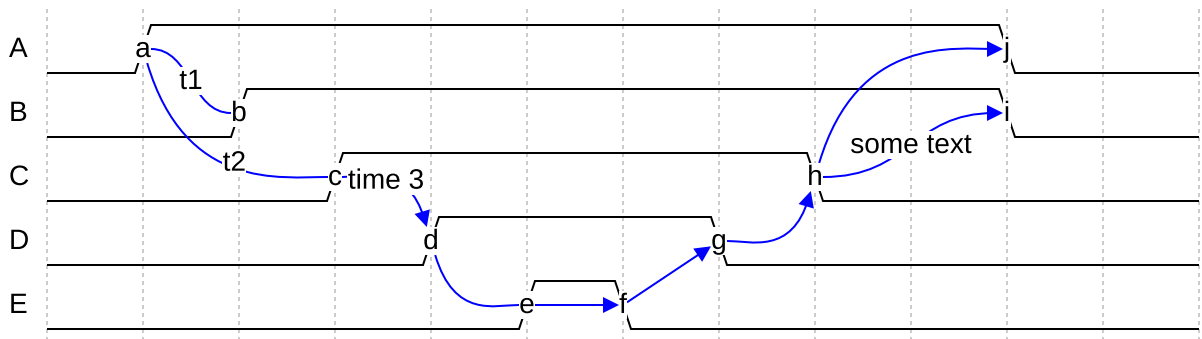
# Graps



# Groups



# Edges



# emoji-cheat-sheet

This cheat sheet is automatically generated from [GitHub Emoji API](#) and [Unicode Full Emoji List](#).

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- [Food & Drink](#)
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- [Symbols](#)
- [Flags](#)
- [GitHub Custom Emoji](#)

## Smileys & Emotion

- [Face Smiling](#)
- [Face Affection](#)
- [Face Tongue](#)
- [Face Hand](#)
- [Face Neutral Skeptical](#)
- [Face Sleepy](#)
- [Face Unwell](#)
- [Face Hat](#)
- [Face Glasses](#)
- [Face Concerned](#)

- [Face Negative](#)
- [Face Costume](#)
- [Cat Face](#)
- [Monkey Face](#)
- [Heart](#)
- [Emotion](#)

Face Smiling

ico	shortcode	ico	shortcode
	grinning		smiley
	smile		grin
	laughing		sweat_smile
	rofl		joy
	slightly_smiling_face		upside_down_face
	wink		blush
	innocent		

Face Affection

ico	shortcode	ico	shortcode
	smiling_face_with_three_hearts		heart_eyes
	star_struck		kissing_heart
	kissing		relaxed
	kissing_closed_eyes		kissing_smiling_eyes
	smiling_face_with_tear		

Face Tongue

ico	shortcode	ico	shortcode
	yum		stuck_out_tongue
	stuck_out_tongue_winking_eye		zany_face
	stuck_out_tongue_closed_eyes		money_mouth_face

Face Hand

ico	shortcode	ico	shortcode
	hugs		hand_over_mouth
	shushing_face		thinking

Face Neutral Skeptical

ico	shortcode	ico	shortcode
	zipper_mouth_face		raised_eyebrow
	neutral_face		expressionless
	no_mouth		face_in_clouds
	smirk		unamused
	roll_eyes		grimacing
	face_exhaling		lying_face

Face Sleepy

ico	shortcode	ico	shortcode
	relieved		pensive
	sleepy		drooling_face
	sleeping		



Face Unwell

ico	shortcode	ico	shortcode
	mask		face_with_thermometer
	face_with_head_bandage		nauseated_face
	vomiting_face		sneezing_face
	hot_face		cold_face
	woozy_face		dizzy_face
	face_with_spiral_eyes		exploding_head

Face Hat

ico	shortcode	ico	shortcode
	cowboy_hat_face		partying_face
	disguised_face		

Face Glasses

ico	shortcode	ico	shortcode
	sunglasses		nerd_face
	monocle_face		

Face Concerned

ico	shortcode	ico	shortcode
	confused		worried
	slightly_frowning_face		frowning_face
	open_mouth		hushed
	astonished		flushed

ico	shortcode	ico	shortcode
	pleading_face		frowning
	anguished		fearful
	cold_sweat		disappointed_relieved
	cry		sob
	scream		confounded
	persevere		disappointed
	sweat		weary
	tired_face		yawning_face

Face Negative

ico	shortcode	ico	shortcode
	triumph		pout rage
	angry		cursing_face
	smiling_imp		imp
	skull		skull_and_crossbones

Face Costume

ico	shortcode	ico	shortcode
	hankey poop shit		clown_face
	japanese_ogre		japanese_goblin
	ghost		alien
	space_invader		robot

Cat Face

ico	shortcode	ico	shortcode
	smiley_cat		smile_cat
	joy_cat		heart_eyes_cat
	smirk_cat		kissing_cat
	scream_cat		crying_cat_face
	pouting_cat		

Monkey Face

ico	shortcode	ico	shortcode
	see_no_evil		hear_no_evil
	speak_no_evil		

Heart

ico	shortcode	ico	shortcode
	love_letter		cupid
	gift_heart		sparkling_heart
	heartpulse		heartbeat
	revolving_hearts		two_hearts
	heart_decoration		heavy_heart_exclamation
	broken_heart		heart_on_fire
	mending_heart		heart
	orange_heart		yellow_heart
	green_heart		blue_heart
	purple_heart		brown_heart
	black_heart		white_heart

Emotion

ico	shortcode	ico	shortcode
	kiss		100
	anger		boom collision
	dizzy		sweat_drops
	dash		hole
	speech_balloon		eye_speech_bubble
	left_speech_bubble		right_anger_bubble
	thought_balloon		zzz

People & Body

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## Hand Fingers Open

ico	shortcode	ico	shortcode
	wave		raised_back_of_hand
	raised_hand_with_fingers_splayed		hand raised_hand
	vulcan_salute		

## Hand Fingers Partial

ico	shortcode	ico	shortcode
	ok_hand		pinched_fingers
	pinching_hand		v
	crossed_fingers		love_you_gesture
	metal		call_me_hand

## Hand Single Finger

ico	shortcode	ico	shortcode
	point_left		point_right
	point_up_2		fu middle_finger
	point_down		point_up

## Hand Fingers Closed

ico	shortcode	ico	shortcode
	+1 thumbsup		-1 thumbsdown

ico	shortcode	ico	shortcode
	fist fist_raised		facepunch fist_oncoming punch
	fist_left		fist_right

Hands

ico	shortcode	ico	shortcode
	clap		raised_hands
	open_hands		palms_up_together
	handshake		pray

Hand Prop

ico	shortcode	ico	shortcode
	writing_hand		nail_care
	selfie		

Body Parts

ico	shortcode	ico	shortcode
	muscle		mechanical_arm
	mechanical_leg		leg
	foot		ear
	ear_with_hearing_aid		nose
	brain		anatomical_heart
	lungs		tooth
	bone		eyes

ico	shortcode	ico	shortcode
	eye		tongue
	lips		

Person

ico	shortcode	ico	shortcode
	baby		child
	boy		girl
	adult		blond_haired_person
	man		bearded_person
	man_beard		woman_beard
	red_haired_man		curly_haired_man
	white_haired_man		bald_man
	woman		red_haired_woman
	person_red_hair		curly_haired_woman
	person_curly_hair		white_haired_woman
	person_white_hair		bald_woman
	person_bald		blond_haired_woman blonde_woman
	blond_haired_man		older_adult
	older_man		older_woman

Person Gesture

ico	shortcode	ico	shortcode
	frowning_person		frowning_man
	frowning_woman		pouting_face
	pouting_man		pouting_woman

ico	shortcode	ico	shortcode
	no_good		ng_man no_good_man
	ng_woman no_good_woman		ok_person
	ok_man		ok_woman
	information_desk_person tipping_hand_person		sassy_man tipping_hand_man
	sassy_woman tipping_hand_woman		raising_hand
	raising_hand_man		raising_hand_woman
	deaf_person		deaf_man
	deaf_woman		bow
	bowing_man		bowing_woman
	facepalm		man_facepalming
	woman_facepalming		shrug
	man_shrugging		woman_shrugging

Person Role

ico	shortcode	ico	shortcode
	health_worker		man_health_worker
	woman_health_worker		student
	man_student		woman_student
	teacher		man_teacher
	woman_teacher		judge
	man_judge		woman_judge
	farmer		man_farmer
	woman_farmer		cook



ico	shortcode	ico	shortcode
	man_cook		woman_cook
	mechanic		man_mechanic
	woman_mechanic		factory_worker
	man_factory_worker		woman_factory_worker
	office_worker		man_office_worker
	woman_office_worker		scientist
	man_scientist		woman_scientist
	technologist		man_technologist
	woman_technologist		singer
	man_singer		woman_singer
	artist		man_artist
	woman_artist		pilot
	man_pilot		woman_pilot
	astronaut		man_astronaut
	woman_astronaut		firefighter
	man_firefighter		woman_firefighter
	cop police_officer		policeman
	policewoman		detective
	male_detective		female_detective
	guard		guardsman
	guardswoman		ninja
	construction_worker		construction_worker_man
	construction_worker_woman		prince
	princess		person_with_turban
	man_with_turban		woman_with_turban
	man_with_gua_pi_mao		woman_with_headscarf

ico	shortcode	ico	shortcode
	person_in_tuxedo		man_in_tuxedo
	woman_in_tuxedo		person_with_veil
	man_with_veil		bride_with_veil
	pregnant_woman		woman_with_veil
	woman_feeding_baby		breast_feeding
	person_feeding_baby		man_feeding_baby

Person Fantasy

ico	shortcode	ico	shortcode
	angel		santa
	mrs_claus		mx_claus
	superhero		superhero_man
	superhero_woman		supervillain
	supervillain_man		supervillain_woman
	mage		mage_man
	mage_woman		fairy
	fairy_man		fairy_woman
	vampire		vampire_man
	vampire_woman		merperson
	merman		mermaid
	elf		elf_man
	elf_woman		genie
	genie_man		genie_woman
	zombie		zombie_man
	zombie_woman		

Person Activity

ico		shortcode		ico		shortcode	
		message				message_man	
		message_woman				haircut	
		haircut_man				haircut_woman	
		walking				walking_man	
		walking_woman				standing_person	
		standing_man				standing_woman	
		kneeling_person				kneeling_man	
		kneeling_woman				person_with_probing_cane	
		man_with_probing_cane				woman_with_probing_cane	
		person_in_motorized_wheelchair				man_in_motorized_wheelchair	
		woman_in_motorized_wheelchair				person_in_manual_wheelchair	
		man_in_manual_wheelchair				woman_in_manual_wheelchair	
		runner				running_man	
		running					
		running_woman				dancer	
						woman_dancing	
		man_dancing				business_suit_levitating	
		dancers				dancing_men	
		dancing_women				sauna_person	
		sauna_man				sauna_woman	
		climbing				climbing_man	
		climbing_woman					

Person Sport

ico	shortcode	ico	shortcode
	person_fencing		horse_racing
	skier		snowboarder
	golfing		golfing_man
	golfing_woman		surfer
	surfing_man		surfing_woman
	rowboat		rowing_man
	rowing_woman		swimmer
	swimming_man		swimming_woman
	bouncing_ball_person		basketball_man bouncing_ball_man
	basketball_woman bouncing_ball_woman		weight_lifting
	weight_lifting_man		weight_lifting_woman
	bicyclist		biking_man
	biking_woman		mountain_bicyclist
	mountain_biking_man		mountain_biking_woman
	cartwheeling		man_cartwheeling
	woman_cartwheeling		wrestling
	men_wrestling		women_wrestling
	water_polo		man_playing_water_polo
	woman_playing_water_polo		handball_person
	man_playing_handball		woman_playing_handball
	juggling_person		man_juggling
	woman_juggling		

Person Resting

ico	shortcode	ico	shortcode
	lotus_position		lotus_position_man
	lotus_position_woman		bath
	sleeping_bed		

Family

ico	shortcode	ico	shortcode
	people_holding_hands		two_women_holding_hands
	couple		two_men_holding_hands
	couplekiss		couplekiss_man_woman
	couplekiss_man_man		couplekiss_woman_woman
	couple_with_heart		couple_with_heart_woman_n
	couple_with_heart_man_man		couple_with_heart_woman_v
	family_man_woman_boy		family_man_woman_girl
	family_man_woman_girl_boy		family_man_woman_boy_boy
	family_man_woman_girl_girl		family_man_man_boy
	family_man_man_girl		family_man_man_girl_boy
	family_man_man_boy_boy		family_man_man_girl_girl
	family_woman_woman_boy		family_woman_woman_girl
	family_woman_woman_girl_boy		family_woman_woman_boy_bc
	family_woman_woman_girl_girl		family_man_boy
	family_man_boy_boy		family_man_girl
	family_man_girl_boy		family_man_girl_girl
	family_woman_boy		family_woman_boy_boy
	family_woman_girl		family_woman_girl_boy

ico	shortcode	ico	shortcode
	family		

Person Symbol

ico	shortcode	ico	shortcode
	speaking_head		bust_in_silhouette
	busts_in_silhouette		people_hugging
	family		footprints

Animals & Nature

- [Animal Mammal](#)
- [Animal Bird](#)
- [Animal Amphibian](#)
- [Animal Reptile](#)
- [Animal Marine](#)
- [Animal Bug](#)
- [Plant Flower](#)
- [Plant Other](#)

Animal Mammal

ico	shortcode	ico	shortcode
	monkey_face		monkey
	gorilla		orangutan
	dog		dog2
	guide_dog		service_dog
	poodle		wolf
	fox_face		raccoon

ico	shortcode	ico	shortcode
	cat		cat2
	black_cat		lion
	tiger		tiger2
	leopard		horse
	racehorse		unicorn
	zebra		deer
	bison		cow
	ox		water_buffalo
	cow2		pig
	pig2		boar
	pig_nose		ram
	sheep		goat
	dromedary_camel		camel
	llama		giraffe
	elephant		mammoth
	rhinoceros		hippopotamus
	mouse		mouse2
	rat		hamster
	rabbit		rabbit2
	chipmunk		beaver
	hedgehog		bat
	bear		polar_bear
	koala		panda_face
	sloth		otter
	skunk		kangaroo
	badger		feet paw_prints

Animal Bird

ico	shortcode	ico	shortcode
	turkey		chicken
	rooster		hatching_chick
	baby_chick		hatched_chick
	bird		penguin
	dove		eagle
	duck		swan
	owl		dodo
	feather		flamingo
	peacock		parrot

Animal Amphibian

ico	shortcode
	frog

Animal Reptile

ico	shortcode	ico	shortcode
	crocodile		turtle
	lizard		snake
	dragon_face		dragon
	sauropod		t-rex



Animal Marine

ico	shortcode	ico	shortcode
	whale		whale2
	dolphin flipper		seal
	fish		tropical_fish
	blowfish		shark
	octopus		shell

Animal Bug

ico	shortcode	ico	shortcode
	snail		butterfly
	bug		ant
	bee honeybee		beetle
	lady_beetle		cricket
	cockroach		spider
	spider_web		scorpion
	mosquito		fly
	worm		microbe

Plant Flower

ico	shortcode	ico	shortcode
	bouquet		cherry_blossom
	white_flower		rosette
	rose		wilted_flower

ico	shortcode	ico	shortcode
	hibiscus		sunflower
	blossom		tulip

Plant Other

ico	shortcode	ico	shortcode
	seedling		potted_plant
	evergreen_tree		deciduous_tree
	palm_tree		cactus
	ear_of_rice		herb
	shamrock		four_leaf_clover
	maple_leaf		fallen_leaf
	leaves		mushroom

Food & Drink

- [Food Fruit](#)
- [Food Vegetable](#)
- [Food Prepared](#)
- [Food Asian](#)
- [Food Marine](#)
- [Food Sweet](#)
- [Drink](#)
- [Dishware](#)

Food Fruit

ico	shortcode	ico	shortcode
	grapes		melon

ico	shortcode	ico	shortcode
	watermelon		mandarin orange tangerine
	lemon		banana
	pineapple		mango
	apple		green_apple
	pear		peach
	cherries		strawberry
	blueberries		kiwi_fruit
	tomato		olive
	coconut		

Food Vegetable

ico	shortcode	ico	shortcode
	avocado		eggplant
	potato		carrot
	corn		hot_pepper
	bell_pepper		cucumber
	leafy_green		broccoli
	garlic		onion
	peanuts		chestnut

Food Prepared

ico	shortcode	ico	shortcode
	bread		croissant
	baguette_bread		flatbread

ico	shortcode	ico	shortcode
	pretzel		bagel
	pancakes		waffle
	cheese		meat_on_bone
	poultry_leg		cut_of_meat
	bacon		hamburger
	fries		pizza
	hotdog		sandwich
	taco		burrito
	tamale		stuffed_flatbread
	falafel		egg
	fried_egg		shallow_pan_of_food
	stew		fondue
	bowl_with_spoon		green_salad
	popcorn		butter
	salt		canned_food

Food Asian

ico	shortcode	ico	shortcode
	bento		rice_cracker
	rice_ball		rice
	curry		ramen
	spaghetti		sweet_potato
	oden		sushi
	fried_shrimp		fish_cake
	moon_cake		dango
	dumpling		fortune_cookie

ico	shortcode	ico	shortcode
	takeout_box		

Food Marine

ico	shortcode	ico	shortcode
	crab		lobster
	shrimp		squid
	oyster		

Food Sweet

ico	shortcode	ico	shortcode
	icecream		shaved_ice
	ice_cream		doughnut
	cookie		birthday
	cake		cupcake
	pie		chocolate_bar
	candy		lollipop
	custard		honey_pot

Drink

ico	shortcode	ico	shortcode
	baby_bottle		milk_glass
	coffee		teapot
	tea		sake
	champagne		wine_glass
	cocktail		tropical_drink

ico	shortcode	ico	shortcode
	beer		beers
	clinking_glasses		tumbler_glass
	cup_with_straw		bubble_tea
	beverage_box		mate
	ice_cube		

Dishware

ico	shortcode	ico	shortcode
	chopsticks		plate_with_cutlery
	fork_and_knife		spoon
	hocho knife		amphora

Travel & Places

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- [Place Geographic](#)
- [Place Building](#)
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- [Place Other](#)
- [Transport Ground](#)
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- [Hotel](#)
- [Time](#)
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Place Map

ico	shortcode	ico	shortcode
	earth_africa		earth_americas
	earth_asia		globe_with_meridians
	world_map		japan
	compass		

Place Geographic

ico	shortcode	ico	shortcode
	mountain_snow		mountain
	volcano		mount_fuji
	camping		beach_umbrella
	desert		desert_island
	national_park		

Place Building

ico	shortcode	ico	shortcode
	stadium		classical_building
	building_construction		bricks
	rock		wood
	hut		houses
	derelict_house		house
	house_with_garden		office
	post_office		european_post_office
	hospital		bank
	hotel		love_hotel

ico	shortcode	ico	shortcode
	convenience_store		school
	department_store		factory
	japanese_castle		european_castle
	wedding		tokyo_tower
	statue_of_liberty		

Place Religious

ico	shortcode	ico	shortcode
	church		mosque
	hindu_temple		synagogue
	shinto_shrine		kaaba

Place Other

ico	shortcode	ico	shortcode
	fountain		tent
	foggy		night_with_stars
	cityscape		sunrise_over_mountains
	sunrise		city_sunset
	city_sunrise		bridge_at_night
	hotsprings		carousel_horse
	ferris_wheel		roller_coaster
	barber		circus_tent



Transport Ground

ico	shortcode	ico	shortcode
	steam_locomotive		railway_car
	bullettrain_side		bullettrain_front
	train2		metro
	light_rail		station
	tram		monorail
	mountain_railway		train
	bus		oncoming_bus
	trolleybus		minibus
	ambulance		fire_engine
	police_car		oncoming_police_car
	taxi		oncoming_taxi
	car		oncoming_automobile
	red_car		
	blue_car		pickup_truck
	truck		articulated_lorry
	tractor		racing_car
	motorcycle		motor_scooter
	manual_wheelchair		motorized_wheelchair
	auto_rickshaw		bike
	kick_scooter		skateboard
	roller_skate		busstop
	motorway		railway_track
	oil_drum		fuelpump
	rotating_light		traffic_light
	vertical_traffic_light		stop_sign

ico	shortcode	ico	shortcode
	construction		

Transport Water

ico	shortcode	ico	shortcode
	anchor		boat sailboat
	canoe		speedboat
	passenger_ship		ferry
	motor_boat		ship

Transport Air

ico	shortcode	ico	shortcode
	airplane		small_airplane
	flight_departure		flight_arrival
	parachute		seat
	helicopter		suspension_railway
	mountain_cableway		aerial_tramway
	artificial_satellite		rocket
	flying_saucer		

Hotel

ico	shortcode	ico	shortcode
	bellhop_bell		luggage

Time

ico	shortcode	ico	shortcode
	hourglass		hourglass_flowingsand
	watch		alarm_clock
	stopwatch		timer_clock
	mantelpiece_clock		clock12
	clock1230		clock1
	clock130		clock2
	clock230		clock3
	clock330		clock4
	clock430		clock5
	clock530		clock6
	clock630		clock7
	clock730		clock8
	clock830		clock9
	clock930		clock10
	clock1030		clock11
	clock1130		

Sky & Weather

ico	shortcode	ico	shortcode
	new_moon		waxing_crescent_moon
	first_quarter_moon		moon
	full_moon		waning_gibbous_moon
	last_quarter_moon		waning_crescent_moon
	crescent_moon		new_moon_with_face

ico	shortcode	ico	shortcode
	first_quarter_moon_with_face		last_quarter_moon_with_fa
	thermometer		sunny
	full_moon_with_face		sun_with_face
	ringed_planet		star
	star2		stars
	milky_way		cloud
	partly_sunny		cloud_with_lightning_and_
	sun_behind_small_cloud		sun_behind_large_cloud
	sun_behind_rain_cloud		cloud_with_rain
	cloud_with_snow		cloud_with_lightning
	tornado		fog
	wind_face		cyclone
	rainbow		closed_umbrella
	open_umbrella		umbrella
	parasol_on_ground		zap
	snowflake		snowman_with_snow
	snowman		comet
	fire		droplet
			

## Activities

- [Event](#)
- [Award Medal](#)
- [Sport](#)
- [Game](#)
- ◀ • [Arts & Crafts](#) ▶

Event

ico	shortcode	ico	shortcode
	jack_o_lantern		christmas_tree
	fireworks		sparkler
	firecracker		sparkles
	balloon		tada
	confetti_ball		tanabata_tree
	bamboo		dolls
	flags		wind_chime
	rice_scene		red_envelope
	ribbon		gift
	reminder_ribbon		tickets
	ticket		

Award Medal

ico	shortcode	ico	shortcode
	medal_military		trophy
	medal_sports		1st_place_medal
	2nd_place_medal		3rd_place_medal

Sport

ico	shortcode	ico	shortcode
	soccer		baseball
	softball		basketball
	volleyball		football
	rugby_football		tennis

ico	shortcode	ico	shortcode
	flying_disc		bowling
	cricket_game		field_hockey
	ice_hockey		lacrosse
	ping_pong		badminton
	boxing_glove		martial_arts_uniform
	goal_net		golf
	ice_skate		fishing_pole_and_fish
	diving_mask		running_shirt_with_sash
	ski		sled
	curling_stone		

Game

ico	shortcode	ico	shortcode
	dart		yo_yo
	kite		gun
	8ball		crystal_ball
	magic_wand		video_game
	joystick		slot_machine
	game_die		jigsaw
	teddy_bear		pinata
	nesting_dolls		spades
	hearts		diamonds
	clubs		chess_pawn
	black_joker		mahjong
	flower_playing_cards		

## Arts & Crafts

ico	shortcode	ico	shortcode
	performing_arts		framed_picture
	art		thread
	sewing_needle		yarn
	knot		

## Objects

- Clothing
- Sound
- Music
- Musical Instrument
- Phone
- Computer
- Light & Video
- Book Paper
- Money
- Mail
- Writing
- Office
- Lock
- Tool
- Science
- Medical
- Household
- Other Object

Clothing

ico	shortcode	ico	shortcode
	eyeglasses		dark_sunglasses
	goggles		lab_coat
	safety_vest		necktie
	shirt		jeans
	tshirt		
	scarf		gloves
	coat		socks
	dress		kimono
	sari		one_piece_swimsuit
	swim_brief		shorts
	bikini		womans_clothes
	purse		handbag
	pouch		shopping
	school_satchel		thong_sandal
	mans_shoe		athletic_shoe
	shoe		
	hiking_boot		flat_shoe
	high_heel		sandal
	ballet_shoes		boot
	crown		womans_hat
	tophat		mortar_board
	billed_cap		military_helmet
	rescue_worker_helmet		prayer_beads
	lipstick		ring
	gem		



Sound

ico	shortcode	ico	shortcode
	<code>mute</code>		<code>speaker</code>
	<code>sound</code>		<code>loud_sound</code>
	<code>loudspeaker</code>		<code>mega</code>
	<code>postal_horn</code>		<code>bell</code>
	<code>no_bell</code>		

Music

ico	shortcode	ico	shortcode
	<code>musical_score</code>		<code>musical_note</code>
	<code>notes</code>		<code>studio_microphone</code>
	<code>level_slider</code>		<code>control_knobs</code>
	<code>microphone</code>		<code>headphones</code>
	<code>radio</code>		

Musical Instrument

ico	shortcode	ico	shortcode
	<code>saxophone</code>		<code>accordion</code>
	<code>guitar</code>		<code>musical_keyboard</code>
	<code>trumpet</code>		<code>violin</code>
	<code>banjo</code>		<code>drum</code>
	<code>long_drum</code>		

Phone

ico	shortcode	ico	shortcode
	iphone		calling
	phone telephone		telephone_receiver
	pager		fax

Computer

ico	shortcode	ico	shortcode
	battery		electric_plug
	computer		desktop_computer
	printer		keyboard
	computer_mouse		trackball
	minidisc		floppy_disk
	cd		dvd
	abacus		

Light & Video

ico	shortcode	ico	shortcode
	movie_camera		film_strip
	film_projector		clapper
	tv		camera
	camera_flash		video_camera
	vhs		mag
	mag_right		candle
	bulb		flashlight

ico	shortcode	ico	shortcode
	izakaya_lantern lantern		diya_lamp

Book Paper

ico	shortcode	ico	shortcode
	notebook_with_decorative_cover		closed_book
	book open_book		green_book
	blue_book		orange_book
	books		notebook
	ledger		page_with_curl
	scroll		page_facing_up
	newspaper		newspaper_roll
	bookmark_tabs		bookmark
	label		

Money

ico	shortcode	ico	shortcode
	moneybag		coin
	yen		dollar
	euro		pound
	money_with_wings		credit_card
	receipt		chart

Mail

ico	shortcode	ico	shortcode
	envelope		e-mail email
	incoming_envelope		envelope_with_arrow
	outbox_tray		inbox_tray
	package		mailbox
	mailbox_closed		mailbox_with_mail
	mailbox_with_no_mail		postbox
	ballot_box		

Writing

ico	shortcode	ico	shortcode
	pencil2		black_nib
	fountain_pen		pen
	paintbrush		crayon
	memo pencil		

Office

ico	shortcode	ico	shortcode
	briefcase		file_folder
	open_file_folder		card_index_dividers
	date		calendar
	spiral_notepad		spiral_calendar
	card_index		chart_with_upwards_trend

ico	shortcode	ico	shortcode
	chart_with_downwards_trend		bar_chart
	clipboard		pushpin
	round_pushpin		paperclip
	paperclips		straight_ruler
	triangular_ruler		scissors
	card_file_box		file_cabinet
	basket		

Lock

ico	shortcode	ico	shortcode
	lock		unlock
	lock_with_ink_pen		closed_lock_with_key
	key		old_key

Tool

ico	shortcode	ico	shortcode
	hammer		axe
	pick		hammer_and_pick
	hammer_and_wrench		dagger
	crossed_swords		bomb
	boomerang		bow_and_arrow
	shield		carpentry_saw
	wrench		screwdriver
	nut_and_bolt		gear
	clamp		balance_scale

ico	shortcode	ico	shortcode
	probing_cane		link
	chains		hook
	toolbox		magnet
	ladder		

Science

ico	shortcode	ico	shortcode
	alembic		test_tube
	petri_dish		dna
	microscope		telescope
	satellite		

Medical

ico	shortcode	ico	shortcode
	syringe		drop_of_blood
	pill		adhesive_bandage
	stethoscope		

Household

ico	shortcode	ico	shortcode
	door		elevator
	mirror		window
	bed		couch_and_lamp
	chair		toilet
	plunger		shower

ico	shortcode	ico	shortcode
	bathtub		mouse_trap
	razor		lotion_bottle
	safety_pin		broom
	basket		roll_of_paper
	bucket		soap
	toothbrush		sponge
	fire_extinguisher		shopping_cart

Other Object

ico	shortcode	ico	shortcode
	smoking		coffin
	headstone		funeral_urn
	nazar_amulet		moyai
	placard		

Symbols

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- [Warning](#)
- [Arrow](#)
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- [Zodiac](#)
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- [Keycap](#)
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Transport Sign

ico	shortcode	ico	shortcode
	atm		put_litter_in_its_place
	potable_water		wheelchair
	mens		womens
	restroom		baby_symbol
	wc		passport_control
	customs		baggage_claim
	left_luggage		

Warning

ico	shortcode	ico	shortcode
	warning		children_crossing
	no_entry		no_entry_sign
	no_bicycles		no_smoking
	do_not_litter		non-potable_water
	no_pedestrians		no_mobile_phones
	underage		radioactive
	biohazard		



## Arrow

ico	shortcode	ico	shortcode
	arrow_up		arrow_upper_right
	arrow_right		arrow_lower_right
	arrow_down		arrow_lower_left
	arrow_left		arrow_upper_left
	arrow_up_down		left_right_arrow
	leftwards_arrow_with_hook		arrow_right_hook
	arrow_heading_up		arrow_heading_down
	arrows_clockwise		arrows_counterclockwise
	back		end
	on		soon
	top		

## Religion

ico	shortcode	ico	shortcode
	place_of_worship		atom_symbol
	om		star_of_david
	wheel_of_dharma		yin_yang
	latin_cross		orthodox_cross
	star_and_crescent		peace_symbol
	menorah		six_pointed_star

## Zodiac

ico	shortcode	ico	shortcode
	aries		taurus

ico	shortcode	ico	shortcode
	gemini		cancer
	leo		virgo
	libra		scorpius
	sagittarius		capricorn
	aquarius		pisces
	ophiuchus		

Av Symbol

ico	shortcode	ico	shortcode
	twisted_rightwards_arrows		repeat
	repeat_one		arrow_forward
	fast_forward		next_track_button
	play_or_pause_button		arrow_backward
	rewind		previous_track_button
	arrow_up_small		arrow_double_up
	arrow_down_small		arrow_double_down
	pause_button		stop_button
	record_button		eject_button
	cinema		low_brightness
	high_brightness		signal_strength
	vibration_mode		mobile_phone_off

Gender

ico	shortcode	ico	shortcode
	female_sign		male_sign

ico	shortcode	ico	shortcode
	transgender_symbol		

Math

ico	shortcode	ico	shortcode
	heavy_multiplication_x		heavy_plus_sign
	heavy_minus_sign		heavy_division_sign
	infinity		

Punctuation

ico	shortcode	ico	shortcode
	bangbang		interrobang
	question		grey_question
	grey_exclamation		exclamation heavy_exclamation_mark
	wavy_dash		

Currency

ico	shortcode	ico	shortcode
	currency_exchange		heavy_dollar_sign

Other Symbol

ico	shortcode	ico	shortcode
	medical_symbol		recycle
	fleur_de_lis		trident

ico	shortcode	ico	shortcode
	name_badge		beginner
	o		white_check_mark
	ballot_box_with_check		heavy_check_mark
	x		negative_squared_cross_mark
	curly_loop		loop
	part_alternation_mark		eight_spoked_asterisk
	eight_pointed_black_star		sparkle
	copyright		registered
	TM		

Keycap

ico	shortcode	ico	shortcode
	hash		asterisk
	zero		one
	two		three
	four		five
	six		seven
	eight		nine
	keycap_ten		

Alphanum

ico	shortcode	ico	shortcode
	capital_abcd		abcd
	1234		symbols
	abc		a

ico	shortcode	ico	shortcode
	ab		b
	cl		cool
	free		information_source
	id		m
	new		ng
	o2		ok
	parking		sos
	up		vs
	koko		sa
	u6708		u6709
	u6307		ideograph_advantage
	u5272		u7121
	u7981		accept
	u7533		u5408
	u7a7a		congratulations
	secret		u55b6
	u6e80		

Geometric

ico	shortcode	ico	shortcode
	red_circle		orange_circle
	yellow_circle		green_circle
	large_blue_circle		purple_circle
	brown_circle		black_circle
	white_circle		red_square
	orange_square		yellow_square

ico	shortcode	ico	shortcode
	green_square		blue_square
	purple_square		brown_square
	black_large_square		white_large_square
	black_medium_square		white_medium_square
	black_medium_small_square		white_medium_small_sq
	black_small_square		white_small_square
	large_orange_diamond		large_blue_diamond
	small_orange_diamond		small_blue_diamond
	small_red_triangle		small_red_triangle_dov
	diamond_shape_with_a_dot_inside		radio_button
	white_square_button		black_square_button

## Flags

- [Flag](#)
- [Country Flag](#)
- [Subdivision Flag](#)

### Flag

ico	shortcode	ico	shortcode
	checkered_flag		triangular_flag_on_post
	crossed_flags		black_flag
	white_flag		rainbow_flag
	transgender_flag		pirate_flag

Country Flag

ico	shortcode	ico	shortcode
	ascension_island		andorra
	united_arab_emirates		afghanistan
	antigua_barbuda		anguilla
	albania		armenia
	angola		antarctica
	argentina		american_samoa
	austria		australia
	aruba		aland_islands
	azerbaijan		bosnia_herzegovina
	barbados		bangladesh
	belgium		burkina_faso
	bulgaria		bahrain
	burundi		benin
	st_barthelemy		bermuda
	brunei		bolivia
	caribbean_netherlands		brazil
	bahamas		bhutan
	bouvet_island		botswana
	belarus		belize
	canada		cocos_islands
	congo_kinshasa		central_african_r
	congo_brazzaville		switzerland
	cote_divoire		cook_islands
	chile		cameroon
	cn		colombia

ico	shortcode	ico	shortcode
	clipperton_island		costa_rica
	cuba		cape_verde
	curacao		christmas_island
	cyprus		czech_republic
	de		diego_garcia
	djibouti		denmark
	dominica		dominican_republic
	algeria		ceuta_melilla
	ecuador		estonia
	egypt		western_sahara
	eritrea		es
	ethiopia		eu european_union
	finland		fiji
	falkland_islands		micronesia
	faroe_islands		fr
	gabon		gb uk
	grenada		georgia
	french_guiana		guernsey
	ghana		gibraltar
	greenland		gambia
	guinea		guadeloupe
	equatorial_guinea		greece
	south_georgia_south_sandwich_islands		guatemala
	guam		guinea_bissau
	guyana		hong_kong



ico	shortcode	ico	shortcode
	heard_mcdonald_islands		honduras
	croatia		haiti
	hungary		canary_islands
	indonesia		ireland
	israel		isle_of_man
	india		british_indian_oc
	iraq		iran
	iceland		it
	jersey		jamaica
	jordan		jp
	kenya		kyrgyzstan
	cambodia		kiribati
	comoros		st_kitts_nevis
	north_korea		kr
	kuwait		cayman_islands
	kazakhstan		laos
	lebanon		st_lucia
	liechtenstein		sri_lanka
	liberia		lesotho
	lithuania		luxembourg
	latvia		libya
	morocco		monaco
	moldova		montenegro
	st_martin		madagascar
	marshall_islands		macedonia
	mali		myanmar
	mongolia		macau

ico	shortcode	ico	shortcode
	northern_mariana_islands		martinique
	mauritania		montserrat
	malta		mauritius
	maldives		malawi
	mexico		malaysia
	mozambique		namibia
	new_caledonia		niger
	norfolk_island		nigeria
	nicaragua		netherlands
	norway		nepal
	nauru		niue
	new_zealand		oman
	panama		peru
	french_polynesia		papua_new_guinea
	philippines		pakistan
	poland		st_pierre_miquelon
	pitcairn_islands		puerto_rico
	palestinian_territories		portugal
	palau		paraguay
	qatar		reunion
	romania		serbia
	ru		rwanda
	saudi_arabia		solomon_islands
	seychelles		sudan
	sweden		singapore
	st_helena		slovenia
	svalbard_jan_mayen		slovakia

ico	shortcode	ico	shortcode
	sierra_leone		san_marino
	senegal		somalia
	suriname		south_sudan
	sao_tome_principe		el_salvador
	sint_maarten		syria
	swaziland		tristan_da_cunha
	turks_caicos_islands		chad
	french_southern_territories		togo
	thailand		tajikistan
	tokelau		timor_leste
	turkmenistan		tunisia
	tonga		tr
	trinidad_tobago		tuvalu
	taiwan		tanzania
	ukraine		uganda
	us_outlying_islands		united_nations
	us		uruguay
	uzbekistan		vatican_city
	st_vincent_grenadines		venezuela
	british_virgin_islands		us_virgin_islands
	vietnam		vanuatu
	wallis_futuna		samoa
	kosovo		yemen
	mayotte		south_africa
	zambia		zimbabwe

## Subdivision Flag

ico	shortcode	ico	shortcode
	england		scotland
	wales		

**typst.**

**typst demo.**

▼ demo0\_0.typ

t

```
// demo0
#import "@preview/cetz:0.3.4": canvas, draw
#import draw: line, content, rect, circle, bezier, group
#set page(width: auto, height: auto, margin: 15pt)
#canvas({
 // Core constants and reusable values
 let plot_width = 24
 let timeline_width = 16
 let time_tick_spacing = 1.0
 let step_width = time_tick_spacing
 let step_gap = 0.1
 let box_width_factor = 0.9
 let rect_height = 0.3
 let border_radius = 0.05
 let gap = 0.15
 let strategy_y_positions = (11, 6.5, 2.0) // Further compressed y-
positions
 // Colors - more pastel version for better text readability
 // Grays
 let dark_gray = rgb("#5D6B7A")
 let cpu_color = rgb("#8899AA")
 let gpu_color = rgb("#6A7A8A")
 let section_bg = rgb("#F7FAFC")
 // Define a consistent CPU light gray color for all strategies
 let cpu_light_gray = cpu_color.lighten(70%)
 // Blues with pastel tones
 let light_blue = rgb("#BBDEFB")
 let mid_blue = rgb("#90CAF9")
 let dark_blue = rgb("#64B5F6")
 // Greens with pastel tones
 let light_green = rgb("#C8E6C9")
 let mid_green = rgb("#A5D6A7")
 let dark_green = rgb("#81C784")
 // Oranges with pastel tones
 let light_orange = rgb("#FFE0B2")
 let mid_orange = rgb("#FFCC80")
 let dark_orange = rgb("#FFB74D")
 // Reds with pastel tones
 let light_red = rgb("#FFCDD2")
 let mid_red = rgb("#EF9A9A")
 let dark_red = rgb("#E57373")
 // Helper functions
 let draw_sim_block(x_pos, y_pos, width, color, label, task_label: "",
```

```

opacity: 100%) = {
 rect(
 (x_pos, y_pos - rect_height / 2),
 (x_pos + width, y_pos + rect_height / 2),
 fill: color.transparentize(100% - opacity),
 stroke: color.darken(10%),
 radius: border_radius,
 name: "block-" + label,
)
 if task_label != "" {
 content((x_pos + width / 2, y_pos), text(size: 8pt)[#task_label],
name: "label-" + label)
 }
}
let draw_structure_rect(x_pos, y_pos, width, color, label, struct_name)
= {
 rect(
 (x_pos, y_pos),
 (x_pos + 0.95 * width, y_pos + 0.1),
 fill: color.transparentize(20%),
 stroke: none,
 name: "block-" + struct_name,
)
 content((x_pos + width / 2, y_pos + 0.15), text(size: 8pt)[#label],
name: "label-" + struct_name, anchor: "south")
}
let draw_util_bar(x_pos, y_pos, percentage, label, is_bad: false) = {
 // Bar showing CPU or GPU utilization - consolidated function
 let width = 2.8
 let height = 0.4
 // Background bar
 rect(
 (x_pos, y_pos - height / 2),
 (x_pos + width, y_pos + height / 2),
 fill: rgb("#E2E8F0"),
 stroke: 0.5pt,
 radius: 0.1,
 name: "bar-bg-" + label,
)
 // Filled portion with appropriate color
 let fill_width = width * percentage / 100
 // Determine color based on percentage and is_bad flag
 let fill_color = if is_bad {
 light_red.darken(20%) // Use darker red for explicitly marked "bad"
 }
 utilization
}

```

```

 } else if percentage < 30 {
 light_red
 } else if percentage < 70 {
 light_orange
 } else {
 light_green
 }
 rect(
 (x_pos, y_pos - height / 2),
 (x_pos + fill_width, y_pos + height / 2),
 fill: fill_color,
 stroke: none,
 radius: (north-west: 0.1, south-west: 0.1),
 name: "bar-fill-" + label,
)
 // Label
 content(
 (rel: (0.03, 0), to: "bar-bg-" + label),
 text(size: 8pt, weight: "bold")[#label #percentage%],
 name: "bar-label-" + label,
 anchor: "center",
)
}
// Title, subtitle and legend
let title_y = 15.0
content(
 (plot_width / 2, title_y),
 text(weight: "bold", size: 16pt)[GPU Batching Strategies for Atomistic
Simulations],
 name: "main-title",
)
content(
 (rel: (0, -1), to: "main-title"),
 text(size: 12pt)[Comparison of Unbatched vs. BinningAutoBatcher vs.
InFlightAutoBatcher],
 name: "subtitle",
)
// Add section backgrounds and scenario setups
let strategies = (
 (strategy_y_positions.at(0), "Unbatched\nSimulations", (80, 5)),
 (strategy_y_positions.at(1), "Binning\nAutoBatcher", (40, 60)),
 (strategy_y_positions.at(2), "InFlight\nAutoBatcher", (60, 90)),
)
for (idx, (y_pos, label, (cpu_util, gpu_util))) in
strategies.enumerate() {

```



```
// Background
rect(
 (0.5, y_pos - 2.0),
 (plot_width - 0.5, y_pos + 2.0),
 fill: section_bg,
 stroke: none,
 radius: border_radius * 3,
 name: "section-bg-" + str(idx),
)
// Scenario label
content((2, y_pos + 1.0), text(weight: "bold", size: 12pt)[#label],
name: "scenario-label-" + str(idx))
// Utilization meters
draw_util_bar(.8, y_pos - 0.1, cpu_util, "CPU Utilization")
// GPU utilization - mark as bad for unbatched simulations (idx == 0)
draw_util_bar(.8, y_pos - 1, gpu_util, "GPU Utilization", is_bad: idx
== 0)
// CPU/GPU labels
content(
 (4.6, y_pos + 1.3),
 text(fill: cpu_color, size: 10pt, weight: "bold")[CPU],
 anchor: "east",
 name: "cpu-label-" + str(idx),
)
content(
 (4.6, y_pos - 0.5),
 text(fill: gpu_color, size: 10pt, weight: "bold")[GPU],
 anchor: "east",
 name: "gpu-label-" + str(idx),
)
// Timeline axis with ticks
line(
 (4.8, y_pos - 1.5),
 (4.8 + timeline_width, y_pos - 1.5),
 stroke: 0.8pt,
 mark: (end: "stealth", fill: black, scale: 0.5),
 name: "timeline-" + str(idx),
)
for t in range(1, 17) {
 let x_pos = 4 + t * time_tick_spacing
 if x_pos <= 4 + timeline_width {
 line(
 (x_pos, y_pos - 1.5 - 0.05),
 (x_pos, y_pos - 1.5 + 0.05),
 stroke: 0.8pt,
```

```

 name: "tick-" + str(idx) + "-" + str(t),
)
 content(
 (rel: (0, -0.2), to: "tick-" + str(idx) + "-" + str(t)),
 text(size: 8pt)[t=#t],
 anchor: "north",
 name: "tick-label-" + str(idx) + "-" + str(t),
)
}
}
// CPU/GPU separator
let separator_y = if idx == 0 { y_pos + 0.6 } else { y_pos + 0.95 }
line(
 (4, separator_y),
 (4 + timeline_width, separator_y),
 stroke: (dash: "dotted", paint: dark_gray.lighten(30%)),
 name: "separator-" + str(idx),
)
}
// 1. Unbatched Simulations
let unbatched_cpu_y = strategy_y_positions.at(0) + 1.4
let unbatched_gpu_y = strategy_y_positions.at(0) - 0.5
// Structure rectangles with consistent pattern
let unbatched_structures = (
 (5.0, 2.0, mid_blue, "Structure 1"),
 (7.0, 2.3, mid_green, "Structure 2"),
 (9.3, 1.8, mid_orange, "Structure 3"),
 (11.1, 2.1, mid_red, "Structure 4"),
 (13.2, 1.8, dark_green, "Structure 5"),
 (15.0, 2.0, dark_green.darken(10%), "Structure 6"),
 (17.0, 1.6, dark_green.darken(15%), "Structure 7"),
 (18.6, 1.5, dark_green.darken(20%), "Structure 8"),
)
for (idx, (x_pos, width, color, label)) in
unbatched_structures.enumerate() {
 draw_structure_rect(x_pos, unbatched_cpu_y, width, color, label,
"unbatched-" + str(idx + 1) + "-cpu")
}
 content((21.8, unbatched_cpu_y - 1.9), text(size: 10pt, weight: "bold")
[... continues], anchor: "center")
// CPU operation blocks in a loop
for x in range(22) {
 let x_offset = 5.25 + x * 0.7
 if x_offset < 20 {
 draw_sim_block(

```

```

 x_offset,
 unbatched_cpu_y - 0.4,
 0.4,
 cpu_light_gray,
 "unbatched-cpu-op-" + str(x),
 task_label: "",
 opacity: 90%,
)
}
}
// GPU blocks with structure IDs and positions
let gpu_blocks = (
 (1, dark_blue, (5.5, 6.5)),
 (2, dark_green, (7.5, 8.5)),
 (3, dark_orange, (9.8, 10.6)),
 (4, dark_red, (11.5, 12.5)),
 (5, dark_green, (13.8, 14.6, 15.4)),
 (6, dark_green.darken(10%), (16.2, 16.8, 17.4, 18.0, 18.6)),
 (7, dark_green.darken(15%), (19.2, 19.8)),
)
for (struct_num, color, positions) in gpu_blocks {
 for (idx, pos) in positions.enumerate() {
 draw_sim_block(
 pos,
 unbatched_gpu_y,
 0.3 * box_width_factor,
 color,
 "unbatched-" + str(struct_num) + "-gpu-" + str(idx + 1),
 task_label: "S" + str(struct_num),
)
 }
}
// 2. BinningAutoBatcher
let binning_cpu_y = strategy_y_positions.at(1) + 1.3
let binning_gpu_y = strategy_y_positions.at(1) - 0.85
// CPU processing blocks - simplified to only show prep batch operations
let cpu_blocks = (
 (5.0, "Prep batch"),
 (10.4, "Prep batch"),
)
for (idx, (x_pos, label)) in cpu_blocks.enumerate() {
 draw_sim_block(x_pos, binning_cpu_y, 1.3, cpu_light_gray, "binning-op-"
+ str(idx), task_label: label)
}
// Batch visualization setup

```

```

let s_colors = (
 dark_blue,
 dark_green,
 dark_orange,
 dark_red,
 dark_blue.darken(10%),
)
let s_labels = ("S1", "S2", "S3", "S4", "S5")
// Bin parameters and active patterns
let bins = (
 (
 // Bin 1: 6 steps with completion patterns
 0,
 false,
 (
 (1, 1, 1, 1, 1), // Step 0: all active
 (1, 1, 1, 1, 1), // Step 1: all active
 (0, 1, 1, 1, 1), // Step 2: structure 1 completes earlier
 (0, 0, 1, 1, 1), // Step 3: structures 1,2 complete
 (0, 0, 0, 1, 0), // Step 4: structures 1,2,3,5 complete
 (0, 0, 0, 1, 0), // Step 5: only 4 remains
),
),
 (
 // Bin 2: 6 steps with different completion pattern
 6,
 true,
 (
 (1, 1, 1, 1, 1), // Step 0: all active
 (1, 1, 1, 1, 1), // Step 1: structure 3 completes early
 (0, 1, 0, 1, 1), // Step 2: structures 1,3 complete
 (0, 1, 0, 0, 1), // Step 3: only 4,5 remain
 (0, 1, 0, 0, 0), // Step 4: only 5 remains
 (0, 1, 0, 0, 0), // Step 5: all complete
),
),
)
// Draw both bins with common function
for (start_step, is_second_bin, patterns) in bins {
 for step in range(patterns.len()) {
 let x_pos = 5.0 + (start_step + step) * (step_width - step_gap)
 let box_width = (step_width - step_gap) * box_width_factor
 let box_x_start = x_pos + ((step_width - step_gap) - box_width) / 2
 // Draw each structure slot
 for i in range(5) {

```

```

 let block_y = binning_gpu_y - 0.3 + i * (rect_height + gap)
 let binning_block_name = "binning-block-" + str(start_step) + "-"
+ str(step) + "-" + str(i)
 if patterns.at(step).at(i) == 1 {
 let color = s_colors.at(i)
 let label = if is_second_bin { "S" + str(i + 6) } else {
s_labels.at(i) }
 rect(
 (box_x_start, block_y - rect_height / 2),
 (box_x_start + box_width, block_y + rect_height / 2),
 fill: color,
 stroke: color.darken(20%),
 radius: border_radius,
 name: binning_block_name,
)
 content(
 (box_x_start + box_width / 2, block_y),
 text(size: 7pt)[#label],
 anchor: "center",
 name: binning_block_name + "-label",
)
 } else {
 // Empty placeholder
 rect(
 (box_x_start, block_y - rect_height / 2),
 (box_x_start + box_width, block_y + rect_height / 2),
 fill: light_red.transparentize(90%),
 stroke: (dash: "dotted", paint:
light_red.transparentize(30%)),
 radius: border_radius,
 name: binning_block_name + "-empty",
)
 }
}
// Add completion label for the last step
if step == patterns.len() - 1 {
 content(
 (box_x_start - 0.25, binning_gpu_y - 0.45),
 text(size: 8pt, fill: dark_gray, style: "italic")[Batch #if
is_second_bin [2] else [1] Complete],
 anchor: "center",
 name: "batch-" + str(if is_second_bin { 2 } else { 1 }) + "-
complete-label",
)
}
}

```

```

 }
 }
 for (x_pos, percentage) in ((5.6, 100), (8.0, 60), (9.8, 30), (11, 100),
(13.5, 60), (15.5, 20)) {
 content(
 (x_pos, binning_gpu_y + 2.7),
 text(size: 8pt, fill: dark_gray)[#percentage% GPU],
 anchor: "center",
)
 }
 // Explanatory arrows and labels - removing the isolated red arrow
 content(
 (10.5, binning_gpu_y - 1.5),
 text(size: 7pt, fill: dark_red, style: "italic")[Must wait for batch
to complete, resulting in underutilized GPU],
 anchor: "center",
 name: "must-wait",
)
 let batch_end_x = 5.0 + 6 * (step_width - step_gap)
 let timeline_y = strategy_y_positions.at(1) - 1.5
 line(
 "must-wait.north",
 (batch_end_x, timeline_y),
 stroke: (dash: "dotted", paint: dark_red),
 mark: (end: "stealth", fill: dark_red, scale: 0.4, offset: 0.05),
 name: "must-wait-arrow",
)
 // 3. In-flightAutoBatcher
 let inflight_cpu_y = strategy_y_positions.at(2) + 1.3
 let inflight_gpu_y = strategy_y_positions.at(2) - 0.9
 // Structure colors and labels
 let all_colors = (
 dark_blue,
 dark_green,
 dark_orange,
 dark_red,
 dark_blue.darken(10%),
 dark_green.darken(10%),
 dark_orange.darken(10%),
 dark_red.darken(10%),
 dark_blue.darken(20%),
 dark_green.darken(20%),
)
 let all_labels = ("S1", "S2", "S3", "S4", "S5", "S6", "S7", "S8", "S9",
"S10")

```

```

// Structure placement by time step - fix structure continuity
let structures_by_step = (
 (0, 1, 2, 3, 4), // t=1: S1-S5
 (0, 1, 2, 3, 4), // t=2: S1-S5 (S1 continues for longer)
 (0, 6, 2, 3, 4), // t=3: S1 continues, S6 replaces S2, S3-S5 continue
 (5, 6, 7, 3, 4), // t=4: S5, S6-S8, S4 continue (S1 completed, S5
swapped in)
 (5, 6, 7, 8, 4), // t=5: S5-S9, S4 (S3 completed, S8 swapped in)
 (5, 6, 7, 8, 9), // t=6: S5-S10 (S4 completed, S9 swapped in)
 (5, 6, 7, 8, 9), // t=7: still fully filled
 (5, 6, 7, -1, -1), // t=8: final step, partially filled (about 50%) -
using S6 instead of isolated S4
)
// Place CPU operations equidistant, one for each time step
// Create the same number of prep boxes as structure steps (8 steps)
// Draw CPU operations - one per time step
for step in range(structures_by_step.len()) {
 let x_pos = 5.0 + step * (step_width - step_gap)
 let box_width = (step_width - step_gap) * box_width_factor * 0.7 //
Make boxes 10% wider (70% instead of 60%)
 // Left align by using 10% of remaining space as left margin instead
of centering
 let box_x_start = x_pos + ((step_width - step_gap) - box_width) * 0.1
 // Use the same generic gray for all prep blocks
 draw_sim_block(
 box_x_start,
 inflight_cpu_y,
 box_width,
 cpu_light_gray,
 "inflight-prep-" + str(step),
 task_label: "Prep",
)
}
// Draw all structures by step
for step in range(structures_by_step.len()) {
 let x_pos = 5.0 + step * (step_width - step_gap)
 let box_width = (step_width - step_gap) * box_width_factor
 let box_x_start = x_pos + ((step_width - step_gap) - box_width) / 2
 // Create a batch name for this step
 let batch_name = "inflight-batch-" + str(step)
 // Draw each structure position
 for pos in range(5) {
 let struct_idx = structures_by_step.at(step).at(pos)
 let block_y = inflight_gpu_y - 0.3 + pos * (rect_height + gap)
 let block_name = "inflight-block-" + str(step) + "-" + str(pos)

```

```

 if struct_idx != -1 {
 let color = all_colors.at(struct_idx)
 let label = all_labels.at(struct_idx)
 // Structure box
 rect(
 (box_x_start, block_y - rect_height / 2),
 (box_x_start + box_width, block_y + rect_height / 2),
 fill: color,
 stroke: color.darken(20%),
 radius: border_radius,
 name: block_name,
)
 content(
 (box_x_start + box_width / 2, block_y),
 text(size: 7pt)[#label],
 name: block_name + "-label",
 anchor: "center",
)
 // Swap-in indicator
 if step > 0 {
 let prev_structures = structures_by_step.at(step - 1)
 let prev_block_name = "inflight-block-" + str(step - 1) + "-" +
str(pos)
 if pos < prev_structures.len() and prev_structures.at(pos) !=
struct_idx {
 // Simple diagonal line for swap-in indicator
 line(
 (rel: (0, 0), to: block_name + ".south-west"), // End at the
block edge
 (rel: (-0.25, -0.3), to: block_name + ".south-west"), //
Start from outside
 stroke: (dash: "dotted", paint: dark_green),
 mark: (start: "stealth", fill: dark_green, scale: 0.3),
 name: block_name + "-swap-indicator",
)
 }
 }
 } else {
 // Empty placeholders with dotted lines to indicate unused GPU
cycles
 rect(
 (box_x_start, block_y - rect_height / 2),
 (box_x_start + box_width, block_y + rect_height / 2),
 fill: light_red.transparentize(90%),
 stroke: (dash: "dotted", paint: light_red.transparentize(30%)),

```



```

 radius: border_radius,
 name: block_name + "-empty",
)
}
}
// Mark the last batch specially
if step == structures_by_step.len() - 1 {
 // Store a reference to the last batch for later reference
 let final_block_name = "inflight-block-" + str(step) + "-0"
 // Instead of using absolute positioning, use relative positioning
 // based on the named elements we just created
 content(
 (rel: (box_width + 1.7, 0.2), to: final_block_name),
 text(size: 8pt, fill: dark_gray, style: "italic")[Final batch
partially filled],
 anchor: "center",
 name: "final-batch-label",
)
}
}
// First label with frame
content(
 (plot_width / 2, -0.8),
 text(
 size: 9pt,
 weight: "bold",
 fill: dark_green,
) [In-flight batching achieves highest GPU utilization and maximizes
predictions per unit time],
 frame: "rect",
 fill: light_green.transparentize(70%),
 stroke: dark_green,
 padding: 3pt,
 radius: border_radius,
)
// Summary notes
content(
 (plot_width / 2, -3),
 box(width: 50em)[
 Unbatched: Each simulation runs sequentially with most
calculations on CPU and minimal GPU utilization\
 Binning: Fixed-size batches improve GPU utilization but can't
adapt to varying simulation completion times\
 In-flight: Dynamic reallocation eliminates GPU idle time by
immediately adding new structures when others complete. Color changes

```

```
indicate in-flight structure replacement.
],
frame: "rect",
fill: rgb("#F7FAFC"),
stroke: 0.5pt,
padding: (10pt, 10pt, 0pt),
radius: border_radius,
)
})
```

GPU Batching Strategies for Atomistic Simulations

Comparison of Unbatched vs. BinningAutoBatcher vs. InFlightAutoBatcher



**Unbatched:** Each simulation runs sequentially with most calculations on CPU and minimal GPU utilization  
**Binning:** Fixed-size batches improve GPU utilization but can't adapt to varying simulation completion times  
**In-flight:** Dynamic reallocation eliminates GPU idle time by immediately adding new structures when others complete. Color changes indicate in-flight structure replacement.